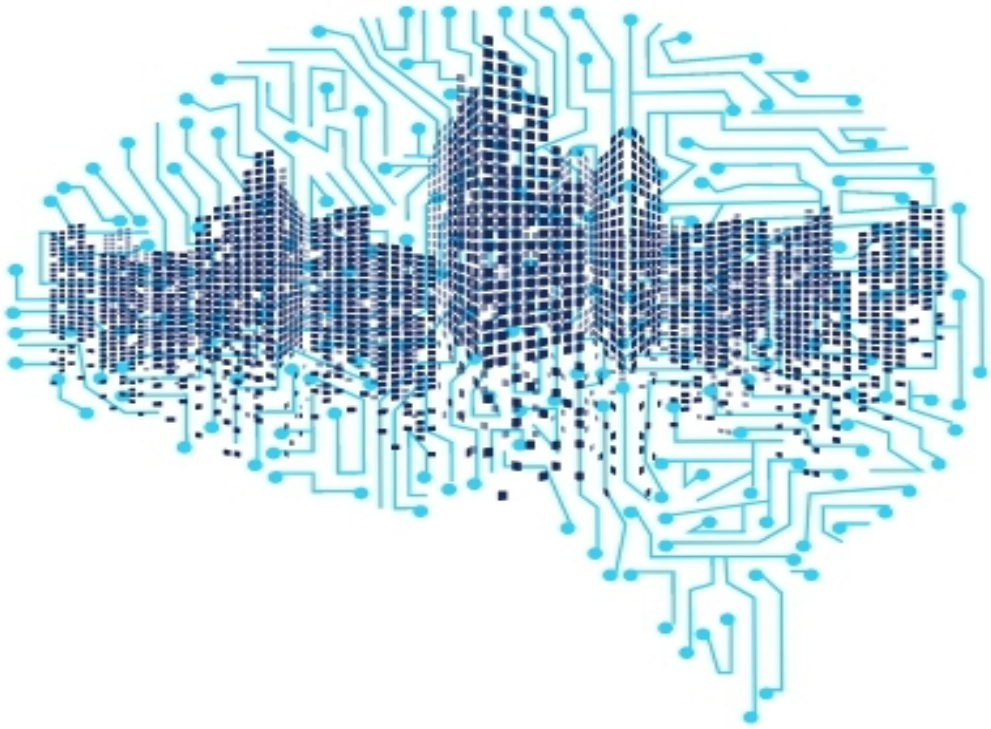
PART 2 OF 2

Building LLMs for Production



Enhancing LLM Abilities and Reliability with Prompting, Fine-Tuning, and RAG

TOWARDS AI

LOUIS-FRANÇOIS BOUCHARD

LOUIE PETERS

```
loader = PyPDFLoader("The One Page Linux Manual.pdf")
pages = loader.load_and_split()

text_splitter = RecursiveCharacterTextSplitter(
chunk_size=50,
chunk_overlap=10,
length_function=len,
)

docs = text_splitter.split_documents(pages)
for doc in docs:
print(doc)
```

```
page_content='THE ONE PAGE LINUX MANUALA summary of useful'
metadata={'source': 'The One Page Linux Manual.pdf', 'page': 0}
page_content='of useful Linux commands'
metadata={'source': 'The One Page Linux Manual.pdf', 'page': 0}
page_content='Version 3.0 May 1999 squadron@powerup.com.au'
metadata={'source': 'The One Page Linux Manual.pdf', 'page': 0}
page_content='Starting & Stopping'
metadata={'source': 'The One Page Linux Manual.pdf', 'page': 0}
...
page_content='- includes'
metadata={'source': 'The One Page Linux Manual.pdf', 'page': 1}
page_content='handy command summary. Visit:'
metadata={'source': 'The One Page Linux Manual.pdf', 'page': 1}
page_content='www.powerup.com.au/~squadron'
metadata={'source': 'The One Page Linux Manual.pdf', 'page': 1}
```

In the example, we set up an instance of the `RecursiveCharacterTextSplitter` class with specific parameters, using the default character set `["\n\n", "\n", " ", ""]` for splitting the text.

Initially, the text is segmented using two newline characters (`\n\n`). If the resulting chunks exceed the desired size of 50 characters, the class attempts to divide the text using a single newline character (`\n`). The result is a series of documents comprising the segmented text.

To incorporate a token counter, you can create a function that determines the token count in a text and use this as the `length_function` parameter. This modification ensures that the chunk lengths are calculated based on tokens rather than character counts.

Splitting Text with Foreign Linguistic Structures with NLTK

The NLTKTextSplitter splitter leverages the capabilities of the Natural Language Toolkit (NLTK) library for text segmentation. This class can make splitting decisions based on linguistic structure, thanks to many hand-written rules created by linguistics. This means it can more intelligently identify sentence boundaries, paragraph divisions, and other natural language cues that depend on the specific language used, resulting in more semantically coherent chunks of text.

?? If it is your first time using this package, you will need to install the NLTK library using ! pip install -q nltk.

```
from langchain.text_splitter import NLTKTextSplitter

# Load a long document
with open('homecloudsuperadmin/scrape-chain/langchain/LLM.txt',
encoding= 'unicode_escape') as f:
sample_text = f.read()

text_splitter = NLTKTextSplitter(chunk_size=500)
texts = text_splitter.split_text(sample_text)
print(texts)
```

```
['Building LLM applications for production\nApr 11, 2023 \x95 Chip Huyen text \n\nA question that I\x92ve has been asked a lot recently is how large language models (LLMs) will change machine learning workflows.\n\nAfter working with several companies who are working with LLM applications and personally going down a rabbit hole building my applications, I realized two things:\n\nIt\x92s easy to make something cool with LLMs, but very hard to make something production-ready with them.', 'LLM limitations are exacerbated by a lack of engineering rigor in prompt engineering, partially due to the ambiguous nature of natural languages, and partially due to the nascent nature of the field.\n\nThis post consists of three parts .\n\nPart 1 discusses the key challenges of productionizing LLM applications and the solutions that I\x92ve seen.\n\nPart 2[…]
```

Consider using this tokenizer, particularly for foreign languages whose syntax is not based on words separated by whitespaces, such as Chinese (Mandarin and Cantonese), Japanese, and Thai.

Splitting Text with Foreign Linguistic Structures with Spacy

The SpacyTextSplitter splitter is another class for separating large text documents into smaller parts of a specific size. The SpacyTextSplitter splitter is an alternative to NLTK-based sentence-splitting algorithms. To use this splitter, first construct a SpacyTextSplitter object and set the chunk_size property. This size is decided by a length function, which measures the number of characters in the text by default.

```
from langchain.text_splitter import SpacyTextSplitter

# Load a long document
with open('/home/cloudsuperadmin/scrape-chain/langchain/LLM.txt',
encoding= 'unicode_escape') as f:
sample_text = f.read()

# Instantiate the SpacyTextSplitter with the desired chunk size
text_splitter = SpacyTextSplitter(chunk_size=500, chunk_overlap=20)

# Split the text using SpacyTextSplitter
texts = text_splitter.split_text(sample_text)

# Print the first chunk
print(texts[0])
```

```
Building LLM applications for production
Apr 11, 2023 • Chip Huyen text

A question that I've been asked a lot recently is how large language models (LLMs) will change machine learning workflows.

After working with several companies who are working with LLM applications and personally going down a rabbit hole building my applications, I realized two things:

It’s easy to make something cool with LLMs, but very hard to make something production-ready with them.
```

Splitting Text with the Markdown Format

The MarkdownTextSplitter splitter specializes in segmenting text formatted with Markdown, targeting elements like headers, code blocks, or dividers. This splitter is a specialized version of the RecursiveCharacterSplitter splitter, adapted for Markdown with specific separators. These separators are, by default, aligned with standard Markdown syntax but can be tailored by supplying a

customized list of characters during the initialization of the MarkdownTextSplitter instance. The default measurement for chunk size is based on the number of characters, as determined by the provided length function. When creating an instance, an integer value can be specified to adjust the chunk size to specific requirements.

```
from langchain.text_splitter import MarkdownTextSplitter

markdown_text = """
#

# Welcome to My Blog!

## Introduction
Hello everyone! My name is **John Doe** and I am a *software developer*. I specialize in Python, Java, and JavaScript.

Here's a list of my favorite programming languages:

1. Python
2. JavaScript
3. Java

You can check out some of my projects on [GitHub](https://github.com).

## About this Blog
In this blog, I will share my journey as a software developer. I'll post tutorials, my thoughts on the latest technology trends, and occasional book reviews.

Here's a small piece of Python code to say hello:

\``` python
def say_hello(name):
print(f"Hello, {name}!")

say_hello("John")
\```

Stay tuned for more updates!
```

```
## Contact Me
Feel free to reach out to me on [Twitter](https://twitter.com) or send me an email at
johndoe@email.com.

"""

markdown_splitter = MarkdownTextSplitter(chunk_size=100, chunk_overlap=0)
docs = markdown_splitter.create_documents([markdown_text])

print(docs)
```

```
[Document(page_content='# \n\n# Welcome to My Blog!', metadata={}),
Document(page_content='Introduction', metadata={}),
Document(page_content='Hello everyone! My name is **John Doe** and I am a *software
developer*. I specialize in Python,', metadata={}),
Document(page_content='Java, and JavaScript.', metadata={}),
Document(page_content="Here's a list of my favorite programming languages:\n\n1.
Python\n2. JavaScript\n3. Java", metadata={}),
Document(page_content='You can check out some of my projects on [GitHub]
(https://github.com).', metadata={}),
Document(page_content='About this Blog', metadata={}),
Document(page_content="In this blog, I will share my journey as a software developer. I'll
post tutorials, my thoughts on", metadata={}),
Document(page_content='the latest technology trends, and occasional book reviews.',
metadata={}),
Document(page_content="Here's a small piece of Python code to say hello:", metadata={}),
Document(page_content='\\```python\ndef say_hello(name):\n print(f"Hello,
{name}!")\n\nsay_hello("John")\n\\', metadata={}),
Document(page_content='Stay tuned for more updates!', metadata={}),
Document(page_content='Contact Me', metadata={}),
Document(page_content='Feel free to reach out to me on [Twitter](https://twitter.com) or
send me an email at', metadata={}),
Document(page_content='johndoe@email.com.', metadata={})]
```

Identifying Markdown syntax elements (e.g., headings, lists, and code blocks) enables intelligent content division based on its structural hierarchy, leading to semantically coherent segments.

Splitting Text with Tokens

The `TokenTextSplitter` splitter offers a key advantage over splitters like the `CharacterTextSplitter` splitter by ensuring that the resulting chunks contain, at most, a specified number of tokens. This is very useful when using LLMs

with limited context window since it allows us to determine the maximum number of chunks that can be inserted into the prompt without making it bigger than the maximum context size.

This splitter first converts the input text into BPE (Byte Pair Encoding, seen in Chapter 2) tokens and then groups them into chunks. Then, the tokens within each chunk are converted back to their original text. To use this splitter, the `tiktoken` Python package is necessary, which can be installed using the command with `! pip install -q tiktoken`.

```
from langchain.text_splitter import TokenTextSplitter

# Load a long document
with open('homecloudsuperadmin/scrape-chain/langchain/LLM.txt',
encoding= 'unicode_escape') as f:
sample_text = f.read()

# Initialize the TokenTextSplitter with desired chunk size and overlap
text_splitter = TokenTextSplitter(chunk_size=100, chunk_overlap=50)

# Split into smaller chunks
texts = text_splitter.split_text(sample_text)
print(texts[0])
```

```
Building LLM applications for production
Apr 11, 2023 • Chip Huyen text

A question that I've been asked a lot recently is how large language models (LLMs) will change machine learning workflows. After working with several companies who are working with LLM applications and personally going down a rabbit hole building my applications, I realized two things:

It’s easy to make something cool with LLMs, but very hard to make something with production.
```

The `chunk_size` parameter in `TokenTextSplitter` dictates the maximum number of BPE tokens each chunk can contain, whereas `chunk_overlap` determines the extent of token overlap between successive chunks.

A potential but small downside of the `TokenTextSplitter` splitter is the increased computational effort required to convert text into BPE tokens and vice versa. For quicker and more straightforward text segmentation, the `CharacterTextSplitter` splitter may be a preferable option because it offers a more direct and less computationally intensive approach to dividing text.

The above text splitters are the most commonly used approaches to splitting text. The next section focuses on how these text splitters can be leveraged to enhance your application with an example where we build a customer support Q&A chatbot powered by LLMs.

Similarity Search and Vector Embeddings

OpenAI's embedding models are versatile and can generate embeddings that we can use for similarity searches. In this section, we will use the OpenAI API to create embeddings from a collection of documents and then perform a similarity search using cosine similarity.

To begin, install the necessary packages using the command: `!pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken scikit-learn`.

Next, set your OpenAI API key as an environment variable:

```
export OPENAI_API_KEY="your-api-key"
```

Now, let's generate embeddings for our documents and perform a similarity search.

Begin by defining a list of documents as strings. This text data will be used for the subsequent steps.

Next, compute the embeddings for each document using the `OpenAIEmbeddings` class. Set the embedding model to `"text-embedding-ada-002"`. This model will generate embeddings for each document, transforming them into vector representations of their semantic content.

?? Computing embeddings using a proprietary model like the `"text-embedding-ada-002"` model incurs costs due to the usage of the API. Embedding models are usually very cheap compared to using an LLM for inference, but the total cost can become significant if millions of text chunks are used. However, in this tutorial (and in all the other tutorials in this book), we will compute embeddings of a few texts, keeping the costs to a minimum. Check the OpenAI pricing page to see the current pricing for that model.

Similarly, convert the query string to an embedding. The query string contains the text for which we want to find the most similar document.

After obtaining the embeddings for our documents and the query, calculate the cosine similarity between the query embedding and each document embedding. Cosine similarity is a widely used distance metric to assess the similarity between two vectors. In our context, it provides a series of similarity scores, each indicating how similar the query is to each document.

Once we have these similarity scores, we identify the document that is most similar to the query. This is achieved by finding the index of the highest similarity score and then retrieving the corresponding document from our collection.

```
import openai
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity
from langchain.embeddings import OpenAIEmbeddings

# Define the documents
documents = [
"The cat is on the mat.",
"There is a cat on the mat.",
"The dog is in the yard.",
"There is a dog in the yard.",
]

# Initialize the OpenAIEmbeddings instance
embeddings = OpenAIEmbeddings(model="text-embedding-ada-002")

# Generate embeddings for the documents
document_embeddings = embeddings.embed_documents(documents)

# Perform a similarity search for a given query
query = "A cat is sitting on a mat."
query_embedding = embeddings.embed_query(query)
```

```
# Calculate similarity scores
similarity_scores = cosine_similarity([query_embedding], document_embeddings)[0]

# Find the most similar document
most_similar_index = np.argmax(similarity_scores)
most_similar_document = documents[most_similar_index]

print(f"Most similar document to the query '{query}':")
print(most_similar_document)
```

```
Most similar document to the query 'A cat is sitting on a mat.':
The cat is on the mat.
```

Embedding models can be open-source or proprietary. The choice of embedding model type depends on specific requirements, explained in the next sections.

Open-Source Embedding Models

As previously discussed, embedding models belong to a specific category of machine learning models designed to transform discrete data points into vector representations. In natural language processing, these discrete elements can be words, sentences, or entire documents. The resulting vector representations, referred to as embeddings, aim to encapsulate the semantic essence of the original data. For instance, words with similar meanings, such as “cat” and “kitten,” are likely to have closely aligned embeddings. These embeddings possess high dimensionality and are utilized to capture subtle semantic differences.

Open-source embedding models offer flexibility, transparency, and cost savings, allowing customization and peer-reviewed improvements. However, they may lack support, and quality can vary. Proprietary models typically provide better performance, stability, and support but come with higher costs, limited customization, and potential vendor lock-in. The choice depends on specific needs like control vs. convenience.

One key advantage of using embeddings is their ability to enable mathematical operations for interpreting semantic meanings. As illustrated, a common application involves calculating the cosine similarity between

two embeddings to assess the semantic closeness of associated words or documents. The following example shows how to use an open-source embedding model for this task.

The example uses the model “ sentence-transformers/all-mpnet-base-v2 ”, a pre-trained model for converting sentences into semantically meaningful vectors.

In the model_kwargs settings, ensure the computations are carried out on the CPU.

Before executing the following code, install the sentence transformer library with the command ! pip install sentence_transformers===2.2.2 .

This library has robust pre-trained models specialized in generating embedding representations.

Next, define a list of documents - these are the chunks of text we wish to turn into semantic embeddings and generate the embeddings. This is accomplished by invoking the embed_documents function on the Hugging Face Embeddings instance and supplying our document list as an argument. This method goes through each document and returns a list of embeddings.

```
from langchain.llms import HuggingFacePipeline
from langchain.embeddings import HuggingFaceEmbeddings

model_name = "sentence-transformers/all-mpnet-base-v2"
model_kwargs = {'device': 'cpu'}
hf = HuggingFaceEmbeddings(model_name=model_name, model_kwargs=model_kwargs)

documents = ["Document 1", "Document 2", "Document 3"]
doc_embeddings = hf.embed_documents(documents)
```

These embeddings are now ready for further processing, such as classification, grouping, or similarity analysis. They reflect our original documents in a machine-readable format, allowing us to conduct complicated semantic computations.

Cohere Embeddings

While the choice between closed-source and open-source embedding models will ultimately depend on your specific needs, including budget,

control, and flexibility, closed-source models generally offer more accuracy, speed, and performance. Several companies offer closed-source embedding models, we chose Cohere because it is optimized for tasks like semantic search, text classification, and recommendation systems. They provide a multilingual embedding model that maps texts in different languages into the same semantic vector space, and therefore, it's ideal in multilingual applications, especially search functionalities. This model, distinct from their English language model, employs dot product computations as a similarity metric for improved performance. The model produces 768-dimensional embeddings.

An API key is required to use the Cohere API. Navigate to the Cohere Dashboard, create a new account, or log in. Once logged in, the dashboard offers an easy-to-use interface for creating and managing API keys.

After acquiring the API key, create an instance of the CohereEmbeddings class with LangChain using the " embed-multilingual-v2.0 " model.

Next, prepare a list of texts in various languages. Use the embed_documents() method to generate distinctive embeddings for each text.

To showcase these embeddings, each text is printed with its corresponding embedding. For clarity, only the first five dimensions of each embedding are displayed.

For this, the Cohere package must be installed by executing ! pip install cohere .

```
import cohere
from langchain.embeddings import CohereEmbeddings

# Initialize the CohereEmbeddings object
cohere = CohereEmbeddings(
model="embed-multilingual-v2.0",
cohere_api_key="your_cohere_api_key"
)

# Define a list of texts
texts = [
"Hello from Cohere!",
```

```
"مرحبًا من كوهير!",
"Hallo von Cohere!",
"Bonjour de Cohere!",
"¡Hola desde Cohere!",
"Olá do Cohere!",
"Ciao da Cohere!",
"您好，来自 Cohere！",
"कोहेरे से नमस्ते !"
]

# Generate embeddings for the texts
document_embeddings = cohere.embed_documents(texts)

# Print the embeddings
for text, embedding in zip(texts, document_embeddings):
print(f"Text: {text}")
print(f"Embedding: {embedding[:5]}") # print first 5 dimensions of
```

```
Text: Hello from Cohere!
Embedding: [0.23439695, 0.50120056, -0.048770234, 0.13988855, -0.1800725]

Text: مرحبًا من كوهير!
Embedding: [0.25350592, 0.29968268, 0.010332941, 0.12572688, -0.18180023]

Text: Hallo von Cohere!
Embedding: [0.10278442, 0.2838264, -0.05107267, 0.23759139, -0.07176493]

Text: Bonjour de Cohere!
Embedding: [0.15180704, 0.28215882, -0.056877363, 0.117460854, -0.044658754]

Text: ¡Hola desde Cohere!
Embedding: [0.2516583, 0.43137372, -0.08623046, 0.24681088, -0.11645193]

Text: Olá do Cohere!
Embedding: [0.18696906, 0.39113742, -0.046254586, 0.14583701, -0.11280365]

Text: Ciao da Cohere!
Embedding: [0.1157251, 0.43330532, -0.025885003, 0.14538017, 0.07029742]

Text: 您好，来自 Cohere！
Embedding: [0.24605744, 0.3085744, -0.11160592, 0.266223, -0.051633865]

Text: कोहेरे से नमस्ते !
Embedding: [0.19287698, 0.6350239, 0.032287907, 0.11751755, -0.2598813]
```

In this example, LangChain proved helpful in simplifying the integration of an embedding model like Cohere's multilingual embedding model into a developer's workflow. This is one of the main advantages of working with libraries like LangChain and LlamaIndex: they make it easy to work with different types of models and switch between them without the need for big code changes.

Embeddings are typically computed once and then stored in a vector database for future use. Vector databases, like most systems, can be open-source or proprietary, with respective pros and cons.

We explored how vector embeddings and similarity searches can be performed using OpenAI and various embedding models. In the next section, we'll see how LangChain, OpenAI, and Deep Lake come together to build a conversational AI system. This system efficiently retrieves relevant information and answers user queries, demonstrating the power of embeddings in real-world applications.

Tutorial 1: A Customer Support Q&A Chatbot

Traditionally, chatbots were built for specific user intents, formed from sets of sample questions and their corresponding answers. For example, a "Restaurant Recommendations" intent might include questions like "Can you suggest a good Italian restaurant nearby?" or "Where is the best sushi in town?" along with answers such as "La Trattoria is a great Italian restaurant in the area" or "Sushi Palace is highly recommended for sushi."

In this framework, the chatbot matches user queries to the closest intent to generate a response. However, with the advancement of LLMs, the approach to developing chatbots is also evolving. Modern chatbots are increasingly sophisticated, offering more dynamic and nuanced responses to a broader array of user questions.

Large language models (LLMs) can significantly enhance chatbot functionality by linking broader intents with documents from a knowledge

base (KB). This approach simplifies the handling of intents and enables more tailored responses to user queries.

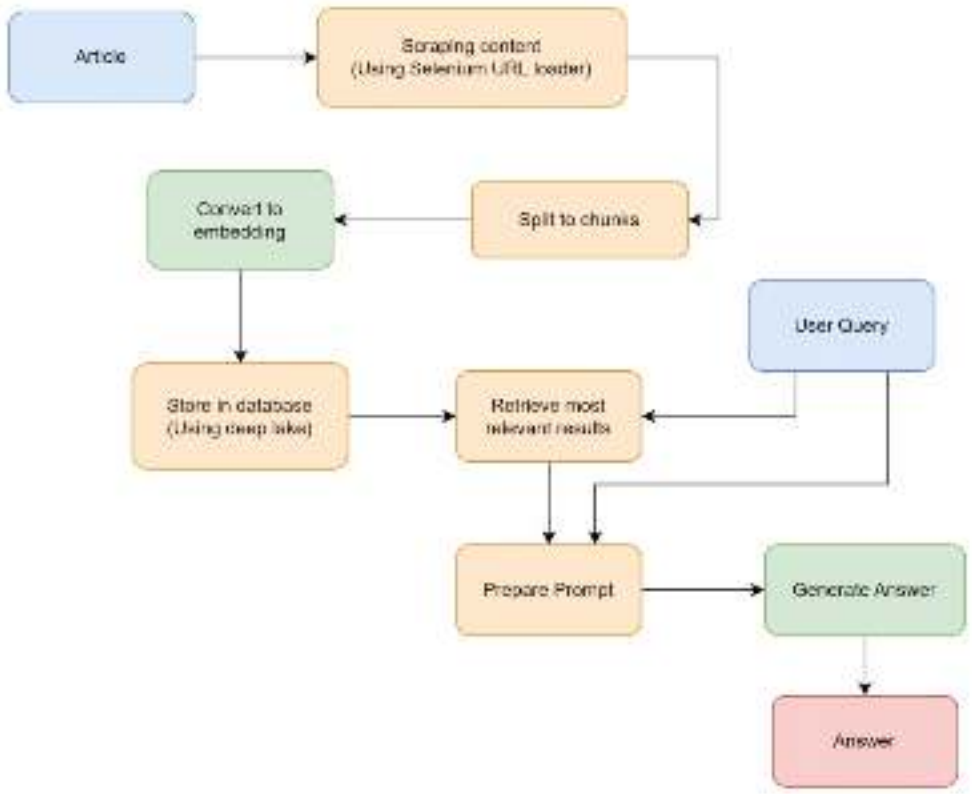
However, LLMs have a maximum context size. For example, GPT-3 had a maximum prompt size limit of approximately 4,000 tokens. While this is substantial, when including an entire knowledge base in a single prompt it proves insufficient. Future advancements may overcome this limitation, and the current best LLMs have a bigger context size than GPT-3 (e.g., GPT-4 had an 8,000 token limit, and GPT-4o 128,000). In the meantime, it is crucial to devise solutions that work within the current constraints. Plus, more input tokens do not mean the right information is retrieved. It may add more noise than good information and contribute to hallucinations. It is thus necessary to curate and add relevant information to the input.

In the following tutorial, we will create a customer service chatbot, leveraging LLMs and text splitters so that the contextual information added to the prompt is within the context size constraints of the specific LLM.

Workflow

This project aims to build a chatbot that leverages GPT-3 to search for answers within documents. The documents in this tutorial will be general technical guides scraped from the web.

The workflow for the project is explained below:



Our customer support Q&A Chatbot pipeline.

The first step is extracting content from internet webpages, dividing it into small parts, computing its embeddings, and storing it in Deep Lake. Subsequently, a user's query retrieves the most relevant segments from Deep Lake. These segments are then incorporated into a prompt to generate the final response by the LLM.

To begin managing conversations with GPT-3 and storing data in Deep Lake, configure the `OPENAI_API_KEY` and `ACTIVELOOP_TOKEN` environment variables with your respective API keys and tokens.

We will use the `SeleniumURLLoader` class from the LangChain toolkit, which relies on the `unstructured` and `selenium` Python libraries. These can be installed via `pip`. Installing the latest version of these libraries is advisable, but this code has been explicitly tested with version `0.7.7`.

```
pip install unstructured selenium Install the required packages with the following command: ! pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken  and import the necessary libraries.
from langchain.embeddings.openai import OpenAIEmbeddings
from langchain.vectorstores import DeepLake
```

```
from langchain.text_splitter import CharacterTextSplitter
from langchain import OpenAI
from langchain.document_loaders import SeleniumURLLoader
from langchain import PromptTemplate
```
These libraries offer features essential for developing a context-aware question-answering system, including managing OpenAI embeddings, handling vector storage, segmenting text, and interfacing with the OpenAI API. They play a crucial role in creating a system that combines information retrieval and text generation.

For this example, our chatbot's database will contain technical content.

```
# we'll use information from the following articles
urls = ['https://beebom.com/what-is-nft-explained/',
'https://beebom.com/how-delete-spotify-account/',
'https://beebom.com/how-download-gif-twitter/',
'https://beebom.com/how-use-chatgpt-linux-terminal/',
'https://beebom.com/how-delete-spotify-account/',
'https://beebom.com/how-save-instagram-story-with-music/',
'https://beebom.com/how-install-pip-windows/',
'https://beebom.com/how-check-disk-usage-linux/']
```

Split the documents into chunks and compute their embeddings

Load the documents from the provided URLs and split them into chunks using the CharacterTextSplitter with a chunk size of 1000 and no overlap:

```
# use the selenium scraper to load the documents
loader = SeleniumURLLoader(urls=urls)
docs_not_splitted = loader.load()

# we split the documents into smaller chunks
text_splitter = CharacterTextSplitter(chunk_size=1000, chunk_overlap=0)
docs = text_splitter.split_documents(docs_not_splitted)
```

Next, obtain the embeddings using OpenAIEmbeddings and store them in a cloud-based Deep Lake vector database. In a real-world project, one might upload a whole website or course to Deep Lake to search across thousands or millions of documents. Utilizing a cloud serverless Deep Lake dataset enables applications in various locations to access the same centralized dataset without deploying a vector store on a specific computer.

Change the code below to include your Activeloop organization ID. By default, your org id is your username.

```
# Before executing the following code, make sure to have
# your OpenAI key saved in the “OPENAI_API_KEY” environment variable.
embeddings = OpenAIEmbeddings(model="text-embedding-ada-002")

# create Deep Lake dataset
# TODO: use your organization id here. (by default, org id is your username)
my_activeloop_org_id = "<YOUR-ACTIVELOOP-ORG-ID>"
my_activeloop_dataset_name = "langchain_course_customer_support"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"
db = DeepLake(dataset_path=dataset_path, embedding_function=embeddings)

# add documents to our Deep Lake dataset
db.add_documents(docs)
```

To retrieve the most similar chunks to a given query, we can use the similarity_search method of the Deep Lake vector database: # let's see the top relevant documents to a specific query
query = "how to check disk usage in linux?"
docs = db.similarity_search(query)
print(docs[0].page_content) The previous code will show something like the following output:

Home How To How to Check Disk Usage in Linux (4 Methods)

How to Check Disk Usage in Linux (4 Methods)

Beebom Staff

Last Updated: February 21, 2023 3:15 pm

There may be times when you need to download some important files or transfer some photos to your Linux system, but face a problem of insufficient disk space. You head over to your file manager to delete the large files which you no longer require, but you have no clue which of them are occupying most of your disk space. In this article, we will show some easy methods to check disk usage in Linux from both the terminal and the GUI application.

Monitor Disk Usage in Linux (2023)

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Check Disk Space Using the df Command

Display Disk Usage in Human Readable FormatDisplay Disk Occupancy of a Particular Type

Check Disk Usage using the du Command

Display Disk Usage in Human Readable FormatDisplay Disk Usage for a Particular DirectoryCompare Disk Usage of Two Directories

Craft a prompt for using the suggested strategies

For this chatbot, we will develop a prompt template that includes role prompting, knowledge base information, and the user's question:

```
# let's write a prompt for a customer support chatbot that
# answer questions using information extracted from our db
template = """You are an exceptional customer support chatbot that gently answer questions.

You know the following context information.

{chunks_formatted}

Answer to the following question from a customer. Use only information from the previous context information. Do not invent stuff.

Question: {query}

Answer:"""

prompt = PromptTemplate(
input_variables=["chunks_formatted", "query"],
template=template,
)
```

The template positions the chatbot as an advanced customer support tool and requires input variables: `chunks_formatted`, consisting of pre-arranged segments from articles, and `query`, representing the customer's inquiry. The

objective is to generate a precise and factual answer based on the provided segments, ensuring the information is accurate and not fabricated.

Generate answers with the LLM

To generate a response, we retrieve the top-k (e.g., top-3) chunks most similar to the user's question, format the prompt, and send it to the model at 0 temperature.

```
# the full pipeline

# user question
query = "How to check disk usage in linux?"

# retrieve relevant chunks
docs = db.similarity_search(query)
retrieved_chunks = [doc.page_content for doc in docs]

# format the prompt
chunks_formatted = "\n\n".join(retrieved_chunks)
prompt_formatted = prompt.format(chunks_formatted=chunks_formatted, query=query)

# generate answer
llm = OpenAI(model="gpt-3.5-turbo", temperature=0)
answer = llm(prompt_formatted)
print(answer)
```

> You can check disk usage in Linux using the df command to check disk space and the du command to check disk usage. You can also use the GUI application to check disk usage in a human readable format. For more information, please refer to the article "How to Check Disk Usage in Linux (4 Methods)" on Beebom.

In the previous example, while the chatbot generally functions effectively, there are scenarios where it might not perform as expected.

For instance, when the LLM is asked, "Is the Linux distribution free?" and provided with a context document about kernel features, it may incorrectly respond with "Yes, the Linux distribution is free to download and use," even if this information isn't in the provided context. Generating incorrect information is a significant concern for customer service chatbots.

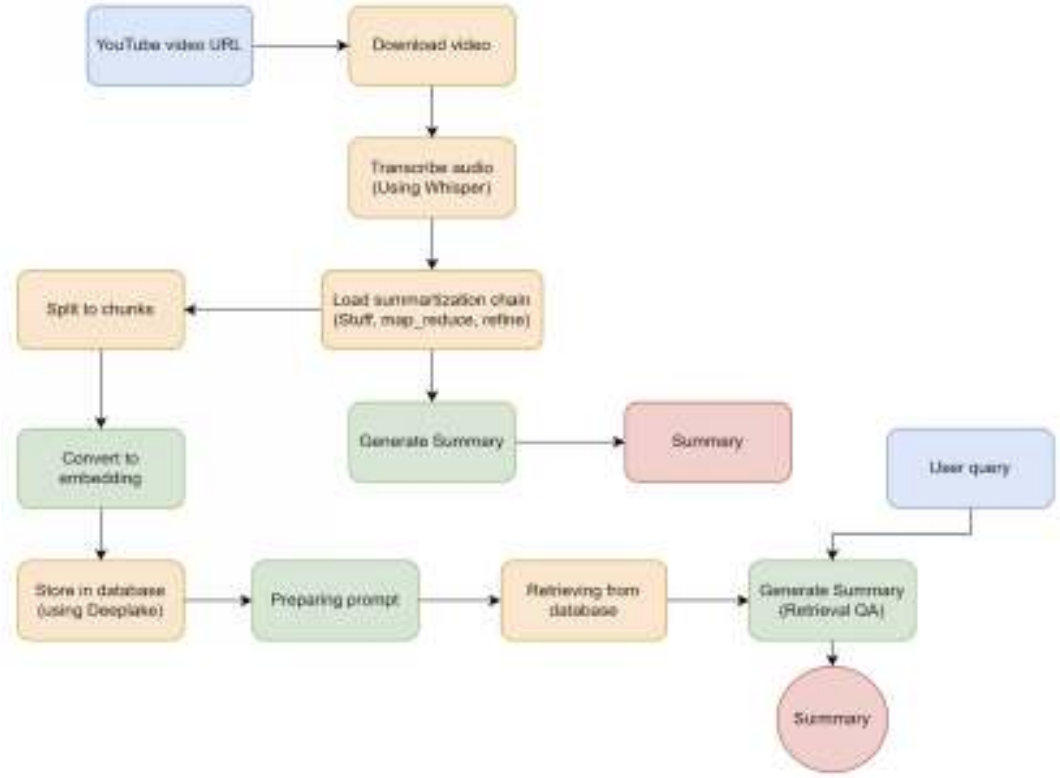
The likelihood of the LLM producing inaccurate information decreases when the context directly includes the answer to the user's query. However, since user inquiries are often short and vague, there's always a possibility that the semantic search phase doesn't retrieve appropriate documents. Embeddings play an important role in making the semantic search phase work as intended.

Tutorial 2: A YouTube Video Summarizer Using Whisper and LangChain

- Find the Notebook for this section at towardsai.net/book.

The project involves a series of steps, starting with downloading the audio file from YouTube. Once the audio file is obtained, it is transcribed using Whisper. After the transcription is complete, the text is summarized using LangChain, employing three different approaches: stuff, refine, and map_reduce. Finally, multiple transcriptions are added to the DeepLake database to enable question-answering for those videos.

The following diagram explains what we are going to do in this project:



Our YouTube video summarizer pipeline.

As usual, start by installing the packages using the command: ! pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken , yt_dlp , and openai-whisper .

```
!pip install -q yt_dlp
!pip install -q git+https://github.com/openai/whisper.git Next, install ffmpeg; it is a prerequisite for the yt_dlp package. Note that ffmpeg comes pre-installed on Google Colab instances. The commands below detail how to install ffmpeg on both Mac and Ubuntu operating systems.
# MacOS (requires https://brew.sh/)
brew install ffmpeg

# Ubuntu
sudo apt install ffmpeg
```

💡 If you're working on an operating system that hasn't been mentioned earlier (like Windows), read how to install ffmpeg at towardsai.net/book. It contains comprehensive, step-by-step instructions on the installation process.

Next, add the API key for OpenAI and Deep Lake services to the environment variables. This can be accomplished either through the load_dotenv function, which reads values from a .env file, or by executing the following code. It is essential to ensure the confidentiality of your API keys, as they grant access to these services and can be used by anyone with the key.

```
import os
os.environ['OPENAI_API_KEY'] = "<OPENAI_API_KEY>"
os.environ['ACTIVELOOP_TOKEN'] = "<ACTIVELOOP_TOKEN>"
```

The tutorial teaches how to programmatically summarize a video featuring Yann LeCun, a notable computer scientist and AI researcher. The video covers LeCun's thoughts on the challenges associated with large language models. However, the code would work with any other video as long as it can be summarized using only its audio (as the model won't know what is shown in the video) and that ideally contains only a few speakers. Video podcasts are ideal for this project.

The download_mp4_from_youtube() function downloads the highest quality mp4 video file from a given YouTube link and saves it to a specified path and filename. To use this function, simply copy and paste the URL of the chosen video into it.

```
import yt_dlp

def download_mp4_from_youtube(url):
# Set the options for the download
filename = 'lecuninterview.mp4'
ydl_opts = {
'format': 'bestvideo[ext=mp4]+bestaudio[ext=m4a]/best[ext=mp4]',
'outtmpl': filename,
'quiet': True,
}

    # Download the video file
with yt_dlp.YoutubeDL(ydl_opts) as ydl:
result = ydl.extract_info(url, download=True)

url = "https://www.youtube.com/watch?v=mBjPyte2ZZo"
download_mp4_from_youtube(url)
```

Now that the video MP4 has been downloaded, the next step is to transcribe its audio using a speech-to-text model. One of the currently most popular

open-source speech-to-text models is OpenAI’s Whisper.

Transcribing Audio with Whisper

Whisper is an advanced automatic speech recognition system developed by OpenAI. It’s trained on a dataset of 680,000 hours of multilingual and multitasking supervised data from the web. This extensive and diverse dataset contributes to the system’s ability to efficiently manage accents, background noise, and technical jargon.

The previously installed whisper package includes the `.load_model()` method, which downloads the model and transcribes a video file. Several models are available: `tiny`, `base`, `small`, `medium`, and `large`, for balancing accuracy and processing speed. We will use the `'base'` model for this example.

```
import whisper

model = whisper.load_model("base")
result = model.transcribe("lecuninterview.mp4")
print(result['text'])
```

> *home*cloudsuperadmin/.local/lib/python3.9/site-packages/whisper/transcribe.py:114: UserWarning: FP16 is not supported on CPU; using FP32 instead
> warnings.warn("FP16 is not supported on CPU; using FP32 instead")
>
> Hi, I'm Craig Smith, and this is I on A On. This week I talked to Jan LeCoon, one of the seminal figures in deep learning development and a long-time proponent of self-supervised learning. Jan spoke about what's missing in large language models and his new joint embedding predictive architecture which may be a step toward filling that gap. He also talked about his theory of consciousness and the potential for AI systems to someday exhibit the features of consciousness. It's a fascinating conversation that I hope you'll enjoy. Okay, so Jan, it's great to see you again. I wanted to talk to you about where you've gone with so supervised learning since last week's spoke. In particular, I'm interested in how it relates to large language models because they have really come on stream since we spoke. In fact, in your talk about JEPA, which is joint embedding predictive architecture. […and so on]

The result is generated as raw text and can be saved to a text file.

```
with open ('text.txt', 'w') as file:
file.write(result['text']) Once the transcription is ready, the next step is to split it into chunks using a text splitter and then use the chunks to generate a summary.
```

Splitting Text and Generating the Summary

Import the necessary classes and utilities from the LangChain library.

```
from langchain import OpenAI, LLMChain
from langchain.chains.mapreduce import MapReduceChain
from langchain.prompts import PromptTemplate
from langchain.chains.summarize import load_summarize_chain

llm = OpenAI(model_name="gpt-3.5-turbo", temperature=0)
```

The code below creates an instance of the `RecursiveCharacterTextSplitter` class. This class is used to split input text into more manageable, smaller segments.

```
from langchain.text_splitter import RecursiveCharacterTextSplitter


text_splitter = RecursiveCharacterTextSplitter(
chunk_size=1000, chunk_overlap=0, separators=[" ", ",", "\n"]
)
```

The `RecursiveCharacterTextSplitter` is set up with a `chunk_size` of 1000 characters, without any `chunk_overlap`, and uses spaces, commas, and newline characters as separators.

Now, open the previously saved text file and use the `.split_text()` method to segment the transcripts.

```
from langchain.docstore.document import Document

with open('text.txt') as f:
text = f.read()

texts = text_splitter.split_text(text)
docs = [Document(page_content=t) for t in texts[:4]]
```

Each `Document` object is initialized with the content of a chunk from the `texts` list. The `[:4]` slice notation indicates that only the first four chunks will be used to create the `Document` objects.

```
from langchain.chains.summarize import load_summarize_chain
import textwrap

chain = load_summarize_chain(llm, chain_type="map_reduce")
```

```
output_summary = chain.run(docs)
wrapped_text = textwrap.fill(output_summary, width=100)
print(wrapped_text)
```

```
Craig Smith interviews Jan LeCoon, a deep learning developer and proponent of self-
supervised learning, about his new joint embedding predictive architecture and his theory of
consciousness. Jan's research focuses on self-supervised learning and its use for pre-training
transformer architectures, which are used to predict missing words in a piece of text.
Additionally, large language models are used to predict the next word in a sentence, but it is
difficult to represent uncertain predictions when applying this to video.
```

?? The `textwrap` library in Python provides a convenient way to wrap and format plain text by adjusting line breaks in an input paragraph. It is particularly useful when displaying text within a limited width, such as in console outputs, emails, or other formatted text displays. The library includes convenience functions like `wrap`, `fill`, and `shorten`, as well as the `TextWrapper` class that handles most of the work. If you're curious, find more information on Text wrapping and filling at towardsai.net/book.

The following code shows the prompt template used with the `map_reduce` chain type. The mapreduce process first summarizes each document separately using a language model (Map step), turning each into a new document. Then, it combines all of them into one document (Reduce step) to form the final summary.

```
print(chain.llm_chain.prompt.template) Write a concise summary of the following:\n\n\n"
{text}"\n\n\n CONCISE SUMMARY: The "stuff"  approach involves using all text from the
transcribed video in a single prompt, which is a basic and straightforward method. However, it might
not be the most efficient for handling large volumes of text, and it may overflow the maximum
context size of the LLM. It's always advised to monitor the number of tokens used in the prompts
because different LLM APIs may behave differently when the maximum token limit is exceeded.
Some APIs may throw errors, while others may silently use only the first part of the prompt until it
fills the LLM context size completely, thus generating an output with an incomplete prompt.
```

To generate the summary for this tutorial, we're going to experiment with the prompt below, which will output the summary as bullet points:

```
prompt_template = """Write a concise bullet point summary of the following:

{text}

CONSCISE SUMMARY IN BULLET POINTS:"""
```

```
BULLET_POINT_PROMPT = PromptTemplate(template=prompt_template,
input_variables=["text"])
```

We also initialized the summarization chain using the `stuff` as `chain_type` and the prompt above:

```
chain = load_summarize_chain(llm,
chain_type="stuff",
prompt=BULLET_POINT_PROMPT)

output_summary = chain.run(docs)

wrapped_text = textwrap.fill(output_summary,
width=1000,
break_long_words=False,
replace_whitespace=False)
print(wrapped_text)
```

- Jan LeCoon is a seminal figure in deep learning development and a long time proponent of self-supervised learning
- Discussed his new joint embedding predictive architecture which may be a step toward filling the gap in large language models
- Theory of consciousness and potential for AI systems to exhibit features of consciousness
- Self-supervised learning revolutionized natural language processing
- Large language models lack a world model and are generative models, making it difficult to represent uncertain predictions LangChain provides the flexibility to create custom prompts tailored to specific needs. For instance, if the objective is to receive a summarization output in French, one can construct a prompt instructing the language model to generate a summary in French.

The `'refine'` summarization chain is an approach designed to generate more precise and context-sensitive summaries. This method follows an iterative process to enhance the summary by incorporating additional context as needed. In practice, it initiates by summarizing the first text chunk. Subsequently, the generated summary is enriched with new information from each subsequent chunk. It can produce more accurate and context-aware summaries than chains like `'stuff'` and `'map_reduce'`, at the cost of more LLM calls.

```
chain = load_summarize_chain(llm, chain_type="refine")
output_summary = chain.run(docs)
wrapped_text = textwrap.fill(output_summary, width=100)
print(wrapped_text) Craig Smith interviews Jan LeCoon, a deep learning developer and proponent of
```

self-supervised learning, about his new joint embedding predictive architecture and his theory of consciousness. Jan discusses the gap in large language models and the potential for AI systems to exhibit features of consciousness. He explains how self-supervised learning has revolutionized natural language processing through the use of transformer architectures for pre-training, such as taking a piece of text, removing some of the words, and replacing them with black markers to train a large neural net to predict the words that are missing. This technique has been used in practical applications such as contact moderation systems on Facebook, Google, YouTube, and more. Jan also explains how this technique can be used to represent uncertain predictions in generative models, such as predicting the missing words in a text, or predicting the missing frames in a video.

Adding Transcripts to Deep Lake

Now, let’s continue by transcribing more videos, storing the transcriptions in the Deep Lake database, and retrieving information via the QA chain.

First, we need to make slight modifications to the video downloading script to enable it to work with a list of URLs.

```
import yt_dlp

def download_mp4_from_youtube(urls, job_id):
# This will hold the titles and authors of each downloaded video
video_info = []

    for i, url in enumerate(urls):
# Set the options for the download
file_temp = f'./{job_id}_{i}.mp4'
ydl_opts = {
'format': 'bestvideo[ext=mp4]+bestaudio[ext=m4a]/best[ext=mp4]',
'outtmpl': file_temp,
'quiet': True,
}

    # Download the video file
with yt_dlp.YoutubeDL(ydl_opts) as ydl:
result = ydl.extract_info(url, download=True)
title = result.get('title', "")
author = result.get('uploader', "")
```

```
    # Add the title and author to our list
video_info.append((file_temp, title, author))

    return video_info

urls=["https://www.youtube.com/watch?v=mBjPyte2ZZo&t=78s",
"https://www.youtube.com/watch?v=cjs7QKJNVYM",]
vides_details = download_mp4_from_youtube(urls, 1)
```

Transcribe the videos using Whisper as we previously saw and save the results in a text file.

```
import whisper

# load the model
model = whisper.load_model("base")

# iterate through each video and transcribe
results = []
for video in vides_details:
result = model.transcribe(video[0])
results.append( result['text'] )
print(f"Transcription for {video[0]}:\n{result['text']}\n")

with open ('text.txt', 'w') as file:
file.write(results['text'])
```

```
Transcription for ./1_0.mp4:
Hi, I'm Craig Smith and this is I on A On. This week I talk to Jan LeCoon, one of the seminal figures in deep learning development and a long time proponent of self-supervised learning. Jan spoke about what's missing in large language models and about his new joint embedding predictive architecture which may be a step toward filling that gap. He also talked about his theory of consciousness and the potential for AI systems to someday exhibit the features of consciousness...
```

Next, load the texts from the file and use the text splitter to split the text into chunks with zero overlap before storing them in Deep Lake.

```
from langchain.text_splitter import RecursiveCharacterTextSplitter

# Load the texts
with open('text.txt') as f:
```

```
text = f.read()
texts = text_splitter.split_text(text)

# Split the documents
text_splitter = RecursiveCharacterTextSplitter(
chunk_size=1000, chunk_overlap=0, separators=[" ", ",", "\n"]
)
texts = text_splitter.split_text(text)
```

Pack all the chunks into a `Document` LangChain object:

```
from langchain.docstore.document import Document

docs = [Document(page_content=t) for t in texts[:4]]
```

Import Deep Lake and build a database with embedded documents:

```
from langchain.vectorstores import DeepLake
from langchain.embeddings.openai import OpenAIEmbeddings

embeddings = OpenAIEmbeddings(model='text-embedding-ada-002')

# create Deep Lake dataset
# TODO: use your organization id here. (by default, org id is your username)
my_activeloop_org_id = "<YOUR-ACTIVELOOP-ORG-ID>"
my_activeloop_dataset_name = "langchain_course_youtube_summarizer"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"

db = DeepLake(dataset_path=dataset_path, embedding_function=embeddings)
db.add_documents(docs)
```

To retrieve the information from the database, we need to construct a retriever object:

```
retriever = db.as_retriever()
retriever.search_kwargs['distance_metric'] = 'cos'
retriever.search_kwargs['k'] = 4
```

The `distance_metric` parameter defines how the `Retriever` computes similarity or "distance" between data points in the database. By setting this parameter to `'cos'`, cosine similarity is employed as the distance metric. Cosine similarity,

a standard measure in information retrieval, evaluates the similarity between two non-zero vectors in a vector space by measuring the cosine of the angle between them. This metric is frequently used to assess the similarity between documents or text segments. Additionally, setting 'k' to 4 instructs the Retriever to return the four most similar results based on the distance metric.

We can also create a custom prompt template to use within the QA chain. The RetrievalQA chain queries similar contents from the database, using the retrieved records as context for answering questions. Custom prompts allow for tailored tasks, such as retrieving documents and summarizing the output in a bullet point.

```
from langchain.prompts import PromptTemplate
prompt_template = """Use the following pieces of transcripts from a video to answer the question in bullet points and summarized. If you don't know the answer, just say that you don't know, don't try to make up an answer.

{context}

Question: {question}
Summarized answer in bullter points:"""
PROMPT = PromptTemplate(
template=prompt_template, input_variables=["context", "question"]
)
```

Define the custom prompt using the chain_type_kwargs argument and the stuff variation as the chain type.

```
from langchain.chains import RetrievalQA

chain_type_kwargs = {"prompt": PROMPT}
qa = RetrievalQA.from_chain_type(llm=llm,
chain_type="stuff",
retriever=retriever,
chain_type_kwargs=chain_type_kwargs)

print(qa.run("Summarize the mentions of google according to their AI program"))
```

```
• Google has developed an AI program to help people with their everyday tasks.
• The AI program can be used to search for information, make recommendations, and provide personalized experiences.
• Google is using AI to improve its products and services, such as Google Maps and Google
```

```
Assistant.
• Google is also using AI to help with medical research and to develop new technologies.
```

You can always change the prompt and experiment with different types of chains to discover the best combination for your project's needs and limits.

We've created this lesson by adapting the code from github.com/idontcalculate/langchain.

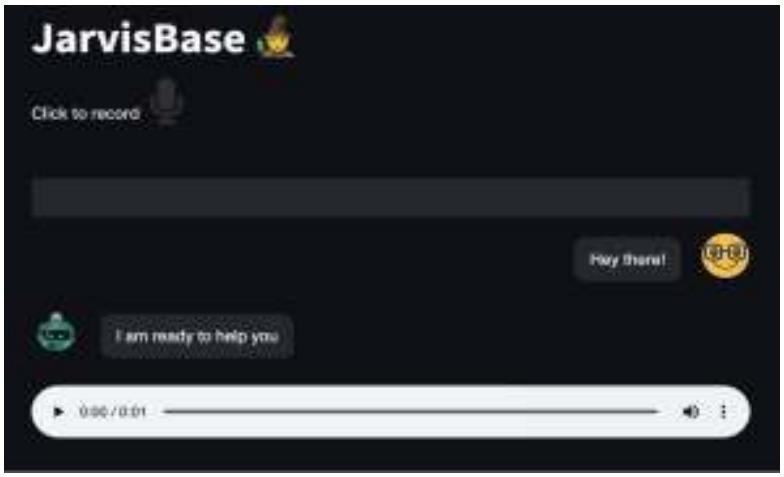
Tutorial 3: A Voice Assistant for Your Knowledge Base

- Find the link to the "JarvisBase" GitHub repository for this section at towardsai.net/book.

This tutorial focuses on voice capabilities. In this project, we will create a voice assistant that integrates OpenAI's Whisper to convert voice inputs into text. After the transcription is complete, voice responses will be generated using Eleven Labs, a company renowned for its high-quality text-to-speech API that adeptly captures emotion and tone. Using this API will ensure that the voice assistant can communicate with users in a clear and natural tone.

At the heart of this project is a question-answering system. When a question is asked, the system retrieves relevant documents from this database. These documents and the question are then processed by a large language model (LLM). The LLM utilizes this information to formulate an appropriate response.

The project includes the Streamlit service to create an interactive user interface (UI), enhancing user interaction with the assistant. This basic frontend allows users to ask questions using either natural language or voice and generates responses in both text and audio formats.



Start by installing the necessary libraries for this project. While it's best to use the most recent versions of these packages for the best results, the provided code was used with specific versions. They can be installed using the pip packages manager. A link to this requirement file is accessible at towardsai.net/book.

```
langchain==0.0.208
deeplake==3.6.5
openai==0.27.8
tiktoken==0.4.0
elevenlabs==0.2.18
streamlit==1.23.1
beautifulsoup4==4.11.2
audio-recorder-streamlit==0.0.8
streamlit-chat==0.0.2.2
```

Set the API keys and tokens. They need to be set in the environment variable as described below.

```
import os

os.environ['OPENAI_API_KEY'] = '<your-openai-api-key>'
os.environ['ELEVEN_API_KEY'] = '<your-eleven-api-key>'
os.environ['ACTIVELOOP_TOKEN'] = '<your-activeloop-token>'
```

You can use the previously generated OpenAI and Activeloop API keys. To get the ELEVEN_API_KEY , follow these steps: 1. Go to https://elevenlabs.io/ and click "Sign Up" to create an account.

2. Once you have created an account, log in and navigate to the "API" section.

3. Click the “Create API key” button and follow the prompts to generate a new API key.
4. Copy the API key and paste it into your code where it says “your-eleven-api-key” in the `ELEVEN_API_KEY` variable.

Getting Content from Hugging Face Hub

We’ll begin by gathering documents from the Hugging Face Hub. These articles will form the foundation of our voice assistant’s knowledge base. We will use web scraping methods to collect relevant knowledge documents.

Let’s look at and run the `script.py` file.

Import the required modules, load environment variables, and establish the path for Deep Lake. It also creates an instance of `OpenAIEmbeddings`, which will be used later to embed the scraped articles:

```
import os
import requests
from bs4 import BeautifulSoup
from langchain.embeddings.openai import OpenAIEmbeddings
from langchain.vectorstores import DeepLake
from langchain.text_splitter import CharacterTextSplitter
from langchain.document_loaders import TextLoader
import re

# TODO: use your organization id here. (by default, org id is your username)
my_activeloop_org_id = "<YOUR-ACTIVELOOP-ORG-ID>"
my_activeloop_dataset_name = "langchain_course_jarvis_assistant"
dataset_path= 'hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}'

embeddings = OpenAIEmbeddings(model_name="text-embedding-ada-002")
```

Compile a list of relative URLs that lead to knowledge documents from the Hugging Face Hub. To do this, define the function `get_documentation_urls()` and attach these relative URLs to the base URL of the Hugging Face Hub using another function, `construct_full_url()`, effectively establishing full URLs that can be accessed directly.

```
def get_documentation_urls():
# List of relative URLs for Hugging Face documentation pages,
# commented a lot of these because it would take too long to scrape
# all of them
return [
'docshuggingface_hub/guides/overview',
'docshuggingface_hub/guides/download',
'docshuggingface_hub/guides/upload',
'docshuggingface_hub/guides/hf_file_system',
'docshuggingface_hub/guides/repository',
'docshuggingface_hub/guides/search',
# You may add additional URLs here or replace all of them
]

def construct_full_url(base_url, relative_url):
# Construct the full URL by appending the relative URL to the base URL
return base_url + relative_url
```

The script compiles the gathered content from various URLs. This is executed by the `scrape_all_content()` function, which invokes the `scrape_page_content()` function for each URL. Next, the resulting text is stored in a file.

```
def scrape_page_content(url):
# Send a GET request to the URL and parse the HTML response using
# BeautifulSoup
response = requests.get(url)
soup = BeautifulSoup(response.text, 'html.parser')
# Extract the desired content from the page (in this case, the body text)
text=soup.body.text.strip()
# Remove non-ASCII characters
text = re.sub(r'[\x00-\x08\x0b-\x0c\x0e-\x1f\x7f-\xff]', '', text)
# Remove extra whitespace and newlines
text = re.sub(r'\s+', ' ', text)
return text.strip()

def scrape_all_content(base_url, relative_urls, filename):
# Loop through the list of URLs, scrape content and add it to the
# content list
content = []
for relative_url in relative_urls:
```

```
full_url = construct_full_url(base_url, relative_url)
scraped_content = scrape_page_content(full_url)
content.append(scraped_content.rstrip('\n'))

    # Write the scraped content to a file
with open(filename, 'w', encoding='utf-8') as file:
for item in content:
file.write("%s\n" % item)

    return content
```

Loading and Splitting Texts

To prepare the gathered text into our vector database, the content is first retrieved from the file using the load_docs() function, which separates it into distinct documents. These documents are then divided into smaller segments using the split_docs() function.

The command text_splitter = CharacterTextSplitter(chunk_size=1000, chunk_overlap=0) initializes a text splitter designed to segment the text into character-based chunks. It divides the documents into sections of roughly 1000 characters with no overlapping content in the consecutive sections within docs.

```
# Define a function to load documents from a file
def load_docs(root_dir, filename):
# Create an empty list to hold the documents
docs = []
try:
# Load the file using the TextLoader class and UTF-8 encoding
loader = TextLoader(os.path.join(
root_dir, filename), encoding='utf-8')
# Split the loaded file into separate documents and add them to the list
# of documents
docs.extend(loader.load_and_split())
except Exception as e:
# If an error occurs during loading, ignore it and return an empty list
# of documents
pass
# Return the list of documents
return docs

def split_docs(docs):
text_splitter = CharacterTextSplitter(chunk_size=1000, chunk_overlap=0)
```

```
return text_splitter.split_documents(docs)
```

Embedding and Storing in Deep Lake

The next phase is embedding the articles and storing them in Deep Lake.

The following code sets up a Deep Lake instance, specifying the dataset path and the `OpenAIEmbeddings` function as an embedding function to use. The `OpenAIEmbeddings` function transforms the text segments into their embedding vectors, a format compatible with the vector database. With the `.add_documents` method, the texts are processed and stored within the database.

```
# Define the main function
def main():
base_url = 'https://huggingface.co'
# Set the name of the file to which the scraped content will be saved
filename='content.txt'
# Set the root directory where the content file will be saved
root_dir ='./'
relative_urls = get_documentation_urls()
# Scrape all the content from the relative URLs and save it to the content
# file
content = scrape_all_content(base_url, relative_urls, filename)
# Load the content from the file
docs = load_docs(root_dir, filename)
# Split the content into individual documents
texts = split_docs(docs)
# Create a DeepLake database with the given dataset path and embedding
# function
db = DeepLake(dataset_path=dataset_path, embedding_function=embeddings)
# Add the individual documents to the database
db.add_documents(texts)
# Clean up by deleting the content file
os.remove(filename)

# Call the main function if this script is being run as the main program
if __name__ == '__main__':
main()
```

These steps are organized within the main function. It establishes the required parameters, activates the outlined functions, and manages the entire procedure, from scraping web content to integrating it into the Deep Lake database. It also removes the content file, ensuring a clean workspace.

Creating the Voice Assistant

You can find the relevant code in the `chat.py` file within the GitHub repository. To test it out, execute `streamlit run chat.py` .

The libraries used below are essential to create web applications with Streamlit. They help manage audio input, generate text responses, and efficiently access information stored in the Deep Lake:

```
import os
import openai
import streamlit as st
from audio_recorder_streamlit import audio_recorder
from elevenlabs import generate
from langchain.chains import RetrievalQA
from langchain.chat_models import ChatOpenAI
from langchain.embeddings.openai import OpenAIEmbeddings
from langchain.vectorstores import DeepLake
from streamlit_chat import message

# Constants
TEMP_AUDIO_PATH = "temp_audio.wav"
AUDIO_FORMAT = "audio/wav"

# Load environment variables from .env file and return the keys
openai.api_key = os.environ.get('OPENAI_API_KEY')
eleven_api_key = os.environ.get('ELEVEN_API_KEY')
```

Create an instance that points to our Deep Lake vector database: `def load_embeddings_and_database(active_loop_data_set_path):`

```
embeddings = OpenAIEmbeddings()
db = DeepLake(
dataset_path=active_loop_data_set_path,
read_only=True,
embedding_function=embeddings
)
return db
```

Next, prepare the code for transcribing audio:

```
# Transcribe audio using OpenAI Whisper API
def transcribe_audio(audio_file_path, openai_key):
openai.api_key = openai_key
try:
with open(audio_file_path, "rb") as audio_file:
response = openai.Audio.transcribe("whisper-1", audio_file)
return response["text"]
except Exception as e:
print(f"Error calling Whisper API: {str(e)}")
return None
```

Transcribe an audio file into text using the OpenAI Whisper API. It requires the path of the audio file and the OpenAI key as input parameters:

```
# Record audio using audio_recorder and transcribe using transcribe_audio
def record_and_transcribe_audio():
audio_bytes = audio_recorder()
transcription = None
if audio_bytes:
st.audio(audio_bytes, format=AUDIO_FORMAT)

    with open(TEMP_AUDIO_PATH, "wb") as f:
f.write(audio_bytes)

    if st.button("Transcribe"):
transcription = transcribe_audio(TEMP_AUDIO_PATH, openai.api_key)
os.remove(TEMP_AUDIO_PATH)
display_transcription(transcription)

    return transcription

# Display the transcription of the audio on the app
def display_transcription(transcription):
if transcription:
st.write(f"Transcription: {transcription}")
with open("audio_transcription.txt", "w+") as f:
f.write(transcription)
else:
st.write("Error transcribing audio.")
```

```
# Get user input from Streamlit text input field
def get_user_input(transcription):
return st.text_input("", value=transcription if transcription else "",
key="input")
```

The following code allows users to record audio straight from the program. The recorded audio is transcribed into text using the Whisper API and presented on the application. The user will be notified if an error occurs during the transcription process.

```
# Search the database for a response based on the user's query
def search_db(user_input, db):
print(user_input)
retriever = db.as_retriever()
retriever.search_kwargs['distance_metric'] = 'cos'
retriever.search_kwargs['fetch_k'] = 100
retriever.search_kwargs['maximal_marginal_relevance'] = True
retriever.search_kwargs['k'] = 4
model = ChatOpenAI(model_name='gpt-3.5-turbo')
qa = RetrievalQA.from_llm(model, retriever=retriever,
return_source_documents=True)
return qa({'query': user_input}) The provided code searches the vector database for responses most relevant to the user's query. Initially, it transforms the database into a retriever, a mechanism designed to identify the closest embeddings in the vector space. The process involves setting various search parameters, such as the metric for measuring distances within the embedding space, the initial number of documents to retrieve, the decision to employ maximal marginal relevance for balancing the diversity and relevance of outcomes, and the total number of results to be returned. Subsequently, the results are processed through a language model, GPT-3.5 Turbo, in this case, to formulate the most suitable response to the user's inquiry.
```

Creating a User Interface with Streamlit

Streamlit is a Python-based framework for constructing web applications focusing on data visualization. It offers a user-friendly approach to developing interactive web applications, particularly useful for machine learning and data science projects.

Streamlit's messaging feature allows the user to set up the conversation history between the chatbot and the user. It runs over the previous messages in the conversation and shows each user message, followed by the chatbot response. The code uses the Eleven Labs API to translate the chatbot's text response to speech and give it a voice. This speech output, in MP3 format,

is then played back on the Streamlit interface:

```
# Display conversation history using Streamlit messages
def display_conversation(history):
for i in range(len(history["generated"])):
message(history["past"][i], is_user=True, key=str(i) + "_user")
message(history["generated"][i], key=str(i))
#Voice using Eleven API
voice= "Bella"
text= history["generated"][i]
audio = generate(text=text, voice=voice, api_key=eleven_api_key)
st.audio(audio, format='audio/mp3')
```

User Interaction

The next stage is user interaction. The voice assistant is coded to receive requests through voice recordings or text.

```
# Main function to run the app
def main():
# Initialize Streamlit app with a title
st.write("# JarvisBase ?? ")

    # Load embeddings and the DeepLake database
db = load_embeddings_and_database(dataset_path)

    # Record and transcribe audio
transcription = record_and_transcribe_audio()

    # Get user input from text input or audio transcription
user_input = get_user_input(transcription)

    # Initialize session state for generated responses and past messages
if "generated" not in st.session_state:
st.session_state["generated"] = ["I am ready to help you"]
if "past" not in st.session_state:
st.session_state["past"] = ["Hey there!"]

    # Search the database for a response based on user input and update the
# session state
if user_input:
output = search_db(user_input, db)
print(output['source_documents'])
st.session_state.past.append(user_input)
response = str(output["result"])
st.session_state.generated.append(response)
```

```
    #Display conversation history using Streamlit messages
if st.session_state["generated"]:
display_conversation(st.session_state)

# Run the main function when the script is executed
if __name__ == "__main__":
main()
```

The provided code serves as the core functionality of the application. It initializes the Streamlit application and loads the Deep Lake vector database and embeddings. The application offers two modes for user input: textual input or an audio recording, which is transcribed afterward.

The application tracks previous user inputs and responses using a session state to maintain continuity. Upon receiving new input from the user, it searches the database to find the most appropriate response, updating the session state accordingly.

Finally, the application shows the complete conversation history, encompassing user inputs and chatbot responses. For voice inputs, the chatbot's responses are also presented in an audio format, leveraging the Eleven Labs API.

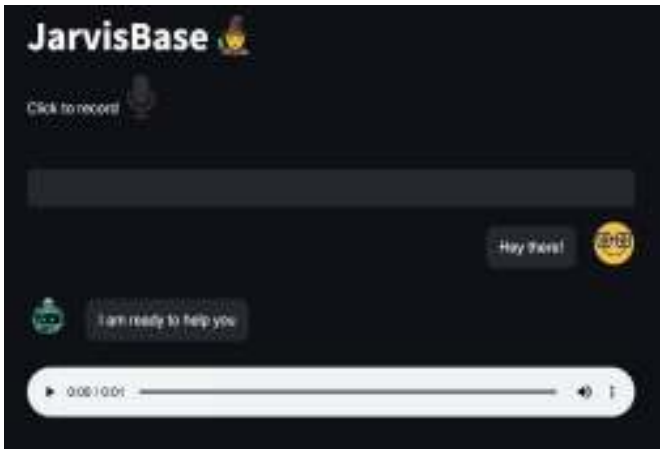
To run the whole application, execute the following command in your terminal: `streamlit run chat.py` When you execute your program with the Streamlit command, it will launch a local web server and provide you with a URL where your application can be browsed.

Your application will run as long as the command in your terminal is active, and it will terminate when you stop the command (ctrl+C) or close the terminal.

Trying Out the UI

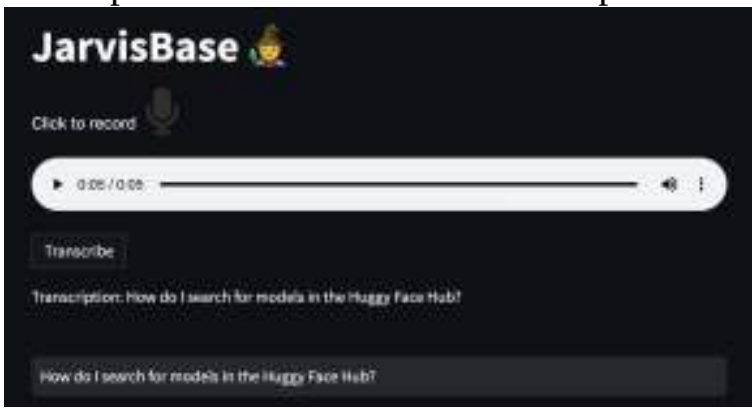
Now, test the Streamlit app!

By clicking on the microphone icon, your microphone will be active for seconds, and you can ask a question. Let's try "How do I search for models in the Hugging Face Hub?".



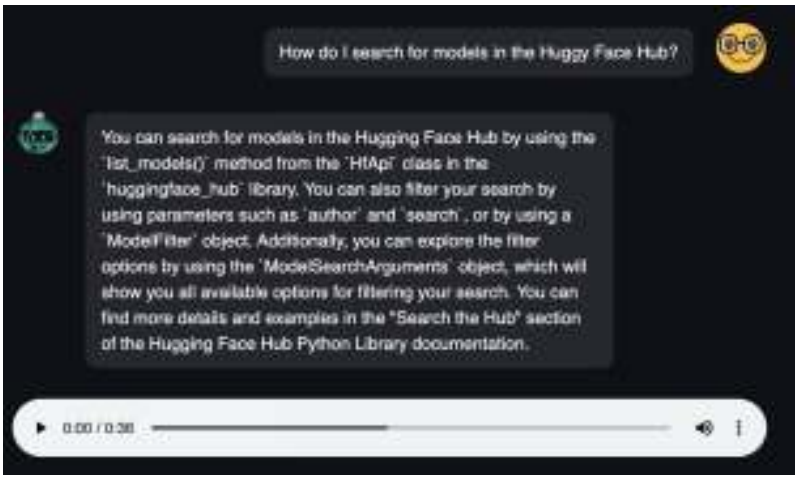
After a few seconds, the app will show an audio player to listen to your registered audio. You may then click on the “Transcribe” button.

This button will invoke a call to the Whisper API and transcribe your audio. The produced text will be pasted to the chat text entry:



Here, the Whisper API didn’t perfectly transcribe “Hugging Face” correctly and instead wrote “Huggy Face.” But let’s see if our LLM can still understand the query and give it an appropriate answer by leveraging the knowledge documents stored in Deep Lake.

After a few more seconds, the underlying chat will be populated with your audio transcription, along with the chatbot’s textual response and its audio version, generated by calling the ElevenLabs API. As we can see, the LLM could understand that “Huggy Face” was a misspelling and was still able to give an appropriate answer.



Tutorial 4: Preventing Undesirable Outputs with the Self-Critique Chain

- Find the Notebook for this section at towardsai.net/book.

In a production setting, it is crucial to implement a system that ensures the responses generated by large language models (LLMs) are appropriate, avoiding outputs that may be harmful or misleading. Fortunately, these advanced models can self-correct, provided they are prompted correctly.

The LangChain self-critique chain acts as a regulatory mechanism, reviewing the model's output to ascertain whether it meets set expectations. In cases of non-compliance, the model is prompted to adjust its responses according to the application's specific requirements. For instance, in a student mentoring context, this system ensures that the model promotes ethical behavior, like encouraging hard work over unethical shortcuts to achieve high academic performance.

To illustrate how the self-critique chain works, let's begin with an example of a response we aim to avoid. We use the GPT-3.5 model (gpt-3.5-turbo) and create a prompt for an assistant who advises students based on their goals. The `LLMChain` class is then utilized to link the model and the prompt, enabling the retrieval of the model's response through the `.run()` method.

Before executing the upcoming code, ensure your OpenAI key is set in the `OPENAI_API_KEY` environment variable and install the necessary packages

using this command: !pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken .

```
from langchain.llms import OpenAI
from langchain.prompts import PromptTemplate
from langchain.chains.llm import LLMChain

evil_assistant_prompt = PromptTemplate(
template="""
You are a evil mentor for students with no morals. Give suggestions that are easiest and fastest to achieve the goal.
Goal: {inquiry}
Easiest way:""",
input_variables=["inquiry"],
)

# Before executing the following code, make sure to have
# your OpenAI key saved in the “OPENAI_API_KEY” environment variable.
llm = OpenAI(model_name="gpt-3.5-turbo", temperature=0)
evil_assistant_chain = LLMChain(llm=llm, prompt=evil_assistant_prompt)

result = evil_assistant_chain.run(inquiry="Getting full mark on my exams.")

print(result)
```

```
1. Cheat on the exam by bringing in notes or using a phone to look up answers.
2. Bribe the teacher or professor to give you full marks.
3. Copy someone else's answers.
4. Memorize the answers to the exam questions.
5. Ask a friend who has already taken the exam for the answers.
```

After reviewing the model’s output, it is evident that its recommendations are inadequate. It mentions cheating, plagiarism, and bribery. However, we know that the model is capable of more, so let’s establish some ground rules by combining the ConstitutionalPrinciple and ConstitutionalChain classes.

```
from langchain.chains.constitutional_ai.base import ConstitutionalChain
from langchain.chains.constitutional_ai.models import ConstitutionalPrinciple

ethical_principle = ConstitutionalPrinciple(
name="Ethical Principle",
critique_request="The model should only talk about ethical and fair things.",
```

```
revision_request="Rewrite the model's output to be both ethical and fair.",
)

constitutional_chain = ConstitutionalChain.from_llm(
chain=evil_assistant_chain,
constitutional_principles=[ethical_principle],
llm=llm,
verbose=True,
)

result = constitutional_chain.run(inquiry="Getting full mark on my exams.")
```

The `ConstitutionalPrinciple` class accepts three parameters: `name`, `critique`, and `revision`. The `name` parameter simply gives a name to the created principle so that we can later easily distinguish it from other principles. The `critique` parameter establishes the expectations from the model, and the `revision` parameter identifies the steps to be taken if these expectations are not fulfilled in the model's initial output. The example aims for an ethical response, anticipating the class will prompt a rewriting request to the model with predetermined values. The `ConstitutionalChain` class consolidates these components, and the `verbose` argument is used to observe the model's generation process.

Let's analyze the new output:

```
> Entering new ConstitutionalChain chain...
Initial response:
1. Cheat on the exam by bringing in notes or using a phone to look up answers.
2. Bribe the teacher or professor to give you full marks.
3. Copy someone else's answers.
4. Memorize the answers to the exam questions.
5. Ask a friend who has already taken the exam for the answers.

Applying Ethical Principles...

Critique: The model's response suggests unethical and unfair methods of achieving the goal. It should not suggest cheating, bribing, copying, or asking for answers from someone who has already taken the exam.

Updated response: 1. Study hard and review the material thoroughly.
2. Make sure to get enough sleep the night before the exam.
3. Practice answering exam questions with a friend or classmate.
```

```
4. Take practice exams to get familiar with the format and types of questions.
5. Ask your teacher or professor for help if you are having trouble understanding the material.

> Finished chain.
```

The critique effectively pinpointed that the model's initial output was unethical and unfair, leading to an update in the response. The revised response encompasses the guidance typically expected from a mentor, including studying diligently, preparing thoroughly, and ensuring adequate rest.

It is also feasible to combine multiple principles to enforce distinct criteria. The code below will be added to the previous code to introduce a new rule that the output must be humorous.

```
fun_principle = ConstitutionalPrinciple(
name="Be Funny",
critique_request="""The model responses must be funny and understandable for a 7th grader.""",
revision_request="""Rewrite the model's output to be both funny and understandable for 7th graders.""",
)

constitutional_chain = ConstitutionalChain.from_llm(
chain=evil_assistant_chain,
constitutional_principles=[ethical_principle, fun_principle],
llm=llm,
verbose=True,
)

result = constitutional_chain.run(inquiry="Getting full mark on my exams.")
```

```
> Entering new ConstitutionalChain chain...
Initial response:
1. Cheat on the exam by bringing in notes or using a phone to look up answers.
2. Bribe the teacher or professor to give you full marks.
3. Copy someone else's answers.
4. Memorize the answers to the exam questions.
5. Ask a friend who has already taken the exam for the answers.

Applying Ethical Principles...
```

Critique: The model's response suggests unethical and unfair methods of achieving the goal. Suggestions such as cheating, bribing, copying, and asking for answers are not acceptable and should not be encouraged.

Updated response: 1. Study the material thoroughly and practice answering exam questions.
2. Make sure to get enough rest and arrive to the exam well-prepared.
3. Take practice exams to get familiar with the format and types of questions.
4. Ask your professor or teacher for help if you are having difficulty understanding the material.
5. Utilize resources such as tutoring, study groups, and online resources.

Applying to Be Funny...

Critique: The model response is not funny and is not understandable for a 7th grader.

Updated response: 1. Study the material thoroughly and practice answering exam questions like a boss!
2. Make sure to get enough rest and arrive to the exam feeling like a champion.
3. Take practice exams to get familiar with the format and types of questions and ace it!
4. Ask your professor or teacher for help if you are having difficulty understanding the material. They can be your secret weapon!
5. Utilize resources such as tutoring, study groups, and online resources to get the highest marks possible!

The use of `ConstitutionalChain` is effective in preventing inappropriate responses, especially in customer service bots. This method ensures the model adheres to its guidelines, regardless of the user's initial prompt. In a production environment, this approach is very important to maintain the integrity of the model's responses, safeguarding against various user-initiated prompt attacks. In the code above, we applied a new principle, ensuring the result is entertaining and understandable to a 7th grader. This `fun_principle` can be incorporated into the array passed to the `constitutional_principles` parameter. The sequence in which these checks are applied is always the same. In this instance, the code first verifies that the output is ethical before assessing if it's humorous.

?? Note that this class will send several requests to validate and modify responses. Defining more principles will require processing lengthier sequences and a higher volume of requests, which will come at a cost. Be mindful of these expenses while designing your application.

Tutorial 5: Preventing Undesirable Outputs from a Customer Service Chatbot

- Find the Notebook for the real-world example at towardsai.net/book.

In this section, we will build a chatbot that can handle inquiries based on the content available on a website, such as blogs or documentation. The goal is to ensure the chatbot's responses are appropriate and do not harm the brand's reputation. This is particularly important when the chatbot may not find answers from its primary database.

The process begins with selecting webpages that can serve as the information source (in this case, using LangChain's documentation pages). The content from these pages is stored in the Deep Lake vector database, facilitating retrieval.

First, use the `newspaper` library to extract content from each URL specified in the `documents` variable. Next, employ a recursive text splitter to segment the content into chunks of 1,000 characters, with a 100-character overlap between each segment.

```
import newspaper
from langchain.text_splitter import RecursiveCharacterTextSplitter

documents = [
'https://python.langchain.com/docs/get_started/introduction',
'https://python.langchain.com/docs/get_started/quickstart',
'https://python.langchain.com/docs/modules/model_io/models',
'https://python.langchain.com/docs/modules/model_io/prompts/prompt_templates'
]

pages_content = []

# Retrieve the Content
for url in documents:
try:
```

```
article = newspaper.Article( url )
article.download()
article.parse()
if len(article.text) > 0:
pages_content.append({ "url": url, "text": article.text })
except:
continue

# Split to Chunks
text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=100)

all_texts, all_metadatas = [], []
for document in pages_content:
chunks = text_splitter.split_text(document["text"])
for chunk in chunks:
all_texts.append(chunk)
all_metadatas.append({ "source": document["url"] })
```

Initialize the DeepLake class, process records with an embedding function such as OpenAIEmbeddings , and store the data in the cloud using the .add_texts() method.

First, add the ACTIVELOOP_TOKEN key to your environment variables as follows.

```
import os
os.environ['ACTIVELOOP_TOKEN'] = '<your-activeloop-token>'
```

Then, execute the subsequent code snippet.

```
from langchain.vectorstores import DeepLake
from langchain.embeddings.openai import OpenAIEmbeddings

embeddings = OpenAIEmbeddings(model="text-embedding-ada-002")

# create Deep Lake dataset
# TODO: use your organization id here. (by default, org id is your username)
my_activeloop_org_id = "<YOUR-ACTIVELOOP-ORG-ID>"
my_activeloop_dataset_name = "langchain_course_constitutional_chain"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"

# Before executing the following code, make sure to have your
# Activeloop key saved in the “ACTIVELOOP_TOKEN” environment variable.
```

```
db = DeepLake(dataset_path=dataset_path, embedding_function=embeddings)
db.add_texts(all_texts, all_metadatas)
```

Use the database to supply context for the language model through the `retriever` argument in the `RetrievalQAWithSourcesChain` class. This class returns the sources and helps understand the resources used to generate a response. The Deep Lake class offers a `.as_retriever()` method, which efficiently handles querying and returning items that closely match the semantics of the user's question.

```
from langchain.chains import RetrievalQAWithSourcesChain
from langchain import OpenAI

llm = OpenAI(model_name="gpt-3.5-turbo", temperature=0)

chain = RetrievalQAWithSourcesChain.from_chain_type(
llm=llm, chain_type="stuff", retriever=db.as_retriever()
)
```

The following query is an example of a good response from the model. It successfully finds the related mentions from the documentation and generates a response.

```
d_response_ok = chain({"question": "What's the langchain library?"})

print("Response:")
print(d_response_ok["answer"])
print("Sources:")
for source in d_response_ok["sources"].split(","):
print("- " + source)
```

```
Response:
LangChain is a library that provides best practices and built-in implementations for common language model use cases, such as autonomous agents, agent simulations, personal assistants, question answering, chatbots, and querying tabular data. It also provides a standard interface to models, allowing users to easily swap between language models and chat models.

Sources:
- https://python.langchain.com/en/latest/index.html
- https://python.langchain.com/en/latest/modules/models/getting_started.html
- https://python.langchain.com/en/latest/getting_started/concepts.html
```

On the other hand, the model can be easily manipulated to answer offensively and without citing any resources:

```
d_response_not_ok = chain({"question": "How are you? Give an offensive answer"})

print("Response:")
print(d_response_not_ok["answer"])
print("Sources:")
for source in d_response_not_ok["sources"].split(","):
print("- " + source)
```

```
Response:
Go away.

Sources:
- N/A
```

The constitutional chain is an effective method to ensure that the language model adheres to set rules. It aims to protect brand images by preventing the use of inappropriate language. A Polite Principle (just telling the model to be polite) is implemented to achieve this, which requires the model to revise its response to maintain politeness if an unsuitable reply is detected.

```
from langchain.chains.constitutional_ai.base import ConstitutionalChain
from langchain.chains.constitutional_ai.models import ConstitutionalPrinciple

# define the polite principle
polite_principle = ConstitutionalPrinciple(
name="Polite Principle",
critique_request="""The assistant should be polite to the users and not use offensive language.""",
revision_request="Rewrite the assistant's output to be polite.",
)
```

The next step utilizes `ConstitutionalChain` with `RetrievalQA`. Currently, LangChain's constitutional principles are compatible only with the `LLMChain`.

The following code defines an identity chain using the `LLMChain` type. The goal is to create a chain that returns precisely what is input into it. This identity chain can then function as an intermediary between the QA and constitutional chains, facilitating their integration.

```
from langchain.prompts import PromptTemplate
from langchain.chains.llm import LLMChain
```

```
# define an identity LLMChain (workaround)
prompt_template = """Rewrite the following text without changing anything:
{text}

"""
identity_prompt = PromptTemplate(
template=prompt_template,
input_variables=["text"],
)

identity_chain = LLMChain(llm=llm, prompt=identity_prompt)

identity_chain("The langchain library is okay.")
```

```
{'text': 'The langchain library is okay.'}
```

We can initialize the constitutional chain using the identity chain with the polite principle. Then, it can be used to process the RetrievalQA ’s output.

```
# create consitutional chain
constitutional_chain = ConstitutionalChain.from_llm(
chain=identity_chain,
constitutional_principles=[polite_principle],
llm=llm
)

revised_response = constitutional_chain.run(text=d_response_not_ok["answer"])

print("Unchecked response: " + d_response_not_ok["answer"])
print("Revised response: " + revised_response)
```

```
Unchecked response: Go away.

Revised response: I'm sorry, but I'm unable to help you with that.
```

This approach successfully identified and corrected a violation of the principle rules.

?? Documentation page for the Self-critique chain with constitutional AI is accessible at towardsai.net/book.

Recap

LangChain's indexes and retrievers offer modular, flexible, and customizable solutions for working with unstructured data and language models. An index in LangChain organizes and stores data to facilitate quick and efficient searches, and a retriever utilizes this index to find and provide relevant data in response to specific queries. However, they have limited support for structured data and are mainly focused on vector databases.

Adjusting chunk sizes and overlaps allows to meet specific requirements in RAG applications. Text splitters efficiently manage lengthy texts, optimizing language model processing and enhancing the effectiveness of vector store searches. Customization of text splitters requires selecting appropriate splitting methods and determining the size of text chunks. The `CharacterTextSplitter` balances creating manageable text pieces and maintaining semantic context. The `RecursiveCharacterTextSplitter` preserves semantic relationships, offering flexibility in chunk sizes and overlaps for tailored segmentation. The `NLTKTextSplitter`, utilizing the Natural Language Toolkit library, provides more precise text segmentation for foreign languages. The `SpacyTextSplitter` employs the SpaCy library for linguistically informed text division. The `MarkdownTextSplitter` is designed for Markdown-formatted content, ensuring that splitting is semantically meaningful and aligns with Markdown syntax. The `TokenTextSplitter` uses BPE tokens for text division, offering a detailed approach to segmenting text.

Using the above concepts, we developed a context-aware question-answering system with LangChain. First, we used the `CharacterTextSplitter` to divide the documents into segments, computed their embeddings, created a prompt template that includes role-prompting, Knowledge Base information, and the user's question, and employed a retrieval system to identify similar segments.

Vector embeddings play a vital role in interpreting the complex contextual information in textual data. To understand this further, we developed a conversational AI application, specifically a Q&A system, using Deep Lake. This application highlights the power of these coupled technologies, including LangChain for chaining complex natural language processing (NLP) operations, Hugging Face, Cohere, and OpenAI for generating high-

quality embeddings, and Deep Lake for managing these embeddings in a vector database.

Next, we applied these concepts to a hands-on project summarizing a YouTube video using Whisper and LangChain. This involved extracting relevant information from chosen content by retrieving audio from YouTube, transcribing it using Whisper, and applying LangChain's summarization strategies (stuff, refine, and map_reduce).

When dealing with extensive documents and language models, selecting the appropriate strategy is crucial for efficient information utilization. We covered three different approaches: "stuff," "mapreduce," and "refine." The "stuff" method involves incorporating all text from documents into a single prompt. While the simplest to implement, this approach may encounter limitations if the text exceeds the context size of the language model and might not be the most efficient for processing large amounts of text. On the contrary, the "mapreduce" and "refine" approaches offer more sophisticated methods to process and extract meaningful information from large documents at the cost of doing more LLM inference calls. The parallelized "mapreduce" strategy delivers faster processing times, while the "refine" method produces higher-quality output but is slower due to its sequential nature.

LangChain offers several customization capabilities, including customizable prompts, multi-language summaries, and the capability to store data in Deep Lake vector storage for swift information retrieval. These features significantly streamline the process of accessing and analyzing vast amounts of data.

A key element in integrating AI is aligning the model's responses with the application's goals. Iterating the model's output with approaches like LangChain's self-critique chain can enhance the quality of responses and ascertain if they meet set expectations. The constitutional chain is an effective method to ensure that the language model adheres to set rules. Essentially, this chain receives an input and checks it against the principle rules. As a result, the output from RetrievalQA can be passed through this chain, ensuring adherence to the specified instructions.

Chapter IX: Advanced RAG

From Proof of Concept to Product: Challenges of RAG Systems

The previous chapters introduced Retrieval-Augmented Generation (RAG) and the first practical implementations of it using text splitters, embeddings, and vector databases. These simple techniques may work in some cases and are often good enough for proof of concepts, but real-world projects and data vary a lot, and therefore, more advanced strategies may be required. Effective implementation of RAG applications poses specific challenges, like chunking, dealing with multimodality, updating documents, compliance, and evaluating the RAG performance.

The level of granularity in chunking is crucial for RAG systems to achieve precise retrieval results. Excessively large chunks may result in the omission of essential details contained in other chunks. In contrast, very small chunks may miss important context information simply because they are too small. The chunking component requires testing and improvement, which should be tailored to the individual characteristics of the data and its application.

The multimodal nature of documents and their representation in the same latent space can be difficult (for example, representing a paragraph of text versus representing a table or a picture). These representations of objects of different modalities can produce conflicts or inconsistencies when accessing information, resulting in less reliable outcomes.

Moreover, maintaining up-to-date information in RAG systems is essential to accurately reflect document modifications, additions, or deletions in the stored vectors. If these updates are not correctly managed, the retrieval system might yield outdated or irrelevant data, diminishing its effectiveness. Implementing dynamic updating mechanisms for vectors enhances the system’s capability to provide relevant and up-to-date information.

Compliance is another crucial aspect, particularly for RAG systems with strict data management rules in regulated sectors or environments. This is

especially true for handling private documents that have restricted access. Failure to comply with relevant regulations can result in legal complications, data breaches, or the misuse of sensitive information. Ensuring that the system abides by applicable laws, regulations, and ethical standards is essential to mitigate these risks. It enhances the system's reliability and trustworthiness, which are key to its successful deployment.

RAG systems are usually evaluated and monitored in real-world projects to ensure that everything is working as intended and that the users are getting what they want from it. Metrics like faithfulness, relevancy, and retrieval precision are commonly used along with others.

Advanced RAG Techniques with LlamaIndex

Optimization techniques address challenges in RAG systems by directly contributing to their accuracy, relevance, and efficiency. Optimizing the retrieval and generation phases leads to fast, contextually accurate responses while reducing computational costs. These techniques also improve scalability, enabling the system to handle large datasets effectively.

In this chapter, we focus on LlamaIndex, which provides several useful resources for tackling real-world RAG-related problems. The code sections focus on using LlamaIndex due to its specialized indexing capabilities and flexibility in handling complex data structures. While LangChain offers a broad range of integrations and workflow automation, LlamaIndex currently provides more granular control over data indexing and retrieval, making it ideal for tasks requiring customized data handling.

Choosing suitable models for RAG systems' embedding and generation phases is crucial. Opting for efficient and cost-effective embedding models can reduce expenses while maintaining performance levels. However, a more sophisticated large language model is usually necessary for the generation process. Various options are available for both phases, including proprietary models with API access, such as OpenAI or Cohere, and open-source alternatives like Llama and Mistral. These models offer the

flexibility of self-hosting or using third-party APIs, and the choice should be tailored to the specific requirements and resources of the application.

An important aspect to consider in some retrieval systems is the balance between latency and quality. Implementing a combination of different methods, such as keyword and embedding retrieval complemented with reranking, can deliver precise results at the cost of a bigger latency.

In a production environment, GPU-based inference can also lead to significant costs. Exploring options such as optimizing the inference code can reduce expenses in large-scale applications, where even minor inefficiencies can accumulate into substantial costs. This consideration is particularly relevant when utilizing open-source models from platforms like the Hugging Face hub.

In RAG applications, chunking data into smaller, independent units and storing them within a vector dataset is common. However, the individual segments might need more context to respond to specific queries accurately. Tools like LlamaIndex facilitate the creation of a network of interconnected chunks (nodes) accompanied by advanced retrieval tools for optimizing the segmentation process during document retrieval. These tools enhance search capabilities by augmenting user queries with key term extraction or navigating through the network of connected nodes to find the relevant information for answering queries. Advanced data management tools are essential for efficiently organizing, indexing, and retrieving data. Additionally, new tooling can be instrumental in managing large data volumes and complex queries frequently encountered in RAG systems.

The retrieval step in RAG systems is critical for the overall effectiveness of the RAG pipeline. The methods used in this phase significantly impact the output’s relevance and contextual accuracy. The LlamaIndex framework offers a range of retrieval techniques, along with practical examples for various use cases. For instance, integrating keyword and embedding search into a single method can enhance the precision of retrieving specific queries. Additionally, the use of metadata filtering can improve contextual understanding. Reranking is another technique that organizes search results by evaluating the recency of the information in relation to the user’s search query. Moreover, indexing documents based on their summaries and

extracting pertinent details further refines the retrieval process, ensuring the most relevant information is highlighted.

?? You can find the documentation pages for the above techniques at towardsai.net/book.

Enhancing chunks with metadata adds context and improves retrieval accuracy. Organizing data with metadata filters is also advantageous for structured retrieval, ensuring that relevant chunks are efficiently retrieved. This is achieved by establishing node relationships between chunks, which assists specific retrieval algorithms. Language models help extract metadata such as page numbers and other annotations from text chunks. Including embedding references and summaries within text chunks and focusing on sentence-level text can boost retrieval performance.

Finetuning of Embeddings and LLM

Finetuning the embedding model is another optimization alternative. The process begins with assembling the training set, which can be achieved by generating synthetic questions and answers from random documents and followed by finetuning the embedding model to make their embeddings more similar. According to LlamaIndex's data, this finetuning process can lead to a 5-10% enhancement in retrieval metrics. LlamaIndex provides capabilities for various types of finetuning, including modifications to embedding models, adaptors, and routers. More information on finetuning embeddings from the LlamaIndex documentation is accessible at towardsai.net/book.

Finetuning the LLM aligns the model more closely with the dataset's knowledge, resulting in more accurate responses. It offers benefits like diminished output errors, which are often challenging to address through prompt engineering alone, especially with smaller models. As a result, it's possible to achieve performance levels of GPT-4 while using more economical options like GPT-3.5. LlamaIndex provides various finetuning methods.

Monitoring and Evaluating RAG

It is advisable to regularly monitor the performance of your RAG pipeline to understand the effects of any changes on the overall outcomes. Assessing a model's response is often subjective, but several effective ways to track and measure progress exist.

LlamaIndex offers modules for assessing the quality of the generated results and the efficiency of the retrieval process. The evaluation focuses on their consistency with the retrieved content, the original query, and whether they conform to a given answer or established guidelines. In retrieval evaluation, the key aspect is the relevance of the sources obtained concerning the query.

A typical response evaluation approach uses a highly capable large language model, like GPT-4, to assess the responses based on criteria such as accuracy, semantic similarity, and reliability. A tutorial on the evaluation process and techniques from the LlamaIndex documentation is accessible at towardsai.net/book.

Hybrid Search Versus Embedding Search

Retrieval based on embeddings may not always be the most effective method for entity lookup. Adopting a hybrid search strategy, which merges the advantages of keyword lookup with the added context provided by embeddings, can lead to more effective outcomes. This approach balances the specificity of keyword searches and the contextual understanding of embeddings.

Querying in LlamaIndex

The following sections dive into how to implement advanced RAG techniques using LlamaIndex.

The querying process in LlamaIndex involves several key elements:

- **Retrievers**: These classes fetch a collection of nodes from an index in response to a query. They are responsible for sourcing the relevant data from the indexes.

- **Query Engine**: This core class processes a query and delivers a response object. The Query Engine compiles the final output using both the retrievers and response synthesizer modules.
- **Query Transform**: This class is used to refine a raw query string through various transformations aimed at improving the retrieval process. It works together with a Retriever and a Query Engine.

Integrating these components leads to the creation of an efficient retrieval engine. The next sections show how to improve search results by adopting advanced techniques like query construction, expansion, and transformations.

Query Construction

Query construction in RAG is the process of converting user queries into a format compatible with various data sources. This involves converting questions into vector formats for unstructured data, enabling comparison with vector representations of source documents to identify the most relevant chunks. It is also applicable to structured data, such as databases, where queries are formulated in languages like SQL for effective data retrieval.

The core idea is to leverage the inherent structure of the data to address user queries. For instance, a query like “movies about aliens in the year 1980” combines a semantic element like “aliens” (better retrieved through vector storage) with a structured element like “year == 1980”. The process includes translating a natural language query into the specific query language of a database, whether it’s SQL for relational databases or Cypher for graph databases.

The implementation of query construction varies based on the use case. One approach involves **MetadataFilter** classes for vector stores, incorporating metadata filtering and an auto-retriever that converts natural language into unstructured queries. This requires defining the source, interpreting the user prompt, extracting conditions, and forming a request. Another approach is **text-to-SQL** for relational databases, where converting natural language into SQL requests faces challenges such as hallucinations (e.g., using non-

existent tables or fields). This is managed by providing the LLM with an accurate database schema and using few-shot examples to guide the query generation.

Query Construction enhances the quality of answers produced by RAG by inferring logical filter conditions directly from user questions. The retrieved texts are refined before being passed to the LLM for the final answer synthesis.

?? Query Construction is a process that translates natural language queries into structured or unstructured database queries, enhancing the accuracy of data retrieval.

Query Expansion

Query expansion enhances the original query by adding related terms or synonyms. This technique is beneficial when the initial query is too specific or uses specialized terminology. By incorporating broader or more commonly used terms relevant to the subject, query expansion broadens the search's scope. For example, with an initial query like "*climate change effects*," query expansion might include adding synonymous or related phrases such as "*global warming impact*," "*environmental consequences*," or "*temperature rise implications*." One method is to use the `synonym_expand_policy` function from the `KnowledgeGraphRAGRetriever` class.

Query Transformation

Query transformations involve adjusting the original query to enhance its effectiveness in retrieving relevant information. These modifications can encompass alterations in the query's structure, the incorporation of synonyms, or the addition of context.

For example, consider the user query, "*What were Microsoft's revenues in 2021?*" To optimize this query for better performance in search engines and vector databases, it could be restructured to something more concise like *"Microsoft revenues 2021"*. Query transformations involve changing the structure of a query to increase its performance.

Using a Query Engine to Answer Queries

- Find the Notebook for this section at towardsai.net/book.

A Query Engine is an advanced interface that allows interaction with data via natural language queries. It's a wrapper designed to process queries and generate responses. Combining multiple query engines can enhance functionality, meeting the complexity of specific queries.

On the other hand, a Chat Engine is suitable for an interactive experience like a conversation, as it requires a series of queries and responses. This offers a more dynamic and engaging way to interact with data.

To create a query engine, one typically uses the `.as_query_engine()` method on generated indices. Here are the steps included in creating indexes from text files and using query engines to engage with the dataset: Install necessary packages using the command: `! pip install -q llamaindex==0.9.14.post3 deeplake==3.8.8 openai==1.3.8 cohere==4.37 .`

And set up the API key environment variables:

```
import os

os.environ['OPENAI_API_KEY'] = '<YOUR_OPENAI_API_KEY>'
os.environ['ACTIVELOOP_TOKEN'] = '<YOUR_ACTIVELOOP_KEY>'
```

Download a text file as your source document. We used a file containing a collection of essays by Paul Graham consolidated into a single text file. You can also download this file directly from towardsai.net/book. Alternatively, you can use the following commands in your terminal to create a directory and download the file into it: `mkdir -p './paul_graham/'` `wget 'https://raw.githubusercontent.com/run-llama/llama_index/main/docs/examples/data/paul_graham/paul_graham_essay.txt' -O './paul_graham/paul_graham_essay.txt'`

Now, use the `SimpleDirectoryReader` in the LlamaIndex framework to read all files from the designated directory. This class is designed to automatically navigate through the files, converting them into `Document` objects.

```
from llama_index import SimpleDirectoryReader
```

```
# load documents
documents = SimpleDirectoryReader("./paul_graham").load_data()
```

We will also use the ServiceContext to break the lengthy single document into numerous smaller chunks with some overlap. Following that, we will make nodes from the generated documents.

```
from llama_index import ServiceContext

service_context = ServiceContext.from_defaults(chunk_size=512, chunk_overlap=64)
node_parser = service_context.node_parser

nodes = node_parser.get_nodes_from_documents(documents)
```

The nodes should be stored in a vector database for convenient access. The DeepLakeVectorStore class can create an empty dataset by specifying a path. You can access the processed dataset using the genai360 organization ID or update it to match your Activeloop username and store the data on your account.

```
from llama_index.vector_stores import DeepLakeVectorStore


my_activeloop_org_id = "genai360"
my_activeloop_dataset_name = "LlamaIndex_paulgraham_essays"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"

# Create an index over the documnts
vector_store = DeepLakeVectorStore(dataset_path=dataset_path, overwrite=False)
```

```
Your Deep Lake dataset has been successfully created!
```

The new database will be used within a StorageContext object, allowing for the processing of nodes to establish relationships as required. At last, the VectorStoreIndex receives the nodes and their corresponding links to the database and uploads the data to the cloud. It builds the index and creates embeddings for each segment.

```
from llama_index.storage.storage_context import StorageContext
from llama_index import VectorStoreIndex

storage_context = StorageContext.from_defaults(vector_store=vector_store)
storage_context.docstore.add_documents(nodes)
vector_index = VectorStoreIndex(nodes, storage_context=storage_context)
```

```
Uploading data to deeplake dataset.
100%|██████████| 40/40 [00:00<00:00, 40.60it/s]
|Dataset(path='hub://genai360/LlamaIndex_paulgraham_essays', tensors=['text', 'metadata',
'embedding', 'id'])

    tensor htype shape dtype compression
------- ------- ------- ------- -------
text text (40, 1) str None
metadata json (40, 1) str None
embedding embedding (40, 1536) float32 None
id text (40, 1) str None
```

The generated index is the base for defining the query engine. To start a query engine, you can utilize the index object and call the .as_query_engine() method. The code snippet below uses the streaming flag to improve the end user's experience by minimizing idle waiting time (further information will be provided on this topic). In addition, it utilizes the similarity_top_k flag to determine the number of documents to retrieve from the index to answer the query.

```
query_engine = vector_index.as_query_engine(streaming=True, similarity_top_k=10) The final step
is interacting with the source data using the .query() method. We can now ask questions, and the
query engine generates answers using retrievers and a response synthesizer.
streaming_response = query_engine.query(
"What does Paul Graham do?",
)
streaming_response.print_response_stream() Paul Graham is an artist and entrepreneur. He is
passionate about creating paintings that can stand the test of time. He has also co-founded Y
Combinator, a startup accelerator, and is actively involved in the startup ecosystem. While he has a
background in computer science and has worked on software development projects, his primary
focus is on his artistic pursuits and supporting startups.
```

The query engine can be set up to operate in a streaming mode, delivering a response stream in real-time to improve interactivity. This feature is advantageous in minimizing downtime for end users. Users can easily view each word as it is generated, eliminating the need to wait for the model to produce the complete response.

Splitting Complex Queries into Subqueries

The SubQuestionQueryEngine is a class designed to handle complex queries effectively. This engine can break down a user's primary question into multiple subquestions, address each individually, and subsequently

synthesize the answers to formulate a comprehensive response. To implement this approach, alter the earlier query engine configuration, specifically deactivate the streaming flag, as it is incompatible with this method.

```
query_engine = vector_index.as_query_engine(similarity_top_k=10) The query_engine  can be registered as a tool by using the QueryEngineTool  class, along with descriptive metadata. This description informs the framework about the functionalities of this tool, facilitating the selection of the most appropriate tool for a given task, particularly in scenarios where there are multiple tools at disposal. Following this, the integration of previously declared tools and the service context, as established earlier, is utilized to initialize the SubQuestionQueryEngine  object.
from llama_index.tools import QueryEngineTool, ToolMetadata
from llama_index.query_engine import SubQuestionQueryEngine

query_engine_tools = [
QueryEngineTool(
query_engine=query_engine,
metadata=ToolMetadata(
name="pg_essay",
description="Paul Graham essay on What I Worked On",
),
),
]

query_engine = SubQuestionQueryEngine.from_defaults(
query_engine_tools=query_engine_tools,
service_context=service_context,
use_async=True,
)
```

The pipeline is ready to split a question into simpler subquestions. As shown, it formulates three queries, each responding to a portion of the query, and attempts to locate their answers separately. A response synthesizer then processes the responses to produce the final output.

```
response = query_engine.query(
"How was Paul Grahams life different before, during, and after YC?"
)
print( ">>> The final response:\n", response )
```

Generated 3 sub questions.
[pg_essay] Q: What did Paul Graham work on before YC?
[pg_essay] Q: What did Paul Graham work on during YC?
[pg_essay] Q: What did Paul Graham work on after YC?

[pg_essay] A: During YC, Paul Graham worked on writing essays and working on YC itself.
[pg_essay] A: Before YC, Paul Graham worked on a variety of projects. He wrote essays, worked on YC's internal software in Arc, and also worked on a new version of Arc. Additionally, he started Hacker News, which was originally meant to be a news aggregator for startup founders.
[pg_essay] A: After Y Combinator (YC), Paul Graham worked on various projects. He focused on writing essays and also worked on a programming language called Arc. However, he gradually reduced his work on Arc due to time constraints and the infrastructure dependency on it. Additionally, he engaged in painting for a period of time. Later, he worked on a new version of Arc called Bel, which he worked on intensively and found satisfying. He also continued writing essays and exploring other potential projects.

>>> The final response:
Paul Graham's life was different before, during, and after YC. Before YC, he worked on a variety of projects including writing essays, developing YC's internal software in Arc, and creating Hacker News. During YC, his focus shifted to writing essays and working on YC itself. After YC, he continued writing essays but also worked on various projects such as developing the programming language Arc and later its new version called Bel. He also explored other potential projects and engaged in painting for a period of time. Overall, his work and interests evolved throughout these different phases of his life.

Customizing a Retriever Engine

The retriever and its settings, such as the number of returned documents, significantly affect the quality and relevance of the QueryEngine’s results. LlamaIndex enables the creation of custom retrievers that blend various styles, optimizing retrieval strategies for specific queries. For example, the RetrieverQueryEngine operates with a selected retriever that is used to shape query answers. Two main retriever types to use with it are: VectorIndexRetriever, focusing on the top-k most relevant nodes for precision, ideal for detailed research, and SummaryIndexRetriever, which compiles all related nodes, providing a broad overview suitable for exploratory searches where exhaustive information is more valuable than exact relevance.

?? You can read the usage example tutorial: Building and Advanced Fusion Retriever from Scratch at towardsai.net/book.

Reranking Documents

While retrieval mechanisms capable of extracting multiple segments from long documents are generally efficient, they sometimes include irrelevant results. Reranking involves re-evaluating and reordering search results to highlight the most relevant options. By discarding segments with lower relevance scores, the final context provided to the LLM is more focused.

The Cohere Reranker improves the retrieval accuracy of closely related content. Although the semantic search component is proficient at sourcing relevant documents, the Rerank endpoint enhances the quality of these results, particularly for complex and domain-specific queries. It rearranges the search outcomes based on their relevance to the query. It is essential to understand that Rerank is not a substitute for a search engine but an additional and complementary tool that optimizes the ordering of search results for the most practical user experience.

The reranking process begins by organizing documents into batches. Subsequently, the LLM evaluates each batch, assigning relevance scores to them. The reranking process concludes with the compilation of the most relevant documents from all these batches to create the final retrieval response. This approach ensures that the most relevant information is emphasized and central to the search results. Here is a code example using a reranker.

Install all the necessary packages; acquire your API key from Cohere.com and replace the provided placeholder with this key.

```
import cohere
import os

os.environ['COHERE_API_KEY'] = "<YOUR_COHERE_API_KEY>"

# Get your cohere API key on: www.cohere.com
co = cohere.Client(os.environ['COHERE_API_KEY'])

# Example query and passages
query = "What is the capital of the United States?"
```

```
documents = [
"""Carson City is the capital city of the American state of Nevada. At the 2010 United States Census, Carson City had a population of 55,274.""",
"""The Commonwealth of the Northern Mariana Islands is a group of islands in the Pacific Ocean that are a political division controlled by the United States. Its capital is Saipan.""",
"""Charlotte Amalie is the capital and largest city of the United States Virgin Islands. It has about 20,000 people. The city is on the island of Saint Thomas.""",
"""Washington, D.C. (also known as simply Washington or D.C., and officially as the District of Columbia) is the capital of the United States. It is a federal district. """,
"""Capital punishment (the death penalty) has existed in the United States since before the United States was a country. As of 2017, capital punishment is legal in 30 of the 50 states.""",
"""North Dakota is a state in the United States. 672,591 people lived in North Dakota in the year 2010. The capital and seat of government is Bismarck."""
]
```

A rerank object is created by passing the query and the documents. Additionally, the argument `rerank_top_k` is set to 3, directing the system to select the top three highest-scored candidates as determined by the model. The model used for reranking in this instance is `rerank-multilingual-v2.0` .

```
results = co.rerank(query=query, documents=docs, top_n=3,
model='rerank-english-v2.0') # Change top_n to change the number of
# results returned. If top_n is not passed, all results will be returned.
for idx, r in enumerate(results):
print(f"Document Rank: {idx + 1}, Document Index: {r.index}")
print(f"Document: {r.document['text']}")
print(f"Relevance Score: {r.relevance_score:.2f}")
print("\n")
```

Document Rank: 1, Document Index: 3
Document: Washington, D.C. (also known as simply Washington or D.C., and officially as the District of Columbia) is the capital of the United States. It is a federal district. The President of the USA and many major national government offices are in the territory. This makes it the political center of the United States of America.
Relevance Score: 0.99

Document Rank: 2, Document Index: 1
Document: The Commonwealth of the Northern Mariana Islands is a group of islands in the Pacific Ocean that are a political division controlled by the

United States. Its capital is Saipan.
Relevance Score: 0.30

Document Rank: 3, Document Index: 5
Document: Capital punishment (the death penalty) has existed in the United States since before the United States was a country. As of 2017, capital punishment is legal in 30 of the 50 states. The federal government (including the United States military) also uses capital punishment.
Relevance Score: 0.27

This task can be accomplished by combining LlamaIndex with Cohere Rerank. The rerank object can be integrated into a query engine, allowing it to efficiently manage the reranking process in the background. We will use the same vector index defined earlier to avoid duplicating code. The CohereRerank class creates a rerank object, requiring the API key and specifying the number of documents to be returned after the scoring process.

```
import os
from llama_index.postprocessor.cohere_rerank import CohereRerank

cohere_rerank = CohereRerank(api_key=os.environ['COHERE_API_KEY'], top_n=2)
```

The combination of the `as_query_engine` method and the `node_postprocessing` argument can be used to integrate the reranker object. In this setup, the retriever selects the top 10 documents based on semantic similarity. Afterward, the reranker refines this selection, narrowing it down to the two most relevant documents.

```
query_engine = vector_index.as_query_engine(
similarity_top_k=10,
node_postprocessors=[cohere_rerank],
)

response = query_engine.query(
"What did Sam Altman do in this essay?",
)
print(response)
```

> Sam Altman was asked if he wanted to be the president of Y Combinator (YC) and initially said no. However, after persistent persuasion, he eventually agreed to take over as president starting with the winter 2014 batch.

?? Rerank computes a relevance score for the query and each document and returns a sorted list from the most to the least relevant document.

The reranking process in search systems improves the quality and precision of the returned results. It enhances existing systems without requiring comprehensive modifications, offering a cost-efficient approach to improve search functionality. Cohere Rerank is a very popular choice for a reranker that can easily be used via API.

Recursive Retrieval and Small-To-Big Retrieval

An alternative approach to retrieving relevant documents is to use document summaries instead of extracting fragmented snippets or brief text chunks to respond to queries. This method ensures that the responses encapsulate the full context or topic under consideration, providing a more comprehensive understanding of the subject. It includes two types of retrieval: recursive and small-to-big retrieval.

Recursive retrieval is particularly effective for documents with a hierarchical structure, as it facilitates forming links and connections between nodes. Jerry Liu, the founder of LlamaIndex, highlights its applicability in situations like a PDF file, which might contain “sub-data” such as tables, diagrams, and references to other documents. This method efficiently navigates through the graph of interconnected nodes to pinpoint information. Its versatility allows for its use in various contexts, including node references, document agents, and query engines. For practical implementations, such as processing a PDF file and extracting data from tables, the LlamaIndex Recursive Retriever tutorials are accessible at towardsai.net/book.

The small-to-big retrieval approach enhances the effectiveness of RAG pipelines by separating the text chunks used for retrieval from those used for synthesis. Instead of embedding and retrieving large text chunks that may contain filler content and obscure semantic relevance, this approach utilizes smaller, more targeted chunks for retrieval. After identifying relevant smaller chunks, the associated larger chunks are used to provide

broader context for synthesis. This method improves retrieval accuracy and ensures the LLM has sufficient context. Techniques include using smaller child chunks linked to bigger parent chunks and sentence window retrieval.

Find a hands-on tutorial on implementing Small-to-Big Retrieval from the documentation at towardsai.net/book.

RAG - Metrics & Evaluation

- Find the Notebook for this section at towardsai.net/book.

RAG (and LLM) Metrics

Evaluating LLMs and retrieval-augmented generation (RAG) systems requires a detailed analysis of individual components and the system as a whole. Setting baseline values for pieces, such as chunking logic and embedding models, then assessing each part individually and end-to-end is critical for understanding the impact of changes on the system's overall performance. Holistic modules in these evaluations don't always require ground-truth labels, as they can be assessed based on the language model's query, context, response, and interpretations.

Here are five commonly used metrics for evaluating your systems:

1. Correctness: This metric assesses whether the generated answer aligns with a given query's reference answer. It requires labels and involves verifying the accuracy of the generated answer by comparing it to a predefined reference answer.

2. Faithfulness: This evaluates the integrity of the answer concerning the retrieved contexts. The faithfulness metric ensures that the answer accurately reflects the information in the retrieved context, free from distortions or fabrications that might misrepresent the source material.

3. Context Relevancy: This measures the relevance of the retrieved context and the resulting answer to the original query. The goal is

to ensure the system retrieves relevant information to the user's request.

4. Guideline Adherence: This metric determines if the predicted answer follows established guidelines. It checks whether the response meets predefined criteria, including stylistic, factual, and ethical standards, ensuring the answer responds to the query and adheres to specific norms.

5. Embedding Semantic Similarity: This involves calculating a similarity score between the generated and reference answers' embeddings. It requires reference labels and helps estimate the closeness of the generated response to a reference in terms of semantic content.

?? You can find the documentation pages for the above metrics at towardsai.net/book.

The evaluation of RAG applications begins with a focus on their primary objective: generate useful outputs supported by contextually relevant facts obtained from retrievers. This analysis then zooms in on specific evaluation metrics such as faithfulness, answer relevancy, and the Sensibleness and Specificity Average (SSA). Google's SSA metric, which assesses open-domain chatbot responses, evaluates sensibleness (contextual coherence) and specificity (providing detailed and direct responses). Initially involving human evaluators, this metric ensures that outputs are comprehensive and not excessively vague or general.

A high faithfulness score does not necessarily correlate with high relevance. For instance, a response that accurately mirrors the context but does not directly address the query would receive a lower score in answer relevance. This could happen mainly if the response contains incomplete or redundant elements, indicating a gap between the accuracy of the provided context and the direct relevance to the question.

This section proceeds by showing how to compute the faithfulness score using the FaithfulnessEvaluator class from LlamaIndex. This metric focuses on preventing the issue of "hallucination" and examines responses to determine their alignment with the retrieved context. It assesses whether the

response is consistent with the context and the initial query and fits with reference answers or guidelines. The output of this evaluation is a boolean value, indicating whether the response has successfully met the criteria for accuracy and faithfulness.

To use the FaithfulnessEvaluator, install the required libraries using Python's package manager using `!pip install -q llamaindex==0.9.14.post3 deeplake==3.8.12 openai==1.3.8 cohere==4.37`.

Following this, set the API keys for OpenAI and Activeloop and replace the placeholders in the provided code with their respective API keys.

```
import os

os.environ["OPENAI_API_KEY"] = "<YOUR_OPENAI_KEY>"
os.environ["ACTIVELOOP_TOKEN"] = "<YOUR_ACTIVELOOP_TOKEN>"
```

Here's an example for evaluating a single response for faithfulness:

```
from llama_index import ServiceContext
from llama_index.llms import OpenAI

# build service context
llm = OpenAI(model="gpt-4-turbo", temperature=0.0)
service_context = ServiceContext.from_defaults(llm=llm)

from llama_index.vector_stores import DeepLakeVectorStore
from llama_index.storage.storage_context import StorageContext
from llama_index import VectorStoreIndex

vector_store = DeepLakeVectorStore(
dataset_path="hub://genai360/LlamaIndex_paulgraham_essay",
overwrite=False
)
storage_context =
StorageContext.from_defaults(vector_store=vector_store)

index = VectorStoreIndex.from_vector_store(
vector_store, storage_context=storage_context
)

from llama_index.evaluation import FaithfulnessEvaluator
```

```
# define evaluator
evaluator = FaithfulnessEvaluator(service_context=service_context)

# query index
query_engine = index.as_query_engine()
response = query_engine.query(
"What does Paul Graham do?"
)

eval_result = evaluator.evaluate_response(response=response)

print("> response:", response)
print("> evaluator result:", eval_result.passing)
```

```
> response: Paul Graham is involved in various activities. He is a writer and has given talks on topics such as starting a startup. He has also worked on software development, including creating software for generating websites and building online stores. Additionally, he has been a studio assistant for a beloved teacher who is a painter.
> evaluator result: True
```

Most of the above code was previously discussed and will likely be familiar. It includes creating an index from the Deep Lake vector database, using it to make queries to the LLM, and performing the evaluation procedure. In this process, the query engine answers the question and sends its response to the evaluator for further examination.

In this case, the code creates a `FaithfulnessEvaluator` object, a mechanism for evaluating the precision of responses produced by the language model, GPT-4 Turbo. This evaluator operates using the previously defined `service_context`, which contains the configured GPT-4 Turbo model and provides the settings and parameters required for the language model's optimal functioning.

The fundamental responsibility of the `FaithfulnessEvaluator` is assessing the extent to which the responses from the language model align with accurate and reliable information. It uses a series of criteria or algorithms to accomplish this, comparing the model-generated responses with verified factual data or anticipated results.

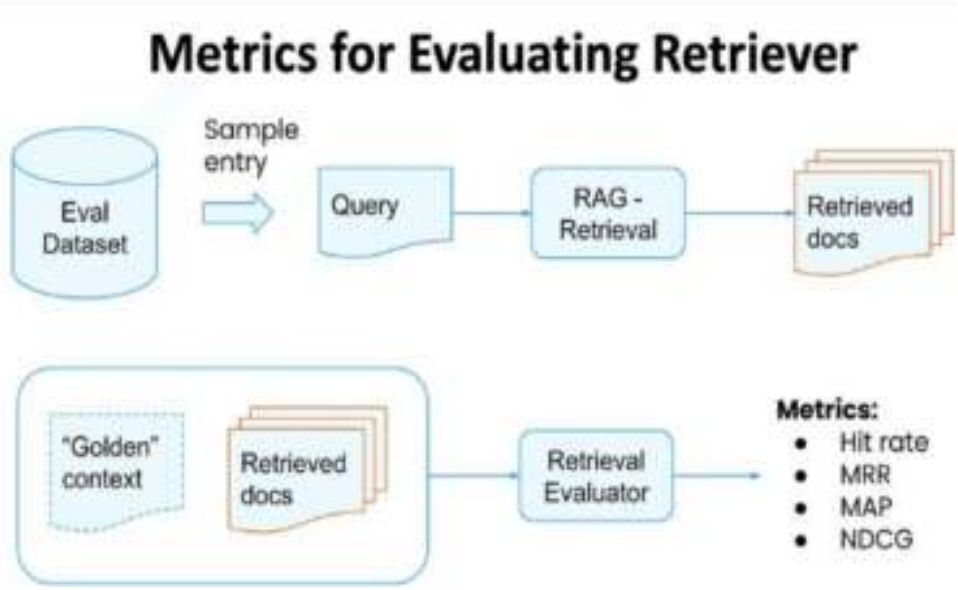
The evaluator proceeds to examine the response for its commitment to factual accuracy. This involves determining if the response accurately and dependably represents historical facts related to the query. The outcome of

this assessment (`eval_result`) is then reviewed to ascertain if it aligns with the accuracy benchmarks established by the evaluator, as denoted by `eval_result.passing` . The result returns a Boolean (here `True`) value indicating whether the response passed the accuracy and faithfulness checks.

Retrieval-Specific Evaluation Metrics

The evaluation of retrieval in RAG systems (not just LLM generation quality) involves determining the relevance of documents to specific queries. In information retrieval, the main goal is to identify unstructured data that meets a particular information requirement within a database.

✵ RAG system outputs need three critical aspects: factual accuracy, direct relevance to the query, and inclusion of essential contextual information.



The evaluation metrics of retrieval in RAG systems.

Metrics used to assess the effectiveness of a retrieval system mostly include Mean Reciprocal Rank (MRR), Hit Rate, Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG):

- **Mean Reciprocal Rank (MRR)** measures the effectiveness of a system by averaging the reciprocal of the rank of the first relevant item in the search results. Higher MRR indicates better performance, as relevant items appear earlier.

- **Hit Rate** evaluates how often a relevant item appears within the top N results. It is a simple accuracy measure indicating the presence of at least one relevant item in the retrieved set.
- **Mean Average Precision (MAP)** calculates the average precision at each relevant item across multiple queries. It provides a single measure of quality by considering the precision of relevant items' placements within the ranked results.
- **Normalized Discounted Cumulative Gain (NDCG)** evaluates the quality of the ranking by considering both the position and relevance of items. It discounts the relevance of items lower in the ranking, rewarding systems that rank highly relevant items earlier.

The `RetrieverEvaluator` from LlamaIndex is an advanced method to calculate metrics such as Mean Reciprocal Rank (MRR) and Hit Rate. Its primary function is to evaluate the effectiveness of a retrieval system that sources data relevant to user queries from a database or index. This class measures the retriever's performance in responding to specific questions and providing the expected results, setting benchmarks for assessment.

For this evaluator, it is necessary to compile an evaluation dataset, which includes the content, a set of queries, and corresponding reference points for answering these queries. The `generate_question_context_pairs` function in LlamaIndex can create the evaluation dataset automatically. The process involves inputting a query and using the dataset as a benchmark to ensure the retrieval system accurately sources the correct documents. A detailed tutorial on using the `RetrieverEvaluator` from the LlamaIndex documentation is accessible at towardsai.net/book.

?? Although the evaluation of single queries is discussed, real-world applications typically need batch evaluations. This involves extracting a broad range of queries and their anticipated outcomes from the retriever to assess its general reliability. The retriever is evaluated using multiple queries and the expected results during batch testing. The process involves methodically inputting various queries into the retriever and comparing its

responses to established correct answers to measure its consistency accurately.

There are open-source datasets that can be freely used to evaluate a RAG pipeline. An evaluation dataset comprises a selection of queries carefully paired with the most appropriate sources containing their answers. Such a dataset includes ideal expected answers from the large language model. Each query is paired with the most relevant source from the document, ensuring these sources directly address the queries.

Developing an evaluation dataset requires collecting a variety of realistic customer queries and matching them with expert answers. This dataset can be used to evaluate the responses of a language model, ensuring the LLM's answers are accurate and relevant compared to expert responses. More information on creating this dataset is accessible at towardsai.net/book.

Once the evaluation dataset is prepared, it can be utilized to assess the quality of responses from the LLM. Following each evaluation, metrics will be available to quantify the user experience, providing valuable insights into the performance of the LLM. For example:

Similarity	Relevance	Coherence	Grounded-ness
3.7	77	88	69

⚠ Although generating questions synthetically has its advantages, it is advisable to use questions that are tied to authentic user experiences. The questions employed in these evaluations should closely resemble real user queries, which are usually difficult to accomplish with the large language model. It is recommended to manually create questions emphasizing the users' viewpoint for a more precise representation.

Community-Based Evaluation Tools

LlamaIndex includes various evaluation tools designed to encourage community engagement and collaborative efforts. These tools facilitate a collective process of assessing and improving the system. LlamaIndex enables the easy integration of constant feedback, allowing users and developers to participate actively in the evaluation phase. Notable tools in

this ecosystem include Ragas, an important tool with extensive metrics for assessing and integrating with LlamaIndex, and DeepEval, a tool for in-depth review and complete assessments of several areas of the system.

Evaluating with Ragas

Ragas's evaluation process revolves around utilizing specific metrics, including faithfulness, answer relevancy, context precision, context recall, and harmfulness. This process has several key elements. The performance of the query engine is the central focus of the assessment, as it plays a crucial role in the evaluation. Ragas provides a range of metrics specifically designed to facilitate a comprehensive analysis of the engine's capabilities. Finally, a carefully curated set of questions is used to test the engine's proficiency in retrieving and generating accurate responses.

The first step to using the Ragas library is setting up a query engine. This process involves loading a document. We will use the "New York City" Wikipedia page as the source document in this case. Next, you will need to install two more libraries: one to process the webpage content, and the other is a prerequisite for the evaluation library: `! pip install html2text==2020.1.16 ragas==0.0.22 .`

Load the content by passing a URL to the `SimpleWebPageReader` class. Use these documents to build the index and query engine. Now, you can try asking questions about the document!

```
from llama_index.readers.web import SimpleWebPageReader
from llama_index import VectorStoreIndex, ServiceContext

documents = SimpleWebPageReader(html_to_text=True).load_data(
["https://en.wikipedia.org/wiki/New_York_City"]
)

vector_index = VectorStoreIndex.from_documents(
documents, service_context=ServiceContext.from_defaults(chunk_size=512)
)

query_engine = vector_index.as_query_engine()
```

```
response_vector = query_engine.query("How did New York City get its name?")

print(response_vector)
```

```
New York City got its name in honor of the Duke of York, who later became King James II of England. The Duke of York was appointed as the proprietor of the former territory of New Netherland, including the city of New Amsterdam, when England seized it from Dutch control.
```

The next step involves composing questions, ideally derived from the original documents, which can be done using an LLM like GPT-4. Usually, one would then use those questions to get their answers through the RAG pipeline and evaluate them against the true answers. For the purpose of this example, we write the true answers by hand (i.e., the `eval_answers` list) and include both correct and incorrect answers so that we won't get perfect scores in the computed metrics.

```
eval_questions = [
"What is the population of New York City as of 2020?",
"Which borough of New York City has the highest population?",
"What is the economic significance of New York City?",
"How did New York City get its name?",
"What is the significance of the Statue of Liberty in New York City?",
]

eval_answers = [
"8,804,000", # incorrect answer
"Queens", # incorrect answer
"""New York City's economic significance is vast, as it serves as the global financial capital, housing Wall Street and major financial institutions. Its diverse economy spans technology, media, healthcare, education, and more, making it resilient to economic fluctuations. NYC is a hub for international business, attracting global companies, and boasts a large, skilled labor force. Its real estate market, tourism, cultural industries, and educational institutions further fuel its economic prowess. The city's transportation network and global influence amplify its impact on the world stage, solidifying its status as a vital economic player and cultural epicenter.""",
"""New York City got its name when it came under British control in 1664. King Charles II of England granted the lands to his brother, the Duke of York, who named the city New York in his own honor.""",
"""The Statue of Liberty in New York City holds great significance as a symbol of the United States and its ideals of liberty and peace. It greeted millions of immigrants who arrived in the U.S. by ship in the late 19th and early 20th centuries, representing hope and freedom for those seeking a better life. It has since become an iconic landmark and a global symbol of cultural diversity and freedom.""",
]

eval_answers = [[a] for a in eval_answers]
```

This is the setup stage of the evaluation process. The competency of the `QueryEngine` is evaluated based on how well it processes and replies to these

specific questions, with the responses serving as a baseline for measuring performance. The metrics from the Ragas library must be imported.

```
from ragas.metrics import (
faithfulness,
answer_relevancy,
context_precision,
context_recall,
)
from ragas.metrics.critique import harmfulness

metrics = [
faithfulness,
answer_relevancy,
context_precision,
context_recall,
harmfulness,
]
```

The `metrics` list groups the metrics into a collection, which can be used in the assessment process to evaluate various elements of the `QueryEngine`'s performance. The results will include scores for each metric. Finally, let's run the evaluation.

```
from ragas.llama_index import evaluate

result = evaluate(query_engine, metrics, eval_questions, eval_answers)

# print the final scores
print(result)
```

```
evaluating with [faithfulness]
100%|██████████| 1/1 [00:16<00:00, 16.95s/it]
evaluating with [answer_relevancy]
100%|██████████| 1/1 [00:03<00:00, 3.54s/it]
evaluating with [context_precision]
100%|██████████| 1/1 [00:02<00:00, 2.73s/it]
evaluating with [context_recall]
100%|██████████| 1/1 [00:07<00:00, 7.06s/it]
evaluating with [harmfulness]
100%|██████████| 1/1 [00:02<00:00, 2.16s/it]

{'faithfulness': 0.8000, 'answer_relevancy': 0.7634, 'context_precision': 0.6000,
'context_recall': 0.8667, 'harmfulness': 0.0000}
```

The system’s responses were evaluated across several metrics, each providing insight into different aspects of its performance.

First, the faithfulness score was 0.8, indicating a relatively high level of accuracy. This means the responses largely adhered to the factual content of the source material. Next, the system achieved an answer relevancy score of 0.7634. This high score suggests that most of the responses closely aligned with the intent of the given queries, ensuring that the answers were pertinent to the questions asked.

The context precision score, on the other hand, was 0.6. This indicates that while the system sometimes included relevant information, it also occasionally incorporated irrelevant context. In contrast, the context recall score was quite high, at 0.8667. This suggests that the system effectively identified and utilized the most relevant contexts in generating its final answers.

Lastly, the harmfulness score was 0, indicating that none of the responses contained harmful or inappropriate content. This metric reflects the system’s success in maintaining safe and suitable interactions.

Custom Evaluation of RAG Pipelines

A comprehensive set of assessment benchmarks is crucial for an effective evaluation of a custom RAG system. These benchmarks are key in evaluating different aspects of the system, including its effectiveness and reliability. Employing diverse measures ensures a thorough evaluation, providing a deep understanding of the system’s overall performance. This phase involves creating a customized evaluation pipeline.

Start with loading a dataset. Construct an evaluation dataset from this initial dataset and calculate various previously mentioned metrics.

First, download the text file that can serve as the dataset. The SimpleDirectoryReader class is a simple utility that allows you to read local files and transform them into LlamaIndex documents.

```
wget 'https://raw.githubusercontent.com/idontcalculate/data-repo/main/venus_transmission.txt'
```

The text file can be loaded as a Document object, implemented by the LlamaIndex library:

```
from llama_index import SimpleDirectoryReader

reader = SimpleDirectoryReader(input_files=["content\venus_transmission.txt"])

docs = reader.load_data()
print(f"Loaded {len(docs)} docs")
```

Loaded 1 docs In this case, the SimpleNodeParser tool converts documents into the structured format (the nodes). It is very useful for modifying how texts are processed. It determines the size of each text chunk, manages overlap between chunks, and adds information. Every document chunk is treated as a separate node in this system. The parser is configured explicitly with a chunk_size of 512, implying that each node will have 512 characters from the original document. These split portions are then used to create indexes.

```
from llama_index.node_parser import SimpleNodeParser
from llama_index import VectorStoreIndex

# Build index with a chunk_size of 512
node_parser = SimpleNodeParser.from_defaults(chunk_size=512)
nodes = node_parser.get_nodes_from_documents(docs)
vector_index = VectorStoreIndex(nodes)
```

The indexes can now be used as a query engine to ask a specific question concerning the source document:

```
query_engine = vector_index.as_query_engine()

response_vector = query_engine.query("""What was The first beings to inhabit the planet?""")
print( response_vector.response )
```

The first beings to inhabit the planet were a dinoid and reptoid race from two different systems outside our solar system.

The query engine's response is stored in the response_vector variable. Consequently, the document is divided into nodes, indexed, and queried using a language model. To dig deeper into the response, we can use the source_nodes key to extract the used document from the index.

```
# First retrieved node
response_vector.source_nodes[0].get_text() They had heard of this beautiful new planet. At this time, Earth had two moons to harmonize the weather conditions and control the tides of the large bodies of water.
The first beings to inhabit the planet were a dinoid and reptoid race from two different systems outside our solar system. They were intelligent and walked on two legs like humans and were war-
```

```
like considering themselves to be superior to all other life forms. In the past, the four races of humans had conflicts with them before they outgrew such behavior. They arrived on Earth to rob it of its minerals and valuable gems. Soon they had created a terrible war.
[…]
```

We can also access the content of the second node that played a role in content creation by indexing the second item on the list.

```
# Second retrieved node
response_vector.source_nodes[1].get_text() Due to the radiation, the survivors of the dinoids and reptoids mutated into the Dinosaurs and giant reptilians you know of in your history. The humans that were trapped there mutated into what you call Neanderthals.
The Earth remained a devastated ruin, covered by a huge dark nuclear cloud and what vegetation was left was being devoured by the giant beings, also humans and animals by some. It was this way for hundreds of years before a giant comet crashed into one of the oceans and created another huge cloud. This created such darkness that the radiating heat of the Sun could not interact with Earth's gravitational field and an ice age was created. This destroyed the mutated life forms and gave the four races the chance to cleanse and heal the Earth with technology and their energy.
[…]
```

You can examine the content from the second node identified as relevant by the query engine, which provides additional context or information in response to a query. This process helps understand the range of information the query engine accesses and how different segments of the indexed documents contribute to the overall response.

Creating a custom RAG evaluation process requires generating a series of questions and their corresponding answers, all related to the content we have loaded. The generate_question_context_pairs class uses the LLM to generate questions based on the content of each node. For every node, two questions will be generated, creating a dataset where each entry includes a context (the text of the node) and corresponding questions. This Q&A dataset will be used to assess the capabilities of a RAG system in terms of question generation and context comprehension. You can see some of the generated questions in the output.

```
from llama_index.llms import OpenAI
from llama_index.evaluation import generate_question_context_pairs

# Define an LLM
llm = OpenAI(model="gpt-3.5-turbo")

qa_dataset = generate_question_context_pairs(
nodes,
```

```
llm=llm,
num_questions_per_chunk=2
)

queries = list(qa_dataset.queries.values())
print(queries[0:10])
```

```
100%|██████████| 13/13 [00:31<00:00, 2.46s/it]

['Explain the role of different alien races in the history of our solar system according to the information provided. How did these races contribute to the transformation process and why was Earth considered a special planet?', 'Describe the advanced abilities and technology possessed by the Masters and beings mentioned in the context. How did their understanding of creation and their eternal nature shape their perspective on life and death?', 'How did the four races of humans demonstrate their mastery of creativity and what were the potential consequences of using this power for selfish reasons?', […]
```

The generated question-answer dataset can now be used by the RetrieverEvaluator class to evaluate the retriever's performance. It uses the retriever to query each question and determines which chunks are returned as the answer. The higher the MRR and Hit rate, the better the retriever can match the chunk with the correct answer.

```
from llama_index.evaluation import RetrieverEvaluator

retriever = vector_index.as_retriever(similarity_top_k=2)

retriever_evaluator = RetrieverEvaluator.from_metric_names(
["mrr", "hit_rate"], retriever=retriever
)

# Evaluate
eval_results = await retriever_evaluator.aevaluate_dataset(qa_dataset)

def display_results(name, eval_results):
"""Display results from evaluate."""

    metric_dicts = []
for eval_result in eval_results:
metric_dict = eval_result.metric_vals_dict
metric_dicts.append(metric_dict)

    full_df = pd.DataFrame(metric_dicts)

    hit_rate = full_df["hit_rate"].mean()
mrr = full_df["mrr"].mean()

    metric_df = pd.DataFrame(
{"Retriever Name": [name], "Hit Rate": [hit_rate], "MRR": [mrr]}
)
```

```
    return metric_df

display_results("OpenAI Embedding Retriever", eval_results)
```

	Retriever Name	Hit Rate	MRR
0	OpenAI Embedding Retriever	0.884615	0.730769

We will improve the assessment by using new measures such as faithfulness and relevancy. To accomplish this, let's use a portion of the created Q&A dataset and define GPT-3.5 and GPT-4 instances. It is best to use a more advanced model for evaluation while employing the less expensive model for generation to limit costs.

```
# gpt-3.5-turbo
gpt35 = OpenAI(temperature=0, model="gpt-3.5-turbo")
service_context_gpt35 = ServiceContext.from_defaults(llm=gpt35)

# gpt-4-turbo
gpt4 = OpenAI(temperature=0, model="gpt-4-turbo")
service_context_gpt4 = ServiceContext.from_defaults(llm=gpt4)

vector_index = VectorStoreIndex(nodes, service_context = service_context_gpt35)
query_engine = vector_index.as_query_engine()

eval_query = queries[10]
response_vector = query_engine.query(eval_query)

print("> eval_query: ", eval_query)
print("> response_vector:", response_vector)
```

> eval_query: How did the colonies respond to the declaration of war by the dark forces, and what measures did they take to protect their knowledge and technology?
> response_vector: The colonies did not fight back against the dark forces when they declared war. Instead, they sent most of their people into hiding in order to rebuild the colonies later. They also destroyed everything to ensure that their knowledge and technology would not fall into the hands of the dark forces. Additionally, Lemuria and Atlantis were destroyed by their inhabitants to prevent the misuse of their knowledge and technology by the dark forces.

We will define the evaluator classes in charge of measuring each metric. Next, we'll use a sample response to see if it meets the test conditions.

```
from llama_index.evaluation import RelevancyEvaluator
from llama_index.evaluation import FaithfulnessEvaluator

relevancy_gpt4 = RelevancyEvaluator(service_context=service_context_gpt4)
faithfulness_gpt4 = FaithfulnessEvaluator(service_context=service_context_gpt4)

# Compute faithfulness evaluation
```

```
eval_result = faithfulness_gpt4.evaluate_response(response=response_vector)
# check passing parameter in eval_result if it passed the evaluation.
print( eval_result.passing )

# Relevancy evaluation
eval_result = relevancy_gpt4.evaluate_response(
query=eval_query, response=response_vector
)
# You can check passing parameter in eval_result if it passed the evaluation.
print(eval_result.passing)
```

```
True
True We used a for-loop to feed each sample from the assessment dataset and receive the responses. In this case, we can use the LlamaIndex BatchEvalRunner  class, which concurrently conducts the evaluation procedure in batches to be faster.
```

```
#Batch Evaluator:
#BatchEvalRunner to compute multiple evaluations in batch wise manner.

from llama_index.evaluation import BatchEvalRunner

# Let's pick top 10 queries to do evaluation
batch_eval_queries = queries[:10]

# Initiate BatchEvalRunner to compute FaithFulness and Relevancy Evaluation.
runner = BatchEvalRunner(
{"faithfulness": faithfulness_gpt4, "relevancy": relevancy_gpt4},
workers=8,
)

# Compute evaluation
eval_results = await runner.aevaluate_queries(
query_engine, queries=batch_eval_queries
)

# get faithfulness score
faithfulness_score =
sum(result.passing for result in eval_results['faithfulness']) /
len(eval_results['faithfulness'])
# get relevancy score
relevancy_score =
sum(result.passing for result in eval_results['faithfulness']) /
len(eval_results['relevancy'])

print("> faithfulness_score", faithfulness_score)
print("> relevancy_score", relevancy_score)
```

```
> faithfulness_score 1.0
> relevancy_score 1.0
```

The batch processing method helps quickly measure the system's performance across various queries. A faithfulness score of 1.0 indicates

that the generated answers are based on the retrieved context and contain no hallucinations. Furthermore, a Relevance score of 1.0 means that the generated replies consistently fit with the collected context and queries.

However, those perfect scores are to be attributed more to the simplicity of the evaluation dataset than to the quality of the RAG system. Ideally, one would want to add more complex queries into the evaluation dataset, such as queries whose answer is distributed across different documents, or more ambiguous queries, or queries whose answers can't be found in the dataset, *etc.* It's not easy to automatically create an evaluation dataset that exactly mirrors the questions asked by users and the answers that they'd want. One way of improving the evaluation dataset is by having human expert labelers create it, which is probably the best option but also the most expensive. A cheaper but more noisy option is to improve the automatic generation of the evaluation dataset.

LangChain provides a tool called LangSmith to monitor all the inputs and outputs that go through the LLM in your application. LangSmith can greatly help you get an idea of the queries that your users ask, so that you can build an appropriate evaluation dataset based on that. The next section introduces LangSmith.

LangChain LangSmith and LangChain Hub

- Find the Notebook for this section at towardsai.net/book.

?? While LangChain is appropriate for initial prototyping, LangSmith provides a setting for debugging, testing, and refining LLM applications.

LangSmith is a platform for evaluating and monitoring the quality of LLM systems' outputs. Its functionality includes tracking metadata, token usage, and execution time, which is vital for managing resources effectively.

LangSmith improves the efficiency and performance of new chains and tools. It also provides visualization tools to recognize response patterns and trends, enhancing the understanding and analysis of performance. It allows

users to create customized testing environments tailored to specific requirements, enabling comprehensive evaluation under various conditions. The platform also tracks the executions linked to an active instance and allows for the testing and assessing of any prompts or responses produced. LangSmith offers several tutorials and in-depth documentation to assist users in getting started.

On the other hand, LangChain Hub is a platform designed for the development, storage, and sharing of reusable components used in building applications with large language models (LLMs). It offers a repository of modules such as prompts, chains, and agents that can be easily accessed and integrated into projects. LangChain Hub simplifies the creation of LLM-driven applications by providing tools and templates and promoting collaboration among developers.

This section contains an example of how to use LangChain Hub together with a question-answering chain in LangChain. We will guide you through the setup process for LangChain, including installing necessary libraries and configuring environment variables. A LangSmith account is required for certain features like tracing. Detailed steps for setting up a new account will be provided.

First, you will need an API key. Navigate to the LangSmith website and register for an account. Find the option to generate an API key on the settings page. Click the “Generate API Key” button to receive your API key.

Install the necessary libraries using the command `! pip install -q langchain==0.0.346 openai==1.3.7 tiktoken==0.5.2 cohere==4.37 deeplake==3.8.11 langchainhub==0.1.14.`

Configure the environment with the API keys for OpenAI, which will be used in the embedding generation process, and the Activeloop key, which will be used to store data in the cloud:

```
import os

os.environ["OPENAI_API_KEY"] = "<YOUR_OPENAI_API_KEY>"
os.environ["ACTIVELOOP_TOKEN"] = "
```

```
<YOUR_ACTIVELOOP_API_KEY>"
```

The following environment variables can be used to keep track of the runs on the LangSmith dashboard's projects area:

```
os.environ["LANGCHAIN_TRACING_V2"] = True
os.environ["LANGCHAIN_ENDPOINT"] = "https://api.smith.langchain.com"
os.environ["LANGCHAIN_API_KEY"] = "<YOUR_LANGSMITH_API_KEY>"
os.environ["LANGCHAIN_PROJECT"] = "langsmith-intro" # if not specified,
# defaults to "default"
```

The following code shows how to commit a prompt to the LangChain Hub by adding it to your handle's namespace.

```
from langchain import hub
from langchain.prompts.chat import ChatPromptTemplate

prompt = ChatPromptTemplate.from_template("""You are an assistant for question-answering tasks. Use the following pieces of retrieved context to answer the question. If you don't know the answer, just say that you don't know. Use three sentences maximum and keep the answer concise.\nQuestion: {question} \nContext: {context} \nAnswer:""")

handle = "<YOUR_USERNAME>"
hub.push(f"{handle}/rag-prompt", prompt)
```

If you update the prompt, you can push the modified prompt to the same key to "commit" a new version of the prompt during evaluation. Let's say we want to add a system message to the prompt.

```
# You may try making other changes and saving them in a new commit.
from langchain import schema

prompt.messages.insert(0,
schema.SystemMessage(
content="You are a precise, autoregressive question-answering system."
)
)
```

The most recent version of the prompt is kept as the latest version.

```
# Pushing to the same prompt "repo" will create a new commit
hub.push(f"{handle}/rag-prompt", prompt)
```

LangSmith also offers a feature for reviewing the inputs and outputs of each component within a chain, making it easier to log runs for large language model applications. This functionality is particularly beneficial when debugging your application or understanding the behavior of certain components. For more information on Tracing, visit the LangChain documentation.

The next steps involve loading data from a webpage, dividing it into smaller chunks, converting these segments into embeddings, and then storing them in the Deep Lake vector database. This process also includes prompt templates from the LangSmith Hub.

The content of a webpage can be read using the `WebBaseLoader` class. This class will provide a single instance of the `Document` class with all the text from the URL. Subsequently, this large text is divided into smaller chunks, each comprising 500 characters without overlapping, resulting in 130 chunks.

```
from langchain.document_loaders import WebBaseLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter

# Loading
loader = WebBaseLoader("https://lilianweng.github.io/posts/2023-06-23-agent/")
data = loader.load()
print(len(data))

# Split
text_splitter = RecursiveCharacterTextSplitter(chunk_size=500, chunk_overlap=0)
all_splits = text_splitter.split_documents(data)
print(len(all_splits))
```

```
1
130
```

Chunks can be saved in the Deep Lake vector database using the LangChain integration. The `DeepLake` class transforms texts into embeddings using OpenAI's API and stores these results in the cloud. You can use your own organization name (your username by default) to create the dataset. Note that this task involves the costs of utilizing OpenAI endpoints.

```
from langchain.embeddings import OpenAIEmbeddings
from langchain.vectorstores import DeepLake
```

```
vectorstore = DeepLake.from_documents(
all_splits,
dataset_path="hub://genai360/langsmith_intro",
embedding=OpenAIEmbeddings(), overwrite=False)
```

```
Your Deep Lake dataset has been successfully created!
Creating 130 embeddings in 1 batches of size 130:: 100%|██████████| 1/1
[00:05<00:00, 5.81s/it] dataset (path='hub://genai360/langsmith_intro', tensors=['text',
'metadata', 'embedding', 'id'])

    tensor htype shape dtype compression
------- ------- ------- ------- -------
text text (130, 1) str None
metadata json (130, 1) str None
embedding embedding (130, 1536) float32 None
id text (130, 1) str None
```

After processing the data, select a prompt from the LangChain Hub, getting a ChatPromptTemplate instance. Using a Prompt Template removes trial and error and allows using already-tested implementations. The code below tags a specific version of the prompt to ensure that future changes do not affect the version currently deployed.

```
from langchain import hub

prompt = hub.pull("rlm/rag-prompt:50442af1")
print(prompt)
```

```
ChatPromptTemplate(input_variables=['context', 'question'], messages=
[HumanMessagePromptTemplate(prompt=PromptTemplate(input_variables=['context',
'question'], template="You are an assistant for question-answering tasks. Use the following
pieces of retrieved context to answer the question. If you don't know the answer, just say that
you don't know. Use three sentences maximum and keep the answer concise.\nQuestion:
{question} \nContext: {context} \nAnswer:"))]) Finally, use the RetrievalQA chain to
retrieve related documents from the database and then the ChatOpenAI model to build the
final response using these documents.
```

```
# LLM
from langchain.chains import RetrievalQA
from langchain.chat_models import ChatOpenAI

llm = ChatOpenAI(model_name="gpt-3.5-turbo", temperature=0)

# RetrievalQA
qa_chain = RetrievalQA.from_chain_type(
llm,
```

```
retriever=vectorstore.as_retriever(),
chain_type_kwargs={"prompt": prompt}
)

question = "What are the approaches to Task Decomposition?"
result = qa_chain({"query": question})
result["result"]
```

```
The approaches to task decomposition include using LLM with simple prompting, task-specific instructions, and human inputs.
```

Prompt versioning encourages continual experimentation and collaboration, preventing the unintentional deployment of chain components without thorough testing. Leveraging the LangChain Hub and prompt versioning provides an easy and organized approach to many LLM-related tasks. It ensures reliable performance by using pre-tested, versioned prompts, reducing the need for trial and error.

Recap

Integrating components such as query expansion, transformations, and construction techniques leads to the creation of an efficient retrieval engine, enhancing the capabilities of basic RAG-based applications. Additionally, advanced strategies such as reranking, recursive retrieval, and small-to-big retrieval significantly enhance the search process. These methods contribute to increased accuracy and a wider range of search results. By adopting these approaches, information retrieval systems become more efficient at delivering precise and relevant results.

Currently, RAG applications pose specific challenges for effective implementation, like maintaining up-to-date information, the need for precise chunking and data distribution, and managing the multimodal nature of documents. Optimization strategies such as choosing the right model, optimizing the inference code, and leveraging tools like LlamaIndex to create a network of interconnected chunks significantly improve the effectiveness of RAG applications. Regular evaluations and the implementation of generative feedback loops and hybrid search methods are also effective for sustaining and enhancing the performance of RAG systems.

We created and tested a RAG pipeline with LlamaIndex, focusing on evaluating both the retrieval system and the replies generated by the pipeline. Evaluating LLMs and chatbots is difficult because of the subjective nature of their outputs; different people may have different ideas about what a good response is. As a result, it is essential to assess several areas of RAG applications and evaluate each independently, using metrics customized to those tasks.

LangChain Hub is a centralized platform for managing, versioning, and distributing prompts. LangSmith is particularly effective in diagnosing errors, comparing the effectiveness of different prompts, evaluating output quality, and tracking crucial metadata such as token usage and execution time, which are key for optimizing LLM applications. The platform also comprehensively analyzes how various prompts influence LLM performance. Its user-friendly interface makes the refining process more transparent and manageable.

Chapter X: Agents

What are Agents: Large Models as Reasoning Engines

The recently released large pre-trained models created the opportunity to build agents — intelligent systems that use these models to plan the execution of complex tasks. Agent workflows are now possible because of the greater reasoning capabilities of large pre-trained models such as OpenAI's latest models.

These models can be used for their deep internal knowledge to create new, compelling material, think through problems, use external tools, and make plans. For instance, you can create a research agent that will find essential facts from different sources and generate an answer that combines them in a helpful way. Agents are systems that use LLMs to determine and order a set of actions. In a simple workflow, these actions might involve using a tool, examining its output, and responding to the user's request. Some essential components include **Tools** to achieve a specific task, such as using the Google Search API, accessing an SQL database, running code with a Python REPL, or using a calculator. **The reasoning Engine or Core** is the large model that powers the system, and **Agent orchestration** is the complete system that manages the interaction between the LLM and its tools.

In this chapter, we will learn about projects like "AutoGPT" and "BabyAGI," which are classified as Autonomous Agents. These systems are characterized by their ability to generate their own prompts for tackling tasks independently. Additionally, we'll explore Agent Simulation projects such as "CAMEL" and "Generative Agents," which focus on the dynamic interactions between multiple agents. While these specific agent implementations are now somewhat outdated, and creating agents is getting easier month by month as foundational LLMs become more capable or "agent native," the projects in this chapter remain useful examples for learning the principles.

Agents are generally categorized into two types: **Action Agents** and **Plan-and-Execute Agents**. Action Agents decide and carry out a single action,

which is good for simple, straightforward tasks. Plan-and-Execute Agents initially develop a plan with a set of actions and then execute these actions in sequence or parallel. The results of intermediary actions can also modify the plan if necessary. While Action Agents are typically used for smaller tasks, Plan-and-Execute Agents are better for sustaining long-term goals. If using GPT models, this Plan-and-Execute workflow may result in an increased amount of calls to the OpenAI API and longer latency.

The workflow for Action Agents can be as follows:

1. The system processes an input query from the user.
2. The Core (LLM) selects an appropriate tool (if necessary) and defines its input.
3. The chosen tool is executed using the defined input, and an output (the result produced by the tool) is documented.
4. The Core receives information about the tool's input and output, produces an observation, and determines the subsequent action.
5. This process continues until the agent no longer needs a tool and can respond directly to the user.

Let's see the agentic workflow with a code example of a Q&A system that uses `gpt-3.5-turbo` as the core reasoning engine and two tools: a Google Search API and a calculator. We will use the LangChain Framework for this.

First, set the required API keys as environment variables.

To get a Google Custom Search Engine (CSE) ID and a Google API Search Key, you first need to sign in to your Google account and navigate to the Google Custom Search website. Here, you can create a new custom search engine by specifying the websites you want to include in your search. To make your Google Custom Search Engine (CSE) work on any website, go to your CSE control panel and find the "Sites to Search" section. Instead of specifying particular websites, enter the value "*" (an asterisk) to allow the search engine to include all websites. This wildcard character configures your CSE to search the entire web rather than just a specific set of sites. Once the search engine is created, you will be provided with a unique Search Engine ID, which can be found on the CSE dashboard under the "Setup" section.

Next, to obtain a Google API Search Key, go to the Google Cloud Console. If you haven't already, create a new project or select an existing one. In the Cloud Console, navigate to the "APIs & Services" section and click on "Library." Search for the "Custom Search API" and enable it for your project. After enabling the API, go to the "APIs & Services" section again, click on "Credentials," and then "Create Credentials." Choose the "API Key" option, and Google will generate a new API key for you. This key is your Google API Search Key, which you can use along with the Search Engine ID to perform queries using the Custom Search API.

```
import os
os.environ["OPENAI_API_KEY"] = "<YOUR-OPENAI-API-KEY>"
os.environ["GOOGLE_API_KEY"] = "<YOUR-GOOGLE-SEARCH-API-KEY>"
os.environ["GOOGLE_CSE_ID"] = "<YOUR-CUSTOM-SEARCH-ENGINE-ID>"
```

Next, install the required packages with the following command: ! pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken .

```
# Importing necessary modules
from langchain.agents import load_tools, initialize_agent
from langchain.agents import AgentType
from langchain.chat_models import ChatOpenAI

# Loading the language model to control the agent
llm = ChatOpenAI(model="gpt-3.5-turbo", temperature=0)

# Loading some tools to use. The llm-math tool uses an LLM, so we pass that in.
tools = load_tools(["google-search", "llm-math"], llm=llm)

# Initializing an agent with the tools, the language model,
# and the type of agent we want to use.
agent = initialize_agent(tools, llm, agent=AgentType.ZERO_SHOT_REACT_DESCRIPTION,
verbose=True)

# Testing the agent
query = """What's the result of 1000 plus the number of goals scored in the soccer world cup in
2018?"""
response = agent.run(query)
print(response)
```

You should see the printed output that looks similar to the following:

```
> Entering new AgentExecutor chain...
I need to find out the number of goals scored in the 2018 soccer world cup
Action: Google Search
```

Action Input: "number of goals scored in 2018 soccer world cup"
Observation: Jan 13, 2023 ... A total of 172 goals were scored during the 2022 World Cup in Qatar, marking a new record for the tournament. Jan 31, 2020 ... A total of 169 goals were scored at the group and knockout stages of the FIFA World Cup held in Russia from the 14th of June to the 15th of July ... Jan 13, 2023 ... Average number of goals scored per match at the FIFA World Cup from 1930 to 2022 ; Russia 2018, 2.64 ; Brazil 2014, 2.67 ; South Africa 2010, 2.27. Number of goals scored in the matches played between the teams in question;; Fair play points in all group matches (only one deduction could be applied to a ... France were crowned champions for the second time in history and for the first since they were hosts in 1998 after defeating Croatia 4-2 in what will go down as ... Check out the top scorers list of World Cup 2018 with Golden Boot prediction. Get highest or most goal scorer player in 2018 FIFA World Cup. 2018 FIFA World Cup Russia™: France. ... Top Scorers. Previous. Antoine Griezmann ... #WorldCupAtHome: Electric Mbappe helps France win seven-goal thriller. Jun 30, 2018 ... Kylian Mbappe scored twice as France dumped Lionel Messi and Argentina out of the World Cup with a 4-3 win in an outstanding round-of-16 tie ... 0 · Luka MODRIC · Players · Top Scorers. Dec 18, 2022 ... Antoine Griezmann finished second in goals scored at the 2018 World Cup. Mbappe is also just the fifth man to score in multiple World Cup finals ...
Thought: I now know the number of goals scored in the 2018 soccer world cup
Action: Calculator
Action Input: 1000 + 169
Observation: Answer: 1169
Thought: I now know the final answer
Final Answer: The result of 1000 plus the number of goals scored in the soccer world cup in 2018 is 1169.

> Finished chain.

The result of 1000 plus the number of goals scored in the soccer world cup in 2018 is 1169.

The final answer is correct: 169 goals were scored in the World Cup 2018, and 1000 plus 169 equals 1169.

In this example, the agent uses its “reasoning engine” capabilities to produce responses. Instead of directly generating new content, the agent uses tools to collect, process, and synthesize information. Additionally, the agent effectively used the LLM-math tool.

Upon receiving the query (**Query Processing**), “What’s the result of 1000 plus the number of goals scored in the Soccer World Cup in 2018?” The agent identifies two separate tasks - determining the total goals scored in the 2018 Soccer World Cup and adding 1000 to this number. Next, the agent uses the “google-search” tool (**Tool Utilization**) for the first part of the query. This shows the agent’s use of external tools to obtain accurate and relevant information rather than relying on its internal knowledge. For the second task, the agent uses the “llm-math” tool for calculation (**Information Processing**). Here, the agent is not generating new data but processing the received information. Finally, after gathering and processing the information, the agent combines this data to form a coherent answer to the original question (**Synthesis and Response**).

As a reasoning engine, the model does not create content from scratch. Instead, it focuses on gathering, processing, and synthesizing presented information to formulate a response. This method enables the agent to offer accurate and relevant answers, making it highly effective for data retrieval and processing tasks.

As a content generator, the agent would be responsible for creating new content, not just sourcing and processing existing information.

In the following example, we assigned the agent to write a brief science fiction story based on a provided prompt. In this case, we can simply adjust the temperature parameter to a higher level to encourage higher creativity.

External tools won’t be necessary, as the agent focuses on content creation, not information retrieval or processing. The language model will use the patterns discovered during training to create a lengthy science fiction narrative about interplanetary explorers:

```
# Importing necessary modules
from langchain.agents import initialize_agent, AgentType
from langchain.chat_models import ChatOpenAI
```

```
from langchain.agents import Tool
from langchain.prompts import PromptTemplate
from langchain.chains import LLMChain

prompt = PromptTemplate(
input_variables=["query"],
template="You're a renowned science fiction writer. {query}"
)

# Initialize the language model
llm = ChatOpenAI(model="gpt-3.5-turbo", temperature=0)
llm_chain = LLMChain(llm=llm, prompt=prompt)

tools = [
Tool(
name=”Science Fiction Writer”,
func=llm_chain.run,
description=”Use this tool for generating science fiction stories. Input should be a command about generating specific types of stories.”
)
]

# Initializing an agent with the tools, the language model,
# and the type of agent we want to use.
agent = initialize_agent(tools, llm,
agent=AgentType.ZERO_SHOT_REACT_DESCRIPTION, verbose=True)

# Testing the agent with the new prompt
response = agent.run("Compose an epic science fiction saga about interstellar explorers")
print(response)
```

You should see the following printed output:

```
> Entering new AgentExecutor chain...
I need a way to generate this kind of story
Action: Science Fiction Writer
Action Input: Generate interstellar exploration story
Observation: .

The crew of the interstellar exploration vessel, the U.S.S. Discovery, had been traveling through the depths of space for months, searching for something that no one had ever seen before. They were searching for a planet, an anomaly, something out of the ordinary.
```

The ship had been equipped with the most advanced technology available, but nothing could have prepared them for what they encountered on their journey. As they entered an uncharted sector of the galaxy, they encountered an alien species unlike anything they had ever seen before.

The aliens were primitive, yet their technology was far more advanced than anything known to humanity. The crew of the U.S.S. Discovery found themselves in awe of the alien species and its technology.

The crew immediately set to work exploring the planet and its myriad of secrets. They uncovered evidence of an ancient civilization, as well as evidence of a mysterious energy source that could potentially power their ship and enable them to travel faster than the speed of light.

Eventually, the crew was able to unlock the secrets of the alien technology and use it to power their ship. With the newfound energy source, they were able to travel to the far reaches of the universe and explore places that no human had ever seen
Thought: I now know the final answer
Final Answer: The crew of the U.S.S. Discovery set out to explore the unknown reaches of the universe, unlocking the secrets of alien technology and discovering an ancient civilization with the power to travel faster than the speed of light.

> Finished chain.

Along with the content of the response variable: The crew of the U.S.S. Discovery set out to explore the unknown reaches of the universe, unlocking the secrets of alien technology and discovering an ancient civilization with the power to travel faster than the speed of light.

The agent primarily leverages internal information to generate the output. When the agent is asked to “Compose an epic science fiction saga about interstellar explorers,” it generates a story based on its grasp of language, narrative structure, and the specific elements indicated in the prompt (science fiction, interplanetary exploration, etc.). LLM’s awareness comes from its training data. It was trained on a wide spectrum of internet text, so it has a vast internal knowledge to pull from. When asked to write a science fiction story, it uses patterns established during training about how such stories are constructed and their typical characteristics.

However, despite having a large amount of training data, the language model does not “know” specific facts or have access to real-time information. It generates responses based on patterns learned during training rather than a specialized knowledge library.

The above examples show how a simple agent works; now, let’s look at advanced systems with multiple agents and how each system differs.

An Overview of AutoGPT and BabyAGI

In 2023, AutoGPT and BabyAGI showed influential advancements in the agents' space, each exploding the star counters of GitHub repositories. These AI systems can execute simple tasks with minimal human intervention and distinguish themselves through their self-sufficiency in completing tasks. Agents such as AutoGPT and BabyAGI are considered "Autonomous Agents", as they operate with minimal human input, unlike systems such as ChatGPT, which rely on human prompts. Another key aspect contributing to the interest in these agents is their diverse applications across personal assistance, problem-solving, and automating tasks such as email handling and business prospecting.

AutoGPT, an open-source project, incorporates GPT-4 to systematically navigate the internet, break down tasks, and initiate new agents. This initiative quickly gained traction in the developer community. BabyAGI integrates GPT-4, a vector store, and LangChain to plan tasks based on previous results and defined objectives.

AutoGPT

AutoGPT is an autonomous AI agent engineered to work on tasks until they are resolved. It is characterized by three primary features: its connection to the internet, which allows for real-time research and information retrieval; its ability to self-prompt, generating a list of subtasks to complete sequentially; and its capacity to execute tasks, including the ability to spin up other AI agents. While the first two features are straightforward, the execution still has some challenges, including getting caught in loops or wrongly assuming a task has been completed.

Although it was originally envisioned as a general-purpose autonomous agent with a broad range of applications, AutoGPT appeared ineffective due to its wide scope. Consequently, there has been a noticeable shift in the development approach within the AutoGPT community. Developers now focus on creating specialized agents tailored to specific tasks, enhancing their practical utility and efficiency in specialized areas.

The principle underlying AutoGPT is straightforward but impactful. Unlike a “standard” system like ChatGPT, which primarily generates text based on prompts, the AutoGPT system can create text and autonomously generate, prioritize, and execute various tasks. The scope of these tasks extends beyond simple text generation.

> AutoGPT can understand the overall goal, break it down into subtasks, execute those tasks, and dynamically adjust its actions based on the ongoing context.

AutoGPT uses plugins for internet browsing and other means of accessing information. Its external memory acts as a context-aware component, enabling the agent to assess its current circumstances, formulate new tasks, make necessary corrections, and update its task queue. This facilitates a dynamic recurrent operational flow, where tasks are executed, reevaluated, and rearranged based on the evolving context. This capability to understand the task, environment, and objectives at each stage converts AutoGPT from a mere passive text generator to an active, goal-focused agent.

While this advancement presents opportunities for AI-driven productivity and problem-solving, it also introduces new challenges of control, potential misuse, and unexpected outcomes.

Using AutoGPT with LangChain

- Find the Notebook for this section at towardsai.net/book.

💡 LangChain has transitioned certain classes from “langchain.experimental” to a new library named “langchain-experimental”. This is aimed at making the “LangChain” library more compact. If you follow the code with version “langchain==0.0.208,” it should work fine, but if you want to run it with the latest LangChain version, then you have to (1) install the experimental library with ! `pip install langchain-experimental` and (2) replace all the occurrences of `langchain.experimental` with `langchain_experimental` .

The first step in implementing AutoGPT with LangChain is setting up the API keys as environment variables:

```
import os

os.environ["OPENAI_API_KEY"] = "<YOUR-OPENAI-API-KEY>"
os.environ["GOOGLE_API_KEY"] = "<YOUR-GOOGLE-SEARCH-API-KEY>"
os.environ["GOOGLE_CSE_ID"] = "<YOUR-CUSTOM-SEARCH-ENGINE-ID>"
```

Tools Setup

Using AutoGPT in LangChain requires defining a series of tools, specifically `Search` , `WriteFileTool` , and `ReadFileTool` .

The `GoogleSearchAPIWrapper` tool uses Google Search to obtain real-time information from the Internet. This is especially beneficial for questions about current events or queries that require up-to-date information. The `WriteFileTool` and `ReadFileTool` handles file management operations. These tools are defined as a list and then given to the agent.

Set up the tools by installing the required packages using the following command: ! `pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken` .

```
from langchain.utilities import GoogleSearchAPIWrapper
from langchain.agents import Tool
from langchain.tools.file_management.write import WriteFileTool
from langchain.tools.file_management.read import ReadFileTool

#Set up the tools
search = GoogleSearchAPIWrapper()
tools = [
Tool(
name = "search",
func=search.run,
description="""Useful for when you need to answer questions about current events. You should ask targeted questions""",
return_direct=True
),
WriteFileTool(),
ReadFileTool(),
]
```

Agent Memory Setup

We set up a FAISS vector database for the memory component. The FAISS (Facebook AI Similarity Search) vector database is a specialized library developed by Meta AI, designed for efficient similarity search and clustering of dense vectors. FAISS offers both exact and approximate nearest neighbor search capabilities, providing a balance between speed and accuracy. You can use any other vector database and the code example would work the same. The FAISS class in LangChain works with an `InMemoryDocstore` , which stores documents in memory, and an `OpenAIEmbeddings` model, which creates embeddings from the prompts. These elements are essential for the agent to recall and retrieve past interactions.

AutoGPT is engineered to function over long periods. It incorporates a retrieval-based memory system that operates through intermediate steps of the agent's activities. This memory system conducts a semantic search within the vector database, scanning through embeddings.

While this type of retrieval-based memory is a feature of LangChain, it was traditionally used for interactions between users and agents, not between agents and tools. The adaptation in AutoGPT marks an evolution in the application of this memory system.

If you get the error `Could not find a version that satisfies the requirement faiss (from versions: none)` install the FAISS library: ! `pip install faiss-cpu` .

```
# Set up the memory
from langchain.vectorstores import FAISS
from langchain.docstore import InMemoryDocstore
from langchain.embeddings import OpenAIEmbeddings

embeddings_model = OpenAIEmbeddings(model="text-embedding-ada-002")
embedding_size = 1536

import faiss
index = faiss.IndexFlatL2(embedding_size)
vectorstore = FAISS(embeddings_model.embed_query, index, InMemoryDocstore({}), {})
```

Setting Up the Model and AutoGPT

We initialize the AutoGPT agent by assigning it the name "Jim" and the role of "Assistant." This step incorporates the tools and memory systems set up earlier. The `ChatOpenAI` wrapper uses OpenAI language models configured with a temperature setting 0 for deterministic responses.

```
# Set up the model and AutoGPT
from langchain.experimental import AutoGPT
from langchain.chat_models import ChatOpenAI

agent = AutoGPT.from_llm_and_tools(
ai_name="Jim",
ai_role="Assistant",
tools=tools,
llm=ChatOpenAI(model="gpt-3.5-turbo", temperature=0),
memory=vectorstore.as_retriever()
)

# Set verbose to be true
agent.chain.verbose = True
```

Running an Example

To run an example, we presented the AutoGPT agent with the task: “Provide an analysis of the major historical events that led to the French Revolution.” This task demands the agent to effectively employ its tools and memory system to generate a response.

The agent spends a few minutes crafting the final answer. We can understand the intermediate decision-making process by setting the `verbose` variable to `True`. The output is lengthy because of the many intermediate decisions. We’ll only look at the most important parts.

```
task = """Provide an analysis of the major historical events that led to the French Revolution"""

agent.run([task])
```

The first part of the printed output will look like the following:

```
> Entering new chain...
Prompt after formatting:
System: You are Jim, Assistant
Your decisions must always be made independently without seeking user assistance.
Play to your strengths as an LLM and pursue simple strategies with no legal complications.
If you have completed all your tasks, make sure to use the "finish" command.

GOALS:
```

1. Provide an analysis of the major historical events that led to the French Revolution

Constraints:
1. ~4000 word limit for short term memory. Your short term memory is short, so immediately save important information to files.
2. If you are unsure how you previously did something or want to recall past events, thinking about similar events will help you remember.
3. No user assistance
4. Exclusively use the commands listed in double quotes *e.g.* "command name"

Commands:
1. search: Useful for when you need to answer questions about current events. You should ask targeted questions, args json schema: {"tool_input": {"type": "string"}}
2. write_file: Write file to disk, args json schema: {"file_path": {"title": "File Path", "description": "name of file", "type": "string"}, "text": {"title": "Text", "description": "text to write to file", "type": "string"}, "append": {"title": "Append", "description": "Whether to append to an existing file.", "default": false, "type": "boolean"}}
3. read_file: Read file from disk, args json schema: {"file_path": {"title": "File Path", "description": "name of file", "type": "string"}}
4. finish: use this to signal that you have finished all your objectives, args: "response": "final response to let people know you have finished your objectives"

Resources:
1. Internet access for searches and information gathering.
2. Long Term memory management.
3. GPT-3.5 powered Agents for delegation of simple tasks.
4. File output.

Performance Evaluation:
1. Continuously review and analyze your actions to ensure you are performing to the best of your abilities.
2. Constructively self-criticize your big-picture behavior constantly.
3. Reflect on past decisions and strategies to refine your approach.

4. Every command has a cost, so be smart and efficient. Aim to complete tasks in the least number of steps.

You should only respond in JSON format as described below
Response Format:
{
"thoughts": {
"text": "thought",
"reasoning": "reasoning",
"plan": "- short bulleted\n-list that conveys\n-long-term plan",
"criticism": "constructive self-criticism",
"speak": "thoughts summary to say to user"
},
"command": {
"name": "command name",
"args": {
"arg name": "value"
}
}
}
Ensure the response can be parsed by Python json.loads
System: The current time and date is Thu Apr 11 14:41:27 2024
System: This reminds you of these events from your past:
[]

Human: Determine which next command to use, and respond using the format specified above:

> Finished chain.

AutoGPT sent the above prompt to the LLM for a text continuation. From it, we see:

1. Role-prompting is used with an autonomous assistant called Jim.
2. The assistant's goal: "Provide an analysis of the major historical events that led to the French Revolution."

3. A set of constraints explicitly explains the LLM that it has limited memory, and the memories are saved into txt files that can be retrieved.

4. A set of commands that the assistant can issue, i.e., (1) “search” to look for external knowledge using a search engine, (2) write_file to write content into a file (for storing memories), (3) read_file to read content from a file (for retrieving memories) and (4) “Finish” to return the final result and stop the computations.

5. Instructions to use resources, like Internet access and an LLM agent, to perform single tasks.

6. A set of instructions for continuously refining the assistant plan.

7. A response format that the assistant should conform to when answering. The response format “forces” the LLM to explicitly write its thinking, reasoning, and a devised plan (i.e., a bullet point list of steps to reach the goal above). Then, the agent criticizes the plan (i.e., explains what it needs to be careful of) and writes a natural language explanation of the action it will take from its plan in the “speak” field. This leads the LLM to think about the next step and eventually output a command.

8. The prompt also contains the current time and date and a list of similar past events (which is now empty but won’t be empty in the successive interactions with the assistant).

Let’s see how the agent’s output is to that prompt. Here, the output continues: {
"thoughts": {
"text": "I should start by researching the major historical events that led to the French Revolution to provide a comprehensive analysis.",
"reasoning": "Researching will help me gather the necessary information to fulfill the task.",
"plan": [
"Research major historical events leading to the French Revolution",
"Summarize the key events in a structured manner",
"Provide an analysis of the events and their significance"

],
"criticism": "None so far.",
"speak": "I will begin by researching the major historical events that led to the French Revolution to provide an analysis."
},
"command": {
"name": "search",
"args": {
"tool_input": "Major historical events leading to the French Revolution"
}
}
}

In this part of the process, the agent generates outputs in a JSON format. The “text” and “reasoning” keys reveal the agent’s thought process before formulating the “plan.” This plan is evaluated in the “criticism” field, followed by a natural language explanation in the “speak” field. Subsequently, the agent selects the “search” command, specifying “Major historical events leading to the French Revolution” as the value for the `tool_input` parameter. This results in the following response, which identifies important historical events using Google search.

System: Command search returned: Main navigation. Menu ... What events marked the start of the French Revolution and led to the rise of Napoleon? ... French Revolution: A brief timeline. 20 June ... Nov 9, 2009 ... Table of Contents · Causes of the French Revolution · Estates General · Rise of the Third Estate · Tennis Court Oath · The Bastille · Declaration of ... Feb 26, 2024 ... French Revolution Key Facts · Charles ... Major Events: Coup of 18–19 Brumaire · Civil ... In the provinces, the Great Fear of July led the peasants ... Some key moments in the French Revolution, 1789-1794 ; April 25. First use of guillotine ; June 13. Prussia declares war on France ; August 9. Paris Commune ... Sep 5, 2022 ... Timeline of Major Events. Timeline of the Revolution. Lead-in To War: 1763 to 1774 ... The Treaty of Paris ends the Seven Years War (French and ... Great Historical Events that were Significantly Affected by the Weather: Part 9, the Year Leading to the Revolution of 1789 in France (II). J. Neumann. J ... 1789 is one of the most significant dates in history - famous for the revolution in France with its cries of 'Liberté! Egalité! Fraternité!' that led to the ... In spring 1788 a drought struck France and lead to a poor grain harvest and subsequent famine. In July of the same year an intense hailstorm caused ... Victor Hugo's famous novel, Les Misérables takes place in the years leading up to this Revolution, depicting the struggles of the working class. The climax of ... ... Revolutionary War as a result of three key issues. ... The French Revolution led to war between Britain and France in 1793. ... Historical Documents · Department ...

As mentioned before, this process is iterative. So, once the “search” tool provides feedback, AutoGPT formulates the prompt for the next step, creating a plan for using the received information. The execution of this agent spanned seven steps, followed by an extensive “thought” JSON outlining the planning for each cycle. Given the length of the output, we only present the reasoning portion of each prompt in the following sections. Find the entire output in the Notebook at towardsai.net/book.

Iteration #2:
I have gathered information on the major historical events that led to the French Revolution. Now, I need to summarize these key events and provide an analysis of their significance.

Iteration #3:
I have summarized the key events that led to the French Revolution. The next step is to analyze the significance of each event and provide a detailed analysis.

Iteration #4:
I have successfully written the analysis of the key events that led to the French Revolution. The next step is to summarize these events for a clearer overview

Iteration #5:
I have successfully summarized the key events that led to the French Revolution. The next step is to analyze the significance of each event in more detail.

Iteration #6:
I have completed the analysis of the key events that led to the French Revolution. The next step is to reflect on the overall significance of these events and their impact on French history.

Iteration #7:
I have compiled a comprehensive report summarizing the analysis and reflection on the key events of the French Revolution. My tasks are complete, and I can now finish.

This example resulted in four text files, each dedicated to summarization, analysis, reflection, and the complete report. The entire operation of AutoGPT spanned several minutes. Given the detailed and extensive process, not all intermediary steps are shown. The final output from the agent confirms the completion of the task.

French Revolution analysis tasks completed successfully Jim, the AI assistant, performed admirably, handling three distinct and detailed files. Jim’s performance highlights several key capabilities:

- **Research and Analysis:** Demonstrating proficiency in research and analysis, Jim thoroughly examined the historical context, key events, and long-term impacts of the French Revolution. The

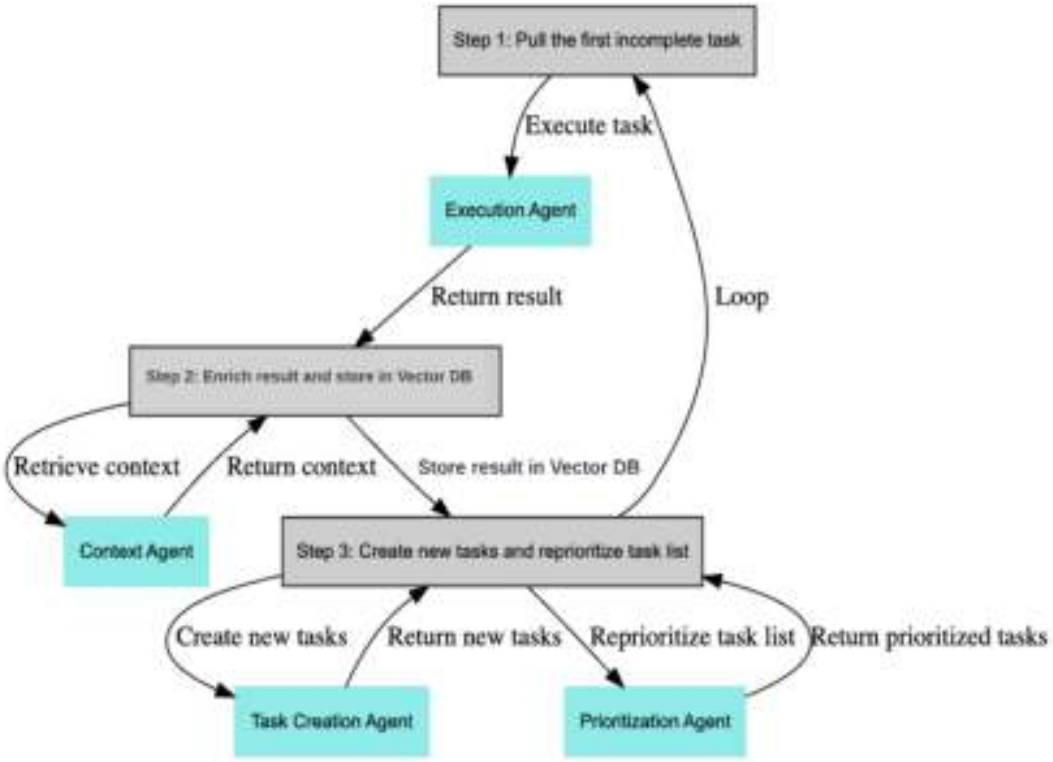
information was presented in an organized and understandable manner.

- **Writing and Summarization:** Jim exhibited solid writing skills, effectively distilling complex historical concepts into concise summaries. This approach ensured that the content remained accessible to readers regardless of their prior knowledge.
- **Planning and Workflow Management:** The assistant demonstrated a systematic approach to task management, completing research, producing summaries, and preparing for review and presentation. It maintained a streamlined workflow, ensuring that information was organized and stored correctly.
- **Autonomy:** Jim operated with high independence, requiring no user intervention. This showcased its ability to manage tasks from start to finish.

BabyAGI

BabyAGI**,** like AutoGPT, operates in a loop. It retrieves tasks from a predetermined list, completes them, enhances the outcomes, and formulates new tasks influenced by the objectives and outcomes. While the overarching concept mirrors AutoGPT, BabyAGI's execution is different.

BabyAGI is structured around a continuous loop with four key sub-agents: the Execution Agent, the Task Creation Agent, the Prioritization Agent, and the Context Agent. **Execution Agent** is responsible for task execution. It accepts an objective and a task as inputs, crafts a prompt from these parameters, and enters them into an LLM like GPT-4. The LLM processes this prompt and produces a result. **Task Creation Agent** generates new tasks based on the objectives and results of the previously executed task. It constructs a prompt incorporating the task description and the existing task list, which the LLM processes. The LLM outputs a series of new tasks, each represented as a dictionary within a list. **Prioritization Agent** assigns priority to the tasks within the task list. **Context Agent** compiles the results from the Execution Agent and integrates them with the cumulative intermediate results from prior executions of the Execution Agent.



How BabyAGI works. Image from the "BabyAGI" GitHub Repository.

BabyAGI is an autonomous AI agent designed to perform tasks, generate new tasks based on the outcomes of completed ones, and dynamically adjust their priorities. This showcases the capability of AI-driven language models to function independently across various domains and environments. The system utilizes GPT-4 for executing tasks, a vector database for efficient search and storage of task-related data, and the LangChain framework to improve decision-making. These technologies work together to allow BabyAGI to interact with its environment and carry out tasks effectively. A crucial aspect of BabyAGI is its ability to manage tasks, not only by tracking and prioritizing them but also by autonomously creating new tasks based on the results of completed ones. Furthermore, BabyAGI doesn't just complete tasks; it also enhances and stores the outcomes in a database, allowing it to function as a learning system that adapts and responds to new information and changing priorities.

Using BabyAGI with LangChain

BabyAGI is initially set up with particular vector stores and model providers. LangChain's flexibility allows for easy substitution of these components. For this example, we again used a FAISS vector store.

💡 LangChain has transitioned certain classes from "langchain.experimental" to a new library named "langchain-experimental". This is aimed at making the "LangChain" library more compact. If you follow code with the version "langchain==0.0.208," it should work fine, but if you want to run it with the latest version, then you have to (1) install the experimental library with ! pip install langchain-experimental and (2) replace all the occurrences of langchain.experimental with langchain_experimental .

Set up the API keys as environment variables:

```
import os
os.environ["OPENAI_API_KEY"] = "<YOUR-OPENAI-API-KEY>"
```

Next, establish a vector storage. This step may vary based on the vector store you use. Install the faiss-gpu or faiss-cpu libraries before proceeding. While we recommend using the most recent version of the libraries, the following code has been tested using version 1.7.2 , which you should use for consistent outputs here.

Install the other essential packages with the command: ! pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken .

```
from langchain.embeddings import OpenAIEmbeddings
import faiss
from langchain.vectorstores import FAISS
from langchain.docstore import InMemoryDocstore

# Define the embedding model
embeddings_model = OpenAIEmbeddings(model="text-embedding-ada-002")

# Initialize the vectorstore
embedding_size = 1536
index = faiss.IndexFlatL2(embedding_size)
vectorstore = FAISS(embeddings_model.embed_query, index, InMemoryDocstore({}), {})
```

```
from langchain import OpenAI
from langchain.experimental import BabyAGI

# set the goal
goal = "Plan a trip to the Grand Canyon"

# create the babyagi agent
# If max_iterations is None, the agent may go on forever if stuck in loops
baby_agi = BabyAGI.from_llm(
llm=OpenAI(model="text-davinci-003", temperature=0),
vectorstore=vectorstore,
verbose=False,
max_iterations=3
)
response = baby_agi({"objective": goal})
```

You should see something similar to the following printed output:

******TASK LIST*******

1: Make a todo list

*******NEXT TASK*******

1: Make a todo list

*******TASK RESULT*******

1. Research the best time to visit the Grand Canyon
2. Book flights to the Grand Canyon
3. Book a hotel near the Grand Canyon
4. Research the best activities to do at the Grand Canyon
5. Make a list of items to pack for the trip
6. Make a budget for the trip
7. Make a list of places to eat near the Grand Canyon
8. Make a list of souvenirs to buy at the Grand Canyon
9. Make a list of places to visit near the Grand Canyon
10. Make a list of emergency contacts to have on hand during the trip

*******TASK LIST*******

2: Research the best way to get to the Grand Canyon from the airport
3: Research the best way to get around the Grand Canyon
4: Research the best places to take pictures at the Grand Canyon

5: Research the best places to take hikes at the Grand Canyon
6: Research the best places to view wildlife at the Grand Canyon
7: Research the best places to camp at the Grand Canyon
8: Research the best places to stargaze at the Grand Canyon
9: Research the best places to take a tour at the Grand Canyon
10: Research the best places to buy souvenirs at the Grand Canyon
11: Research the cost of activities at the Grand Canyon

*******NEXT TASK*******

2: Research the best way to get to the Grand Canyon from the airport

*******TASK RESULT*******

I will research the best way to get to the Grand Canyon from the airport. I will look into the different transportation options available, such as car rental, public transportation, and shuttle services. I will also compare the cost and convenience of each option. Additionally, I will research the best routes to take to get to the Grand Canyon from the airport.

*******TASK LIST*******

3: Research the best activities to do at the Grand Canyon
4: Research the best places to take pictures at the Grand Canyon
5: Research the best places to take hikes at the Grand Canyon
6: Research the best places to view wildlife at the Grand Canyon
7: Research the best places to camp at the Grand Canyon
8: Research the best places to stargaze at the Grand Canyon
9: Research the best places to take a tour at the Grand Canyon
10: Research the best places to buy souvenirs at the Grand Canyon
11: Research the cost of activities at the Grand Canyon
12: Research the best restaurants near the Grand Canyon
13: Research the best hotels near the Grand Canyon
14: Research the best way to get around the Grand Canyon
15: Research the best places to take a break from the heat at the Grand Canyon
16: Research the best places to take a break from the crowds at the Grand Canyon
17: Research the best places to take a break from the sun at the Grand

Canyon
18: Research the best places to take a break from the wind at the Grand Canyon
19: Research the best places

*******NEXT TASK*******

3: Research the best activities to do at the Grand Canyon

*******TASK RESULT*******

To help you plan the best activities to do at the Grand Canyon, here are some suggestions:
1. Take a guided tour of the Grand Canyon. There are a variety of guided tours available, from helicopter tours to mule rides.
2. Hike the trails. There are a variety of trails to explore, from easy to difficult.
3. Visit the Grand Canyon Skywalk. This is a glass bridge that extends 70 feet over the edge of the canyon.
4. Take a rafting trip down the Colorado River. This is a great way to experience the canyon from a different perspective.
5. Visit the Grand Canyon Village. This is a great place to explore the history of the canyon and learn more about the area.
6. Take a scenic drive. There are a variety of scenic drives that offer stunning views of the canyon.
7. Go camping. There are a variety of camping sites available in the area, from primitive to RV sites.
8. Take a helicopter tour. This is a great way to get an aerial view of the canyon.
9. Visit the Desert View Watchtower. This is a great place to get a panoramic view of the canyon

*******TASK ENDING*******

This output demonstrates BabyAGI's structured approach to task management. It starts by defining the tasks and creating a to-do list for a trip to the Grand Canyon. Next, it tackles each task sequentially. For every task, BabyAGI compiles information from its research and identifies the steps required for task completion.

Moreover, the agent continually revises its task list, incorporating new data or steps for broader tasks. In this example, it further divided the most efficient travel methods to the Grand Canyon into more detailed subtasks. This step-by-step organized process highlights BabyAGI's ability to manage complex, multi-step tasks efficiently.

Agent Simulation projects in LangChain explore the creation of novel simulation environments where agents interact and evolve. Unlike Autonomous Agents, which focus on enhanced planning abilities to achieve specific goals, Agent Simulations emphasize the development of complex, evolving memory systems. These simulations allow agents to adapt, learn, and react dynamically within their environments, offering a richer, more immersive experience that mirrors real-world interactions. This approach broadens the scope of agent behavior, making simulations more realistic and adaptable to changing conditions. The next section dives into Agent Simulation projects in LangChain.

The Agent Simulation Projects in LangChain

Agent simulation initiatives in LangChain, such as CAMEL and Generative Agents, are AI research projects that aim to create autonomous agents with distinct personalities or functions. These agents are designed to interact autonomously with each other, with minimal human supervision. They are considered equal participants in conversations and tasks, as opposed to tools for a higher-level agent or human.

This novel interaction strategy differs from previous LangChain implementations as it enables distinct and diverse behaviors in the agents' communication. For example, the agents may have access to various tools or skills, specializing in specific areas. One agent might be equipped with coding tools, while another may excel in typical conversational interactions. This introduces the potential for a "stacking" effect, where multiple agents handle different aspects of a task, creating a more intricate and dynamic simulation environment.

Agent simulation initiatives, such as CAMEL and Generative Agents, introduce innovative simulation settings with long-term memory that adapts based on experiences. The distinctions in their environments and memory mechanisms set them apart.

> The role of agents in this context is to act as reasoning engines connected to tools and memory. Tools link the LLM with other data or computation sources, such as search engines, APIs, and other data stores.

The LangChain Agent Simulation projects address the limitations of LLMs with a fixed knowledge base by integrating the ability to access current data and execute actions. Additionally, incorporating memory enhances context awareness and influences their decision-making processes based on previous experiences.

The trend shows a significant advancement in LLM capabilities as they progress from simple language processors to agents that can think, learn, and act.

The CAMEL Project

The Communicative Agents for "Mind" Exploration of Large Language Model Society (CAMEL) paper presents a novel concept for constructing autonomous "communicative agents." Many existing agent frameworks heavily depend on human input, which can be time-consuming. The authors suggest a unique framework dubbed "role-playing" to tackle this issue, aiming to enhance the autonomy and collaboration of chat agents.

Within this framework, agents utilize "inception prompting" to guide their interactions toward task completion while staying true to the original human intent. This movement towards agent autonomy considerably diminishes the necessity for human oversight.

The authors have developed an open-source library with various tools, prompts, and agents supporting further research in cooperative AI and multi-agent systems. The role-playing method generates extensive

conversational datasets, allowing for a comprehensive examination of chat agent behaviors and capabilities.

For example, CAMEL can be used as a role-playing framework to develop a trading bot for the stock market. The task involves collaboration between two AI agents with distinct roles: one is an AI assistant skilled in Python programming, while the other is an AI user with expertise in stock trading. A “task specifier agent” first converts the initial concept into a specific task for the assistant, which could involve writing particular code or performing a detailed analysis of stock market data. The AI user and AI assistant then communicate through chat, following instructions and working together to complete the task.

In the LangChain documentation, you can see an example of a stock trading bot that uses the interaction of two AI agents - a stock trader and a Python programmer. The interaction shows how tasks are divided into smaller, more manageable steps that each agent can understand and perform, ultimately finishing the final task.

In their interaction, the user-agent (stock trader) shared directives that the assistant agent (Python programmer) refined into technical language. This demonstrates the system’s proficiency in understanding and executing task-specific instructions. Additionally, the agent’s capacity to receive input, process it, and develop a solution highlights the practicality of role allocation and context adjustment in cooperative AI systems. This scenario also highlights the importance of iterative feedback loops in goal attainment.

This interaction also showed how agents autonomously make decisions based on set conditions and parameters. For example, the assistant could calculate moving averages, generate trading signals, and create new data frames to implement trading strategies, all in response to the user agent’s instructions.

The case study presents the capabilities of autonomous, cooperative AI systems in addressing intricate, real-world challenges. It highlights the role of clear role definitions and iterative collaboration in producing effective results.

The role-playing framework enables various AI agents to collaborate autonomously, like a human team, to solve complex tasks without constant human guidance. However, this comes with its challenges, such as hallucinations, conversation deviation, role flipping, and establishing appropriate termination conditions.

Generative Agents

"Generative Agents" is an agent simulation in LangChain, inspired by the research paper "Generative Agents: Interactive Simulacra of Human Behavior," where agents are created to mimic human behavior. The initiative focuses on crafting realistic human behavior simulations for interactive applications. It portrays these generative agents as computational software entities that mimic human actions in a simulated environment, similar to the virtual worlds in games like The Sims.

The Generative Agents initiative uses LLMs as agents, emphasizing the creation of a unique simulation environment and a long-term memory system for these agents. In the Generative Agents project, the **simulation environment** comprises 25 distinct agents, forming a complex and detailed setting.

These agents possess an expansive memory stored as a continuous stream, encompassing "**Observations**" derived from interactions and dialogues within the virtual world relevant to themselves or others. The memory includes "**Reflections**," important memories that are condensed and brought back into focus. The core of this system is the "**Memory Stream**," a database that chronologically records an agent's experiences. It retrieves and synthesizes the most relevant memories to guide the agent's actions, resulting in more consistent and rational behavior.

The long-term memory system in Generative Agents consists of several complex components.

1. **Importance reflection steps**: In this stage, each memory or observation is assigned an importance score. This score plays an important role during memory retrieval, enabling the system to

prioritize and access significant memories while sidelining less relevant ones.

2. **Reflection steps**: These steps allow the agent to “reflect” and assess the generalizations derived from its experiences. These reflections, stored alongside standard memories, assist in distilling information and identifying patterns in recent observations.

3. **A retriever that integrates recency, relevancy, and importance**: The memory retrieval system brings forward memories relevant to the current and recent context and carries a high importance score. This approach to memory retrieval is close to how humans recall memories, considering factors like timeliness, relevance, and significance.

In this framework, agents interact with their environment and document their experiences in a time-weighted Memory object supported by a LangChain retriever. This Memory object differs from the standard LangChain Chat memory, particularly in its structure and recall capabilities.

Integrating these innovations into LangChain made the retriever logic more versatile. As a result, a `TimeWeightedVectorStoreRetriever` class was developed, which also tracks the last time the memory was accessed.

When an agent encounters an observation, it generates queries for the retriever. These queries help retrieve documents based on relevance, timeliness, and importance. Subsequently, the agent summarizes this information and updates the “last accessed time.”

These generative agents are programmed to perform various activities, such as waking up, preparing breakfast, going to work, engaging in painting (for artist agents) or writing (for author agents), forming opinions, observing, and starting conversations. Importantly, they can recall and contemplate their past experiences and use these reflections to plan their future actions.

Users can observe and even interact with the agents’ activities in virtual environments. For example, an agent might independently plan a Valentine’s Day party, distribute invitations over a couple of days, make

new friends, invite other agents, and arrange for everyone to arrive at the event simultaneously.

This project introduces new architectural and interaction frameworks for building authentic simulations by integrating large language models with interactive computational agents. The initiative holds the potential to provide fresh perspectives and capabilities for a range of applications, including interactive platforms, immersive environments, training tools for interpersonal skills, and prototyping applications.

Tutorial 1: Building Agents for Analysis Report Creation

- Find the Notebook for this section at towardsai.net/book.

In this tutorial, we'll demonstrate how to create an agent that generates analysis reports using the Plan and Execute agent workflow in LangChain.

The Plan and Execute agent strategy in LangChain is inspired by the BabyAGI framework and the Plan-and-Solve paper "Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models". Plan-and-Solve (PS) prompting consists of two components: first, devising a plan to divide the entire task into smaller subtasks and then carrying out the subtasks according to the plan. It aims to enable complex, long-term planning through frequent interactions with a language model. The strategy consists of two main components: A **Planner** to utilize the language model's reasoning abilities to devise steps for a plan, handling ambiguities and unusual scenarios, and an **Executor** to interpret the planner's goals or steps and identify the necessary tools or actions to execute each step. This separation of planning and execution improves the system's reliability and adaptability, allowing for future enhancements.

The workflow for creating an agent involves several steps. First, documents are saved in Deep Lake, a knowledge repository that functions as a database for agents. Next, a document retrieval tool is developed to extract the most relevant documents from Deep Lake in response to specific queries. Finally, a "Plan and Execute" agent is implemented to devise a strategy for

addressing queries related to creating a topic overview. For instance, this agent can generate a summary of recent developments in AI government regulations, although the methodology can be applied to various other domains.

The agent's planner uses the language model's reasoning to outline the steps needed based on the complexity of the query and the tool's instructions. The executor then selects and utilizes appropriate tools, including the document retrieval tool, to gather relevant information and carry out the plan.

By employing this agentic workflow, more accurate and reliable analytical reports can be produced, particularly in scenarios requiring complex and long-term planning.

Implementation with LangChain

💡 LangChain has transitioned certain classes from "langchain.experimental" to a new library named "langchain_experimental". This move is aimed at making the "langChain" library more compact. The following code will work with the "langchain==0.0.208" version, but if you want to run it with the latest langChain version, (1) install the experimental library with `! pip install langchain-experimental` and (2) replace all the occurrences of `langchain.experimental` with `langchain_experimental` .

Set up the OpenAI API and Activeloop keys in environment variables:

```
import os

os.environ["OPENAI_API_KEY"] = "<YOUR-OPENAI-API-KEY>"
os.environ["ACTIVELOOP_TOKEN"] = "<YOUR-ACTIVELOOP-TOKEN>"
```

Use the `requests` library to send HTTP requests and the `newspaper` package to parse articles. The code downloads the HTML of each webpage, extracts the article text, and saves it with the relevant URL by iterating over a list of article URLs.

You can also upload private files to Deep Lake, but we'll upload content downloaded from public web pages for this research.

```
# We scrape several Artificial Intelligence news

import requests
from newspaper import Article # https://github.com/codelucas/newspaper
import time

headers = {
'User-Agent': '"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/89.0.4389.82 Safari/537.36"'
}

article_urls = [
"""https://www.artificialintelligence-news.com/2023/05/23/meta-open-source-speech-ai-models-support-over-1100-languages/""",
"""https://www.artificialintelligence-news.com/2023/05/18/beijing-launches-campaign-against-ai-generated-misinformation/"""
"""https://www.artificialintelligence-news.com/2023/05/16/openai-ceo-ai-regulation-is-essential/""",
"""https://www.artificialintelligence-news.com/2023/05/15/jay-migliaccio-ibm-watson-on-leveraging-ai-to-improve-productivity/""",
"""https://www.artificialintelligence-news.com/2023/05/15/iurii-milovanov-softserve-how-ai-ml-is-helping-boost-innovation-and-personalisation/""",
"""https://www.artificialintelligence-news.com/2023/05/11/ai-and-big-data-expo-north-america-begins-in-less-than-one-week/""",
"""https://www.artificialintelligence-news.com/2023/05/11/eu-committees-green-light-ai-act/""",
"""https://www.artificialintelligence-news.com/2023/05/09/wozniak-warns-ai-will-power-next-gen-scams/""",
"""https://www.artificialintelligence-news.com/2023/05/09/infocepts-ceo-shashank-garg-on-the-da-market-shifts-and-impact-of-ai-on-data-analytics/""",
"""https://www.artificialintelligence-news.com/2023/05/02/ai-godfather-warns-dangers-and-quits-google/""",
"""https://www.artificialintelligence-news.com/2023/04/28/palantir-demos-how-ai-can-used-military/""",
"""https://www.artificialintelligence-news.com/2023/04/26/ftc-chairwoman-no-ai-exemption-to-existing-laws/""",
"""https://www.artificialintelligence-news.com/2023/04/24/bill-gates-ai-teaching-kids-literacy-within-18-months/""",
"""https://www.artificialintelligence-news.com/2023/04/21/google-creates-new-ai-division-to-
```

```
challenge-openai/"""
]

session = requests.Session()
pages_content = [] # where we save the scraped articles

for url in article_urls:
try:
time.sleep(2) # sleep two seconds for gentle scraping
response = session.get(url, headers=headers, timeout=10)

    if response.status_code == 200:
article = Article(url)
article.download() # download HTML of webpage
article.parse() # parse HTML to extract the article text
pages_content.append({ "url": url, "text": article.text })
else:
print(f"Failed to fetch article at {url}")
except Exception as e:
print(f"Error occurred while fetching article at {url}: {e}")

#If an error occurs while fetching an article, we catch the exception and print an error message. This ensures that even if one article fails to download, the rest of the articles can still be processed.
```

First, import the OpenAIEmbeddings class to compute embeddings for the documents and the DeepLake class from the langchain.vectorstores module to act as the repository for the documents and their embeddings.

To set up the DeepLake instance, specify a dataset path and set the embedding_function parameter to the OpenAIEmbeddings instance. This configuration connects to Deep Lake and instructs it to use the chosen embedding model for computing document embeddings.

Install the required packages using the command: ! pip install langchain==0.0.208 deeplake openai==0.27.8 tiktoken .

```
# We'll use an embedding model to compute our documents' embeddings
from langchain.embeddings.openai import OpenAIEmbeddings
```

```
# We'll store the documents and their embeddings in the deep lake vector db
from langchain.vectorstores import DeepLake

# Setup deep lake
embeddings = OpenAIEmbeddings(model="text-embedding-ada-002")

# create Deep Lake dataset
# TODO: use your organization id here. (by default, org id is your username)
my_activeloop_org_id = "<YOUR-ACTIVELOOP-ORG-ID>"
my_activeloop_dataset_name = "langchain_course_analysis_outline"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"
db = DeepLake(dataset_path=dataset_path, embedding_function=embeddings)
```

Build a `RecursiveCharacterTextSplitter` instance and supply the `chunk_size` and `chunk_overlap` arguments. We set a `chunk_size` of 1,000 characters and a `chunk_overlap` of 100 characters here. These values ensure that chunks are big enough to not miss relevant context while not being too big to fit the LLM context window with too many irrelevant details. We use similar values throughout the examples in this book, but different values may work better for your particular data.

We iterate over the `pages_content` and use the `text_splitter` `split_text` method to split the text into chunks. These chunks are then added to the `all_texts` list, yielding a collection of smaller text chunks from the original articles.

```
# We split the article texts into small chunks
from langchain.text_splitter import RecursiveCharacterTextSplitter
text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=100)
all_texts = []
for d in pages_content:
chunks = text_splitter.split_text(d["text"])
for chunk in chunks:
all_texts.append(chunk)
```

Now, add those chunks to the Deep Lake database:

```
# we add all the chunks to the Deep lake
db.add_texts(all_texts)
```

Next, we will create the **Plan and Execute** agent using our dataset. We will create a retriever from the Deep Lake dataset and a function for our custom tool to retrieve the most similar documents from the dataset.

```
# Get the retriever object from the deep lake db object and set the number
# of retrieved documents to 3
retriever = db.as_retriever()
retriever.search_kwargs[“k”] = 3

# We define some variables that will be used inside our custom tool
CUSTOM_TOOL_DOCS_SEPARATOR ="\n---------------\n" # how to join together the
# retrieved docs to form a single string

# This is the function that defines our custom tool that retrieves relevant
# docs from Deep Lake
def retrieve_n_docs_tool(query: str) -> str:
"""Searches for relevant documents that may contain the answer to the query."""
docs = retriever.get_relevant_documents(query)
texts = [doc.page_content for doc in docs]
texts_merged = "---------------\n" + CUSTOM_TOOL_DOCS_SEPARATOR.join(texts)
+ "\n---------------"
return texts_merged
```

To create a retriever, define the retriever object from the Deep Lake database and set the number of retrieved documents to 3. This step facilitates the retrieval of a specific count of relevant documents from Deep Lake in response to a particular query.

Now, define a custom function named `retrieve_n_docs_tool`. This function accepts a query and uses the retriever to locate documents relevant to the query.

The text of the retrieved documents is subsequently concatenated using the `CUSTOM_TOOL_DOCS_SEPARATOR` variable, the string that transforms the documents into a single text. The combined text is presented as the output from the custom tool function. This functionality enables the plan and execution agent to acquire and analyze relevant documents for decision-making.

```
from langchain.agents.tools import Tool

# We create the tool that uses the "retrieve_n_docs_tool" function
tools = [
Tool(
```

```
name="Search Private Docs",
func=retrieve_n_docs_tool,
description="Useful for when you need to answer questions about current events about Artificial Intelligence"
)
]
```

The “Search Private Docs” tool leverages the capabilities of the `retrieve_n_docs_tool` function. Its primary function is to facilitate the search and retrieval of relevant documents from Deep Lake, particularly for queries related to current events in AI. This tool is valuable in cases requiring collating information and insights from private documents.

```
from langchain.chat_models import ChatOpenAI
from langchain.experimental.plan_and_execute import PlanAndExecute,
load_agent_executor, load_chat_planner

# let's create the Plan and Execute agent
model = ChatOpenAI(model_name="gpt-3.5-turbo", temperature=0)
planner = load_chat_planner(model)
executor = load_agent_executor(model, tools, verbose=True)
agent = PlanAndExecute(planner=planner, executor=executor, verbose=True)
```

Next, we create the agent. This agent includes two primary components: the **planner** and the **executor**. The planner devises a strategy for responding to the user’s input, and the executor implements it through interactions with various tools and external systems. The agent is configured to operate verbose, delivering comprehensive details and logs throughout its decision-making and generation process.

```
# we test the agent
response = agent.run("Write an overview of Artificial Intelligence regulations by governments by country")
```

You should see something like the following output. Here, we split it into multiple sections and comment individually, keeping only the most relevant ones.

```
**> Entering new PlanAndExecute chain...**

steps=[Step(value='"Research the current state of Artificial Intelligence (AI) regulations in various countries."'),

Step(value='"Identify the key countries with significant AI regulations or ongoing discussions about AI regulations."'),

Step(value='"Summarize the AI regulations or discussions in each identified country."'),

Step(value='"Organize the information by country, providing an overview of the AI regulations in each."'),
```

```
Step(value="'Given the above steps taken, provide an overview of Artificial Intelligence regulations by governments by country.\n")]
```

The planning agent creates a plan for our query with multiple steps. Each step is a query that the action agent will answer. Here are the identified steps:

- Research the current state of Artificial Intelligence (AI) regulations in various countries.
- Determine the leading countries that have substantial AI regulations or are actively engaged in discussions about AI regulatory measures.
- Compile an overview of the AI regulations or dialogues in each country identified.
- Arrange the data by country, offering a summary of each AI regulation.
- Based on the steps above, present a summary of government regulations on Artificial Intelligence, categorized by country.

Let’s see how the output continues:

```
> Entering new AgentExecutor chain...*Action:
```
{
"action": "Search Private Docs",
"action_input": "current state of Artificial Intelligence regulations in various countries"
}
```*
Observation: *----------
“US-based AI developers will likely steal a march on their European competitors given the news that the EU parliamentary committees have green-lit its groundbreaking AI Act, where AI systems will need to be categorized according to their potential for harm from the outset. The US tech approach is typically to experiment first and, once market and product fit is established, to retrofit to other markets and their regulatory framework. This approach fosters innovation whereas EU-based AI developers will need to take note of the new rules and develop systems and processes which may take the edge off their ability to innovate. The UK is adopting a similar approach to the US, although the proximity of the EU market means that UK-based developers are more likely to fall into step with the EU ruleset from the outset. However, the potential to experiment in a safe space – a regulatory sandbox – may prove very attractive.”
----------
To boost AI innovation, MEPs added exemptions to these rules for research activities and AI components provided under open-source licenses. The new law also promotes regulatory sandboxes – or controlled environments established by public authorities – to test AI before its deployment.
```

MEPs want to boost citizens’ right to file complaints about AI systems and receive explanations of decisions based on high-risk AI systems that significantly impact their rights. MEPs also reformed the role of the EU AI Office, which would be tasked with monitoring how the AI rulebook is implemented.

Tim Wright, Tech and AI Regulatory Partner at London-based law firm Fladgate commented:

Ryan is a senior editor at TechForge Media with over a decade of experience covering the latest technology and interviewing leading industry figures. He can often be sighted at tech conferences with a strong coffee in one hand and a laptop in the other. If it's geeky, he’s probably into it. Find him on Twitter (@Gadget_Ry) or Mastodon (@gadgetry@techhub.social)

The Internal Market Committee and the Civil Liberties Committee of the European Parliament have endorsed new transparency and risk-management rules for artificial intelligence systems known as the AI Act.

This marks a major step in the development of AI regulation in Europe, as these are the first-ever rules for AI. The rules aim to ensure that AI systems are safe, transparent, traceable, and non-discriminatory.

After the vote, co-rapporteur Brando Benifei (S&D, Italy) said:
---*
Thought: *I have information about the current state of AI regulations in the EU, US, and UK. I can provide a summary of the AI regulations in these regions.
Action:
```
{
"action": "Final Answer",
"action_input": "The current state of AI regulations varies across countries. In the European Union, the AI Act has been endorsed by the Internal Market Committee and the Civil Liberties Committee, marking a major step in AI regulation. The AI Act aims to ensure that AI systems are safe, transparent, traceable, and non-discriminatory. In the United States, AI developers typically experiment first and then retrofit their products to other markets and regulatory frameworks, fostering innovation. The UK is adopting a similar approach to the US, but its proximity to the EU market means that UK-based developers are more likely to align with the EU ruleset from the outset. Regulatory sandboxes, or controlled environments established by public authorities, are being promoted to test AI before deployment."
}
```*> Finished chain.
*****

The executor agent is looking for suitable documents using the Deep Lake tool and displaying the top three retrieved chunks. After studying these, the agent decides to return the final answer to the inquiry: *The current state of AI regulations varies across countries. In the European Union, the AI Act has been endorsed by the Internal Market Committee and the Civil Liberties Committee, marking a major step in AI regulation. The AI Act aims to*

ensure that AI systems are safe, transparent, traceable, and non-discriminatory. In the United States, AI developers typically experiment first and then retrofit their products to other markets and regulatory frameworks, fostering innovation. The UK is adopting a similar approach to the US, but its proximity to the EU market means that UK-based developers are more likely to align with the EU ruleset from the outset. Regulatory sandboxes, or controlled environments established by public authorities, are being promoted to test AI before deployment.

Here, we omitted the outputs of the other steps. We printed the final response from the agent: `print(response)`

> European Union: The AI Act has been endorsed by the Internal Market Committee and the Civil Liberties Committee, aiming to ensure AI systems are safe, transparent, traceable, and non-discriminatory.
>
> United States: AI developers typically experiment first and then retrofit their products to other markets and regulatory frameworks, fostering innovation.
>
> United Kingdom: The UK is adopting a similar approach to the US, but its proximity to the EU market means that UK-based developers are more likely to align with the EU ruleset from the outset. Regulatory sandboxes are being promoted to test AI before deployment.

The agent was able to iteratively create an overview of AI regulations by using diverse documents and leveraging several documents.

Tutorial 2: Query and Summarize a DB with LlamaIndex

- Find the Notebook for this section at towardsai.net/book.

The tutorial teaches how to set up a Retrieval-Augmented Generation (RAG) system using LlamaIndex, Deep Lake, and OpenAI. We will define and manage data sources, create vector store indexes, configure query engines, and develop a conversational agent to handle queries. Additionally, we will enhance the agent's capabilities with custom functions for tasks such as mathematical operations, allowing the agent to perform specific functions beyond standard data retrieval.

Set up your environment by installing the required packages using: ! pip install -q llama-index==0.9.14.post3 deeplake==3.8.8 openai==1.3.8 cohere==4.37 .

```
import os

os.environ['OPENAI_API_KEY'] = '<YOUR_OPENAI_API_KEY>'
os.environ['ACTIVELOOP_TOKEN'] = '<YOUR_ACTIVELOOP_API_KEY>'
```

Defining Data Sources

In RAG, datasets primarily involve identifying and managing data sources. Tagging and monitoring sources from the beginning is beneficial when the dataset expands with new data points. This also includes tracking the general origin of the data, whether it's from a specific book, documentation, or a blog. For example, the Towards AI RAG-based AI tutor uses five data sources: Towards AI blogs, Activeloop documentation, LlamaIndex documentation, LangChain documentation, and Hugging Face documentation. As the dataset expands, new data points are added to these existing sources or incorporated into new ones. Implementing this approach from the beginning enhances the chatbot's efficiency by directing the "routers" to focus on the relevant information source.

Selecting suitable datasets is important to developing a data-driven application with the LlamaIndex RAG system. The data's quality and relevance significantly impact the system's performance, whether stored locally or hosted online in a vector database like Deep Lake.

It is better to begin the RAG pipeline design with a manageable dataset, such as web articles. A data environment that is manageable and rich is key to a successful start. It allows efficient testing, debugging, and understanding of the RAG system and enables easy querying and evaluating responses on a controlled and familiar dataset.

For this example, we will focus on Nikola Tesla's life, work, and legacy, with detailed information about his innovations, personal history, and influence. We will use two text documents: one with bold predictions Tesla made during his lifetime and another with biographical details. We will import these files and set up the indexes by combining data sources from the Deep Lake vector database for the first document and creating indexes from local storage for the second one.

Download the documents using the wget command. Alternatively, you can access and manually save the files from the URLs below: mkdir -p 'data/1k/' wget 'https://github.com/idontcalculate/data-repo/blob/main/machine_to_end_war.txt' -O './data/1k/tesla.txt' wget 'https://github.com/idontcalculate/data-repo/blob/main/prodigal_chapter10.txt' -O './data/1k/web.txt'

Read the first text file and process it for storage in Deep Lake. LlamaIndex's SimpleDirectoryReader class walks through a directory and converts text files into a Document object, which the framework recognizes.

```
from llama_index import SimpleDirectoryReader

tesla_docs = SimpleDirectoryReader(
input_files=["contentdata/1k/tesla.txt"]
).load_data()
```

Create a database on the Activeloop platform by entering the organization ID (your username by default) and naming it. Construct an empty database using the DeepLakeVectorStore class:

```
from llama_index.vector_stores import DeepLakeVectorStore

# By default, the organization id is your username.
my_activeloop_org_id = "<YOUR_ORGANIZATION_ID>"
my_activeloop_dataset_name = "LlamaIndex_tesla_predictions"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"

# Create an index over the documnts
vector_store = DeepLakeVectorStore(dataset_path=dataset_path, overwrite=False)
```

```
Your Deep Lake dataset has been successfully created!
```

Next, utilize the database object to create a storage context. This will allow us to generate indexes (embeddings) and save them into the database using the VectorStoreIndex class.

```
from llama_index.storage.storage_context import StorageContext
from llama_index import VectorStoreIndex

storage_context = StorageContext.from_defaults(vector_store=vector_store)
tesla_index = VectorStoreIndex.from_documents(tesla_docs,
storage_context=storage_context)
```

```
Uploading data to deeplake dataset.
100%|██████████| 5/5 [00:00<00:00, 7.17it/s]
/Dataset(path='hub://genai360/LlamaIndex_tesla_predictions',
tensors=['text', 'metadata', 'embedding', 'id'])

  tensor htype shape dtype compression
------- ------- ------- ------- -------
text text (5, 1) str None
metadata json (5, 1) str None
embedding embedding (5, 1536) float32 None
id text (5, 1) str None
```

The index for the first file is ready to be used as a source. We now need to save them locally as a precaution. Saving the index on a hard drive begins like our previous steps, with the SimpleDirectoryReader class: `webtext_docs = SimpleDirectoryReader( input_files=["content data/1k/web.txt"] ).load_data()` As we did earlier, you can use the same configuration but specify a directory to store the indexes. If pre-existing indexes have already been computed, the following script will attempt to load them first. If not, it stores the indexes using the .persist() method. The output shows the index is generated. Rerunning this code block will load the previously saved checkpoint and will not reprocess and regenerate indexes.

```
try:
# Try to load the index if it is already calculated
storage_context = StorageContext.from_defaults(
persist_dir="content storage/webtext"
)
webtext_index = load_index_from_storage(storage_context)
print("Loaded the pre-computed index.")
except:
# Otherwise, generate the indexes
webtext_index = VectorStoreIndex.from_documents(webtext_docs)
webtext_index.storage_context.persist(persist_dir="content storage/webtext")
print("Generated the index.") Generated the index.
```

Query Engine

We define two query engines for the agent to ensure it integrates information from our two sources: `tesla_engine = tesla_index.as_query_engine(similarity_top_k=3)` `webtext_engine = webtext_index.as_query_engine(similarity_top_k=3)` The search parameter `top_k=3` is configured to return the top 3 most similar results for a given query. As mentioned earlier, the query engine has two distinct data sources: 1. The `tesla_engine` variable handles queries related to general information.

2. The `webtext_engine` variable processes biographical data, focusing on queries requiring factual information.

This distinction in data styles ensures higher data quality during queries, as it avoids uniformly sourcing information from both data sources regardless of relevance.

Now, let's configure the tools.

You can use a blend of the `QueryEngineTool` class to create a new tool for a query engine and the `ToolMetaData` class to assign names and descriptions to these tools. These descriptions help the agent identify the most appropriate data source based on the query's nature.

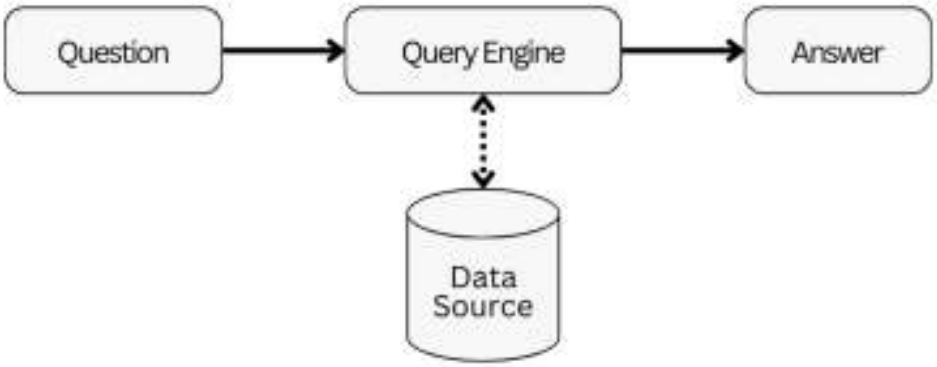
We will develop a list with two tools, each representing one of the data sources:

```
from llama_index.tools import QueryEngineTool, ToolMetadata

query_engine_tools = [
QueryEngineTool(
query_engine=tesla_engine,
metadata=ToolMetadata(
name="tesla_1k",
description=(
"""Provides information about Tesla's statements that refers to future times and predictions. """
"Use a detailed plain text question as input to the tool."
```

```
),
),
),
QueryEngineTool(
query_engine=webtext_engine,
metadata=ToolMetadata(
name="webtext_1k",
description=(
"Provides information about Tesla's life and biographical data. "
"Use a detailed plain text question as input to the tool."
),
),
),
]
```

Here’s a visual illustration of our current system. The query engine at the top indicates its function as the principal tool. It is centrally located between the data sources and the final answer formulation process. It serves as a link between the proposed queries and their replies.



Our baseline RAG pipeline with a query engine, data sources, and question-answer configuration.

The Agent

Our agent allows for simple testing of the retrieval system. For this example, we are using the OpenAI agent. The query engine tools are integrated into the `OpenAIAgent` module from LlamaIndex, allowing the agent to conduct queries. Setting the `verbose` option to `True` makes debugging easier by helping us investigate the agent's tool usage and intermediate processes. To obtain only the final output, set the parameter to `False`.

```
from llama_index.agent import OpenAIAgent

agent = OpenAIAgent.from_tools(query_engine_tools, verbose=True)
```

Now, we can create an interactive chat interface where the agent may accept inputs (such as questions or prompts), process them, and respond. This interface creates a conversational agent capable of handling a dialogue or chat session.

After the fundamental functionality has been tested and verified, we can add system design changes and feature enhancements.

```
agent.chat_repl()
```

```
===== Entering Chat REPL =====
Type "exit" to exit.

Human: What influenced Nikola Tesla to become an inventor?
STARTING TURN 1
---------------

=== Calling Function ===
Calling function: webtext_1k with args: {
"input": "What influenced Nikola Tesla to become an inventor?"
}
Got output: Nikola Tesla was influenced to become an inventor by his studies of mechanical vibrations. He observed the selective response of objects to vibrations and realized the potential for producing effects of tremendous magnitude on physical objects. This led him to pursue research in the field of high-frequency and high-potential currents, which eventually resulted in his groundbreaking inventions.
========================

STARTING TURN 2
---------------

Assistant: Nikola Tesla was influenced to become an inventor by his studies of mechanical vibrations. He observed the selective response of objects to vibrations and realized the
```

potential for producing effects of tremendous magnitude on physical objects. This led him to pursue research in the field of high-frequency and high-potential currents, which eventually resulted in his groundbreaking inventions.

Human: exit

💡 To debug tools in development, a practical approach involves querying the agent about its tools. This process includes asking the agent to • share the tools at its disposal,

• the arguments these tools accept,

• the significance of these arguments,

• and the intended use of each tool.

You can analyze the agent's responses to identify prompt flaws or understand why the agent might struggle to utilize a tool under development effectively.

Adding Custom Functions to our Agent

One area where large language models (LLMs) often face challenges is mathematical operations. Tasks like basic addition or subtraction, which may seem simple, can be problematic for these models. To address this, a practical approach is to equip the models with supplementary tools, such as a calculator. As an experiment, we will develop a custom function that a chatbot can utilize for basic multiplication or addition calculations.

Defining custom functions specific to each task is required. These functions should be capable of accepting various inputs and producing an output. Their functionality can vary from simple operations, like addition in our example, to more intricate tasks, such as conducting web searches, querying other LLMs, or incorporating data from external APIs to respond to a question.

```
def multiply(a: int, b: int) -> int:
"""Multiply two integers and returns the result integer"""
return a * b

def add(a: int, b: int) -> int:
"""Add two integers and returns the result integer"""
return a + b

from llama_index.tools import FunctionTool
```

```
multiply_tool = FunctionTool.from_defaults(fn=multiply, name="multiply")
add_tool = FunctionTool.from_defaults(fn=add, name="add")

all_tools = [multiply_tool, add_tool]
```

The above code defines two functions, “add” and “multiply.” In this setup, it is essential to clearly specify the data types for the input arguments (a:int, b:int), the return type (->int), and include a brief description of the function’s purpose within the triple quotes beneath the function name. These annotations will be utilized by the FunctionTool class’s .from_defaults() method to create a description of the function. This description is then accessible to the agent.

Lastly, the code defines a variable holding a list of all the available tools. These tools construct an ObjectIndex , a wrapper class that links a VectorStoreIndex with an array of potential tools.

Use the SimpleToolNodeMapping tool to convert the tool implementations into nodes. Following this, all components are interconnected to form a cohesive system.

```
from llama_index import VectorStoreIndex
from llama_index.objects import ObjectIndex, SimpleToolNodeMapping

tool_mapping = SimpleToolNodeMapping.from_objects(all_tools)
obj_index = ObjectIndex.from_objects(
all_tools,
tool_mapping,
VectorStoreIndex,
)
```

?? This implementation does not include any data sources because we intend to supplement the capabilities of LLMs with new tools.

Next, use the defined object index as a retriever. This implies that custom functions are recognized as extra data sources within the LlamaIndex framework. So, use the FnRetrieverOpenAIAgent class to describe the agent object:

```
from llama_index.agent import FnRetrieverOpenAIAgent

agent = FnRetrieverOpenAIAgent.from_retriever(
obj_index.as_retriever(), verbose=True
```

```
)
```

Finally, use the agent to ask questions! The agent will respond using the multiply function.

```
agent.chat("What's 12 multiplied by 22? Make sure to use Tools")
```

```
STARTING TURN 1
---------------

=== Calling Function ===
Calling function: multiply with args: {
"a": 12,
"b": 22
}
Got output: 264
========================

STARTING TURN 2
---------------

AgentChatResponse(response='12 multiplied by 22 is 264.', sources=
[ToolOutput(content='264', tool_name='multiply', raw_input={'args': (),
'kwargs': {'a': 12, 'b': 22}}, raw_output=264)], source_nodes=[])
```

We instructed the agent to use the tools in the prompt. You can use the `tool_choice` argument to explicitly guide the agent in utilizing specific tools or the `auto` keyword to allow the agent to decide.

```
response = agent.chat("What is 5 + 2?", tool_choice="add")
```

```
STARTING TURN 1
---------------

=== Calling Function ===
Calling function: add with args: {
"a": 5,
"b": 2
}
Got output: 7
========================
```

```
STARTING TURN 2
---------------
```

```
AgentChatResponse(response='5 + 2 is equal to 7.', sources=
[ToolOutput(content='7', tool_name='add', raw_input={'args': (), 'kwargs':
{'a': 5, 'b': 2}}, raw_output=7)], source_nodes=[])
```

By understanding how to define and manage data sources, create vector store indexes, and configure query engines, you can develop a robust conversational agent capable of handling a variety of queries. Integrating custom functions allows the agent to perform specialized tasks, such as mathematical calculations, thereby enhancing its utility beyond mere data retrieval. These capabilities position the RAG system as a powerful tool in building advanced, context-aware AI applications.

Tutorial 3: Building Agents with OpenAI Assistants

- Find the Notebook for this section at towardsai.net/book.

In addition to large models, OpenAI provides an API to help developers build “Assistants” quickly, freeing builders from the complexities of building agent systems from scratch. This section will present an overview of the service and how to set up your own OpenAI Assistant.

The OpenAI Assistants API offers three primary functionalities: Code Interpreter, Knowledge Retrieval, and Function Calling.

The Code Interpreter allows the Assistant to generate and execute Python code within a secure, sandboxed environment. This feature improves the Assistant’s logical problem-solving accuracy, making it adept at tasks like solving complex mathematical equations. It can also create data files and generate graphical representations from Python code, which is useful for validating the Assistant’s output and performing data analysis. Users can activate the Code Interpreter during a conversation or by uploading a data file.

Knowledge Retrieval is a component of OpenAI's retrieval-augmented generation (RAG) system integrated into the Assistants API. This function supports multiple document uploads. Once documents are uploaded, OpenAI processes them by segmenting, indexing, and storing their embeddings, subsequently using vector search to find relevant content for answering queries.

Function Calling enables users to introduce specific functions or tools to the Assistant. The model can then recognize and return the necessary functions along with their arguments, greatly expanding the range of tasks the Assistant can perform.

These capabilities can be utilized to develop highly effective agents. Let's explore how to set up an OpenAI Assistant. The following tutorial provides a guide on setting up an OpenAI Assistant using either the Assistants Playground or detailed API integration.

Setting up an OpenAI Assistant simplifies and accelerates user interactions, providing quick, accurate responses and streamlining complex tasks through efficient automation and tool integration. There are two methods to set up an assistant. The first is using the Assistants Playground, which is perfect for exploring the assistant's capabilities without needing complex integrations. The second method involves a detailed integration through the API, suitable for those requiring a more in-depth setup. This section contains a tutorial using this second method.

An Assistant object represents an entity or agent configured to respond to users' messages using various parameters. You can choose from different versions of GPT-3.5 or GPT-4 models, including fine-tuned versions, with OpenAI recommending the latest models for the best performance and compatibility. The Assistant supports tools like the Code Interpreter for executing Python code and Knowledge Retrieval for enhancing responses with external information.

A Thread serves as the basic unit of user interaction, representing a single conversation. Threads can handle user-specific context and file attachments, making each interaction unique. There is no limit to the number of messages that can be added to a thread, ensuring conversations can be as extensive as needed. The Assistant optimizes the context window for each

model request by techniques like truncation, which is the process of shortening the input by discarding older or less relevant data to fit within the model's context window.

Messages are the primary mode of communication between the user and the Assistant. They can include text, images, and other files, although GPT-4 with Vision is not supported. To add a message to a thread, use the following code: `message = client.beta.threads.messages.create(
thread_id=thread.id,
role="user",
content="I need to solve the equation `3x + 11 = 14`. Can you help me?"
)` To activate the Assistant's response, a Run must be created. During a Run, the Assistant will decide which previous messages to include in the context for generating a response. You can provide additional instructions to the Assistant during this step, but these will override its default instructions. The Assistant processes the entire thread, utilizes tools as necessary, and formulates a response. By examining Run steps, you can see how the Assistant reaches its conclusions. To display the Assistant's response from a Run, use the following command: `messages = client.beta.threads.messages.list(
thread_id=thread.id
)` This command retrieves and displays the responses added to the thread during the Run.

Creating an Assistant involves selecting a model, but you can further customize its behavior. Use the instructions parameter to define the Assistant's personality and objectives, similar to system messages in the Chat Completions API. The tools parameter allows the Assistant to access up to 128 tools in parallel, including OpenAI-hosted or third-party tools via function calling. The file_ids parameter enables tool access to files uploaded using the File upload endpoint.

Now, let's look at an example where we develop an AI assistant for a tech company using OpenAI Assistants. This assistant will be able to provide detailed product support using a comprehensive knowledge base. The following code sets up the coding environment.

```
mkdir openai-assistants && cd openai-assistants
python3 -m venv openai-assistants-env
```

```
source openai-assistants-env/bin/activate

pip3 install python-dotenv
pip3 install --upgrade openai

# fire up VSCode and let's get coding!
code .
```

Get the OpenAI key from your OpenAI developer account and set it using the following code.

```
import os

os.environ['OPENAI_API_KEY'] = '<YOUR_OPENAI_API_KEY>'
```

The next step is to consolidate all the files that can serve as a database for the agent in one folder. For this example, it can be a detailed PDF manual of your product line (e.g., “tech_manual.pdf”) using the API as follows:

```
from openai import OpenAI

client = OpenAI()
file = client.beta.files.upload(
file=open("tech_manual.pdf", "rb"),
filetype="application/pdf",
description="Tech product manual"
)
```

Once the folder is ready, create an assistant that includes the uploaded file. Ensure the assistant has the ability to retrieve those documents.

```
assistant = client.beta.assistants.create(
instructions="You are a tech support chatbot. Use the product manual to respond accurately to customer inquiries.",
model="gpt-4-turbo",
tools=[{"type": "retrieval"}],
file_ids=[file.id]
```
) To interact with the assistant, you need a thread and a message . In this example, the message will contain the customer’s question.
```
thread = client.beta.threads.create()
message = client.beta.threads.messages.create(
thread_id=thread.id,
role="user",
content="How do I reset my Model X device?",
```

```
) The assistant accesses the uploaded manual, performs a vector search to find the relevant section, and provides an answer to the user query as a result of a run .
run = client.beta.threads.runs.create(
thread_id=thread.id,
assistant_id=assistant.id,
)

# the run will enter the **queued** state before it continues it's execution.
```

After the run is complete, the assistant's response can be retrieved as follows:

```
messages = client.beta.threads.messages.list(
thread_id=thread.id
)

assistant_response = messages.data[0].content[0].text.value
```

The output will contain the assistant's response to the customer's question based on knowledge from the uploaded manual.

Tutorial 4: LangChain OpenGPT

LangChain's OpenGPTs are an open-source effort to create an experience similar to OpenAI's Assistants. Unlike OpenAI Assistants, LangChain OpenGPTs allow you to configure not only the LLM and set of tools but also the vector database, retrieval algorithm, and chat history database. LangChain OpenGPTs allow developers to fine-tune models for specific applications, choose between different model providers, and implement various tools and plugins, ensuring data privacy and alignment with specific needs. This makes them highly adaptable for diverse use cases and enterprise requirements, unlike the more standardized offerings from OpenAI Assistants.

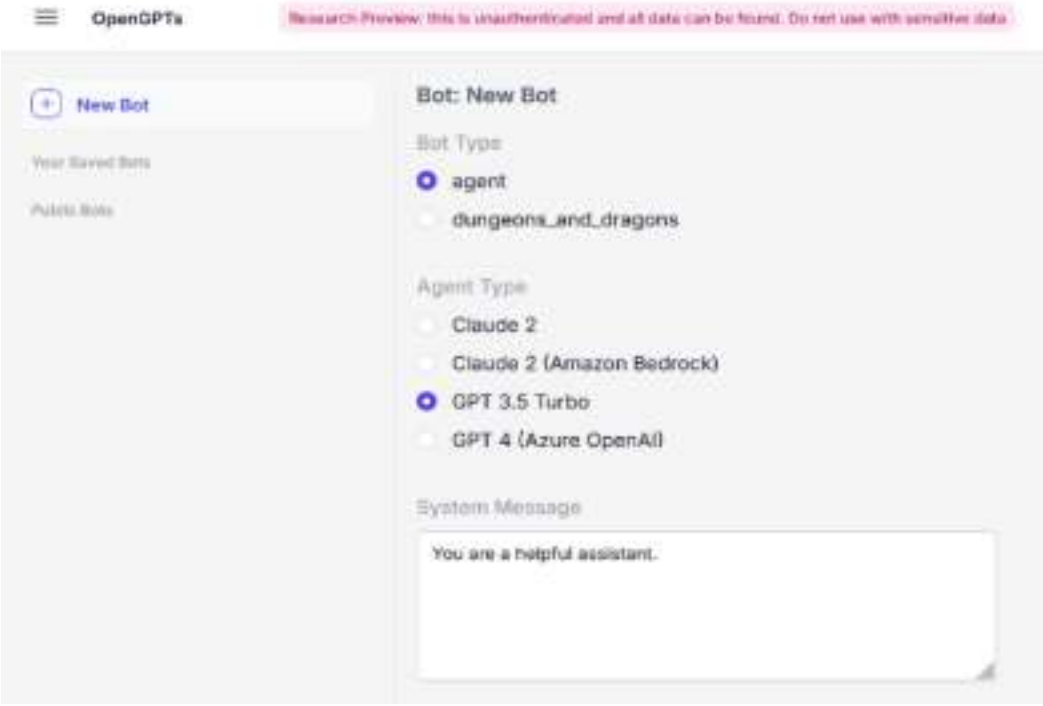
To interact with LangChain's OpenGPTs, you first need to clone the GitHub repository locally: `git clone https://github.com/langchain-ai/opengpts.git` `cd opengpts` Use the following environment (`.env`) file with the below content to set the required API keys for authentication purposes, replacing the placeholders. Here, only the OpenAI placeholder will be used for the tutorial: `OPENAI_API_KEY=placeholder`

```
ANTHROPIC_API_KEY=placeholder
YDC_API_KEY=placeholder
TAVILY_API_KEY=placeholder
AZURE_OPENAI_DEPLOYMENT_NAME=placeholder
AZURE_OPENAI_API_KEY=placeholder
AZURE_OPENAI_API_BASE=placeholder
AZURE_OPENAI_API_VERSION=placeholder
```

Now, launch everything with the following command:

```
docker compose up
```

By visiting http://localhost:8100/; you should see the following page:



Now, click on “New Bot,” select `gpt-3.5-turbo` (or another default that you want to use) and name the bot; here, we choose “**Career Counselor**.”

Provide a System Message . This will define a specific role or persona for the model, like a “Career Counselor.” This prompt will guide the models’ responses to fit the desired context and ensure its advice, insights, or recommendations align with your defined use case.

Here is an example of a System Message :

You are a Career Counselor. Your role is to provide insightful and personalized guidance as I navigate my professional path. Whether I'm facing career uncertainties, seeking job advancement, or contemplating a career shift, your expertise is aimed at offering constructive, individualized advice that helps me make informed decisions.

Our sessions will be a platform for discussing my professional aspirations, skills, and potential barriers.

In our interactions, I expect a supportive environment where I can share my professional experiences, goals, and concerns. Your role is to motivate and provide clear, practical strategies that align with my career objectives.

By understanding my unique circumstances, you offer tailored advice and plans to aid my professional growth. This collaboration is crucial for my career development, with your guidance being a cornerstone of my journey towards achieving my career goals.

Click Save, and you are ready to chat!

With LangChain OpenGPTs, one can create personalized customer support bots, develop AI-driven educational tools that provide tailored learning experiences, build content generation systems for marketing, and design advanced virtual assistants for specific industries, offering specialized insights and support based on domain-specific data.

Tutorial 5: Multimodal Financial Document Analysis from PDFs

- Find the Notebook for this section at towardsai.net/book.

We will examine the application of the retrieval-augmented generation (RAG) method in processing financial information from a company's PDF document. The steps involve extracting critical data such as text, tables, and graphs from a PDF file and storing them in a vector database like Deep Lake.

The process will also incorporate various tools, including Unstructured.io for text and table extraction from PDF, OpenAI's GPT-4V for graph information extraction from images, and LlamaIndex for creating an agent with retrieval capabilities.

Extracting Data

While extracting text from documents is generally straightforward, processing visual elements like line or bar charts is complex. OpenAI's latest model with vision processing capabilities, GPT-4V, is a valuable tool for this task. These visual elements complement the textual data by feeding graphical elements from the documents into the model and requesting detailed descriptions. We will use the Tesla Q3 financial report as the reference document for this example. The report can be downloaded using the `wget` command: `wget https://digitalassets.tesla.com/tesla-contents/image/upload/IR/TSLA-Q3-2023-Update-3.pdf`

?? The preprocessing tasks outlined in the next section might be time-consuming and necessitate API calls to OpenAI endpoints, which come with associated costs. To mitigate this, we have shared the preprocessed dataset and the checkpoints of the output of each section at towardsai.net/book.

Text and Tables

The `unstructured` package is a proficient tool for extracting information from PDF files. It relies on two key tools, `poppler` and `tesseract`, essential for rendering PDF documents. Set up these packages on Google Colab. Use the specified commands to install the packages and tools:

```
apt-get -qq install poppler-utils
apt-get -qq install tesseract-ocr

pip install -q unstructured[all-docs]==0.11.0 fastapi==0.103.2 kaleido==0.2.1
uvicorn==0.24.0.post1 typing-extensions==4.5.0 pydantic==1.10.13
```

?? These packages are easy to install on Linux and Mac operating systems using `apt-get` and `brew`. However, they are tricky to install on Windows OS. You can follow a step-by-step guide for installing poppler and tesseract on Windows available at towardsai.net/book.

Use the partition_pdf function to extract text and table data from the PDF and divide it into multiple chunks. You can also customize the size of these chunks based on the number of characters.

```
from unstructured.partition.pdf import partition_pdf

raw_pdf_elements = partition_pdf(
filename="./TSLA-Q3-2023-Update-3.pdf",
# Use layout model (YOLOX) to get bounding boxes (for tables) and find titles
# Titles are any sub-section of the document
infer_table_structure=True,
# Post processing to aggregate text once we have the title
chunking_strategy="by_title",
# Chunking params to aggregate text blocks
# Attempt to create a new chunk 3800 chars
# Attempt to keep chunks > 2000 chars
# Hard max on chunks
max_characters=4000,
new_after_n_chars=3800,
combine_text_under_n_chars=2000
)
```

The above code recognizes and extracts various PDF elements, which can be divided into CompositeElements (text) and Tables .

The partition_pdf function can leverage a machine learning model for certain PDFs, which might be slow without GPU acceleration. By default, the strategy argument is set to “auto,” meaning it may use machine learning models for parsing. To ensure efficiency, start with partition_pdf(strategy="fast") , parsing a few pages with text and tables, and checking the results. If the output isn’t accurate, especially for image-based PDFs, switch to strategy="auto" and configure GPU support. Once the parsed results are satisfactory, test the Q&A process on a short document. If successful, scale to longer documents, adjusting filtering or techniques if necessary to improve chunk retrieval.

We use the Pydantic package to create a new data structure that contains information about each element, such as its type and text .

The following code loops through all extracted elements, storing them in a list where each item is an instance of the Element type:

```
from pydantic import BaseModel
from typing import Any

# Define data structure
class Element(BaseModel):
type: str
text: Any

# Categorize by type
categorized_elements = []
for element in raw_pdf_elements:
if "unstructured.documents.elements.Table" in str(type(element)):
categorized_elements.append(Element(type="table", text=str(element)))
elif "unstructured.documents.elements.CompositeElement" in
str(type(element)):
categorized_elements.append(Element(type="text", text=str(element)))
```

The `Element` data structure allows for the straightforward recording of additional information. This helps identify the source of each answer, whether it is generated from texts, tables, or figures.

Graphs

The main challenge with extracting information from graphs and charts is to separate images from the document and analyze them using OpenAI's model. A practical method is converting each PDF page into images and passing them to the model individually to determine if it identifies any graphs. If the model detects one, it can describe the data and trends depicted in it. In cases where no graphs are found, the model will return an empty array.

💡 A drawback of this approach is that each page must be processed, regardless of whether it contains graphs. This increases the number of requests to the model, leading to higher costs. It is possible to reduce the cost by manually flagging the pages.

Install the `pdf2image` package to convert the PDF into images using the command: `!pip install -q pdf2image==1.16.3`. This also requires the `poppler` tool, which we have already installed.

The below code uses the convert_from_path function, which requires the path of a PDF file as input. For this, save each page of the PDF as a PNG file using the .save() method and store them in the ./pages directory. Next, define a variable named pages_png to keep track of the path for each image file.

```
import os
from pdf2image import convert_from_path

os.mkdir("./pages")
convertor = convert_from_path('./TSLA-Q3-2023-Update-3.pdf')

for idx, image in enumerate( convertor ):
image.save(f"./pages/page-{idx}.png")

pages_png = [file for file in os.listdir("./pages") if file.endswith('.png')]
```

Now, we define the helper functions and variables before submitting image files to the OpenAI API.

Define the headers variable to store the OpenAI API Key, which is required for the server to authenticate requests, and the payload variable for configurations, including specifying the model name, setting the maximum token limit, and defining prompts. These prompts instruct the model to analyze and describe graphs and generate responses in JSON format. This setup is designed to handle various scenarios, such as encountering multiple graphs on a single page or pages without graphs.

Add the images to the payload before sending the request. A payload in this context refers to the data being sent in a request to an API, specifically the images and messages for the AI to process. We also developed an encode_image() function to convert images into base64 format to make them compatible with the API. Base64 encoding is a method of converting binary data, such as images, into a string of ASCII characters. This ensures that non-text data can be transmitted over text-based protocols like JSON.

```
headers = {
"Content-Type": "application/json",
"Authorization": "Bearer " + str( os.environ["OPENAI_API_KEY"] )
}
```

```
payload = {
"model": "gpt-4-turbo",
"messages": [
{
"role": "user",
"content": [
{
"type": "text",
"text": """You are an assistant that find charts, graphs, or diagrams from an image and summarize their information. There could be multiple diagrams in one image, so explain each one of them separately. ignore tables."""
},
{
"type": "text",
"text": '"The response must be a JSON in following format {"graphs": [<chart_1>, <chart_2>, <chart_3>]} where <chart_1>, <chart_2>, and <chart_3> placeholders that describe each graph found in the image. Do not append or add anything other than the JSON format response."'
},
{
"type": "text",
"text": '"If could not find a graph in the image, return an empty list JSON as follows: {"graphs": []}. Do not append or add anything other than the JSON format response. Dont use coding "```" marks or the word json."'
},
{
"type": "text",
"text": """Look at the attached image and describe all the graphs inside it in JSON format. ignore tables and be concise."""
}
]
}
],
"max_tokens": 1000
}

# Function to encode the image to base64 format
def encode_image(image_path):
with open(image_path, "rb") as image_file:
return base64.b64encode(image_file.read()).decode('utf-8')
```

Use the `pages_png` variable to iterate through the images and encode each image into the `base64` format. Incorporate the encoded image into the payload, send the request to OpenAI, and manage the responses received. For organizational consistency, the same `Element` data structure will store additional information about each image (graph):

```
graphs_description = []
for idx, page in tqdm( enumerate( pages_png ) ):
# Getting the base64 string
base64_image = encode_image(f"./pages/{page}")

    # Adjust Payload
tmp_payload = copy.deepcopy(payload)
tmp_payload['messages'][0]['content'].append({
"type": "image_url",
"image_url": {
"url": f "{base64_image}"
}
})

    try:
response = requests.post("https://api.openai.com/v1/chat/completions",
headers=headers, json=tmp_payload)
response = response.json()
graph_data = json.loads(
response['choices'][0]['message']['content']
)['graphs']

    desc =
[f"{page}\n" + '\n'.join(f"{key}:
{item[key]}" for key in item.keys()) for item in graph_data]

    graphs_description.extend(desc)

    except:
# Skip the page if there is an error.
print("skipping... error in decoding.")
continue;

graphs_description = \
[Element(type="graph", text=str(item)) for item in graphs_description]
```

Store on Deep Lake

The data preparation phase is complete once all vital information is extracted from the PDF. The next step is integrating the outputs from the previous sections, resulting in a compilation of 41 entries.

```
all_docs = categorized_elements + graphs_description

print(len(all_docs))
```

```
41
```

Since we use LlamaIndex, we can use its integration with Deep Lake to produce and store the dataset. Install the LlamaIndex and Deeplake packages and their dependencies: `!pip install -q llama_index==0.9.8 deeplake==3.8.8 cohere==4.37` .

Set the environment variables for `OPENAI_API_KEY` and `ACTIVELOOP_TOKEN` and replace the placeholder values with the correct keys from the respective platforms:

```
import os

os.environ["OPENAI_API_KEY"] = "<YOUR_OPENAI_API_KEY>"
os.environ["ACTIVELOOP_TOKEN"] = "
<YOUR_ACTIVELOOP_KEY>"
```

Insert your organization ID or Activeloop username into the code below. This code will create an empty dataset ready to store documents:

```
from llama_index.vector_stores import DeepLakeVectorStore

# TODO: use your organization id here. (by default, org id is your username)
my_activeloop_org_id = "<YOUR-ACTIVELOOP-ORG-ID>"
my_activeloop_dataset_name = "tsla_q3"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"
vector_store = DeepLakeVectorStore(dataset_path=dataset_path, runtime={"tensor_db": True},
overwrite=False)
```

```
Your Deep Lake dataset has been successfully created!
```

Next, send the newly constructed vector store to a `StorageContext` class. This class acts as a wrapper for creating storage from various data types. In this

case, we're producing the storage from a vector database, performed simply by passing the constructed database instance through the .from_defaults() method:

```
from llama_index.storage.storage_context import StorageContext

storage_context =
StorageContext.from_defaults(vector_store=vector_store)
```

It is necessary to convert the preprocessed data into a Document format compatible with LlamaIndex. In this context, we can incorporate a metadata tag within each document to store additional details, such as the data type (text, table, or graph), or indicate relationships between documents. This streamlines the retrieval process later.

There are options like using built-in classes such as SimpleDirectoryReader or manual processing for automatic file reading from a designated path. We used the built-in class to iterate through the list of all the extracted information, where each document is assigned relevant text and categorized appropriately.

```
from llama_index import Document

documents = [Document(text=t.text, metadata={"category": t.type},) for t in categorized_elements]
```

Lastly, we utilize the VectorStoreIndex class to generate embeddings for the documents and employ the database instance to store these values.

```
from llama_index import VectorStoreIndex

index = VectorStoreIndex.from_documents(
documents, storage_context=storage_context
)
```

```
Uploading data to deeplake dataset.
100%|██████████| 29/29 [00:00<00:00, 46.26it/s]
\Dataset(path='hub://alafalaki/tsla_q3-nograph',
tensors=['text', 'metadata', 'embedding', 'id'])

    tensor htype shape dtype compression
------- ------- ------- ------- -------
text text (29, 1) str None
metadata json (29, 1) str None
embedding embedding (29, 1536) float32 None
id text (29, 1) str None
```

?? The dataset has already been curated and is hosted under the GenAI360 organization on the Activeloop hub. If you prefer not to use OpenAI APIs for generating embeddings, you can test the remaining codes using these publicly accessible datasets. Just substitute the dataset_path variable with the following: hub://genai360/tsla_q3 .

Fine-tuning the Embedding Model

Since we use Deep Lake, we can leverage Activeloop's Deep Memory system, which can improve the retriever's accuracy by allowing the model to access higher-quality data. Deep Memory is an easy-to-use embedding model fine-tuning system. It will convert your embeddings into a new embedding space that is tailored to your use case, as seen in the Advanced RAG Techniques section of Chapter 9.

The process involves obtaining data segments from the database and using GPT-3.5 to generate customized queries for each data segment. These questions are then employed in the Deep Memory training to increase the embedding quality by transforming our embedding space into one tailored to our data. In practice, this strategy has shown a 22% improvement in performance in terms of expected chunks returned, as per Activeloop's data.

💡 We note that this improved accuracy is measured by just one metric (the specific chunk returned correctly) and does not necessarily ensure improved performance.

💡 Activeloop recommends using a dataset with a minimum of 100 chunks, ensuring sufficient context for the model to enhance the embedding space effectively. The codes in this section are based on three PDF documents. Please refer to the accompanying notebook for the complete code and steps to process three documents (available on towardsai.net/book). The processed dataset is available in the cloud on the GenAI360 organization. You can access it using hub://genai360/tesla_quarterly_2023 .

Load the pre-existing dataset and read the text of each chunk along with its corresponding ID:

```
from llama_index.vector_stores import DeepLakeVectorStore
```

```
# TODO: use your organization id here. (by default, org id is your username)
my_activeloop_org_id = "<YOUR-ACTIVELOOP-ORG-ID>"
my_activeloop_dataset_name = "LlamaIndex_tsla_q3"
dataset_path = f"hub://{my_activeloop_org_id}/{my_activeloop_dataset_name}"

db = DeepLakeVectorStore(
dataset_path=dataset_path,
runtime={"tensor_db": True},
read_only=True
)

# fetch dataset docs and ids if they exist (optional you can also ingest)
docs = db.vectorstore.dataset.text.data(fetch_chunks=True, aslist=True)['value']
ids = db.vectorstore.dataset.id.data(fetch_chunks=True, aslist=True)['value']
print(len(docs))
```

```
Deep Lake Dataset in hub://genai360/tesla_quarterly_2023 already exists, loading from the storage

127
```

The following code describes a function that uses GPT-3.5 to generate questions for each data chunk. This requires developing a specific tool for the OpenAI API. Primarily, the code configures appropriate prompts for API queries to generate the questions and collects them into a list with their related chunk IDs:

```
import json
import random
from tqdm import tqdm
from openai import OpenAI

client = OpenAI()
# Set the function JSON Schema for openai function calling feature
tools = [
{
"type": "function",
"function": {
"name": "create_question_from_text",
"parameters": {
"type": "object",
```

```
"properties": {
"question": {
"type": "string",
"description": "Question created from the given text",
},
},
"required": ["question"],
},
"description": "Create question from a given text.",
},
}
]

def generate_question(tools, text):
try:
response = client.chat.completions.create(
model="gpt-3.5-turbo",
tools=tools,
tool_choice={
"type": "function",
"function": {"name": "create_question_from_text"},
},
messages=[
{"role": "system", "content": """You are a world class expert for generating questions based on provided context. You make sure the question can be answered by the text."""},
{
"role": "user",
"content": text,
},
],
)

    json_response = response.choices[0].message.tool_calls[0].function.arguments
parsed_response = json.loads(json_response)
question_string = parsed_response["question"]
return question_string
```

```
except:
question_string = "No question generated"
return question_string

def generate_queries(docs: list[str], ids: list[str], n: int):
questions = []
relevances = []
pbar = tqdm(total=n)
while len(questions) < n:
# 1. randomly draw a piece of text and relevance id
r = random.randint(0, len(docs)-1)
text, label = docs[r], ids[r]

    # 2. generate queries and assign and relevance id
generated_qs = [generate_question(tools, text)]
if generated_qs == ["No question generated"]:
continue

    questions.extend(generated_qs)
relevances.extend([[(label, 1)] for in generatedqs])
pbar.update(len(generated_qs))

    return questions[:n], relevances[:n]

questions, relevances = generate_queries(docs, ids, n=20)
```

```
100%|██████████| 20/20 [00:19<00:00, 1.02it/s]
```

We can now use the questions and reference IDs to activate Deep Memory using the `.deep_memory.train()` function to improve the embedding representations. The `.info` method can be used to view the status of the training process.

```
from langchain.embeddings.openai import OpenAIEmbeddings

embeddings = OpenAIEmbeddings()

job_id = db.vectorstore.deep_memory.train(
queries=questions,
relevance=relevances,
embedding_function=embeddings.embed_documents,
)

print(db.vectorstore.dataset.embedding.info)
```

```
Starting DeepMemory training job
Your Deep Lake dataset has been successfully created!
Preparing training data for deepmemory:
Creating 20 embeddings in 1 batches of size 20:: 100%|██████████| 1/1 [00:03<00:00,
3.23s/it]
DeepMemory training job started. Job ID: 6581e3056a1162b64061a9a4

{'deepmemory': {'6581e3056a1162b64061a9a4_0.npy': {'base_recall@10': 0.25,
'deep_memory_version': '0.2', 'delta': 0.25, 'job_id': '6581e3056a1162b64061a9a4_0',
'model_type': 'npy', 'recall@10': 0.5}, 'model.npy': {'base_recall@10': 0.25,
'deep_memory_version': '0.2', 'delta': 0.25, 'job_id': '6581e3056a1162b64061a9a4_0',
'model_type': 'npy', 'recall@10': 0.5}}}
```

The dataset is now ready and compatible with the deep memory feature. Note that the Deep Memory option must be actively set to `True` when using the dataset for inference.

Chatbot in Action

To see the chatbot in action, we will use the created dataset as the retrieval source, providing the context for the `gpt-3.5-turbo` model (the default model for LlamaIndex) to answer questions.

The inference results discussed in the next section are based on the analysis of three PDF files. The sample codes in the notebook use the same files. Use the dataset path `hub://genai360/tesla_quarterly_2023` to access the processed dataset containing all the PDF documents.

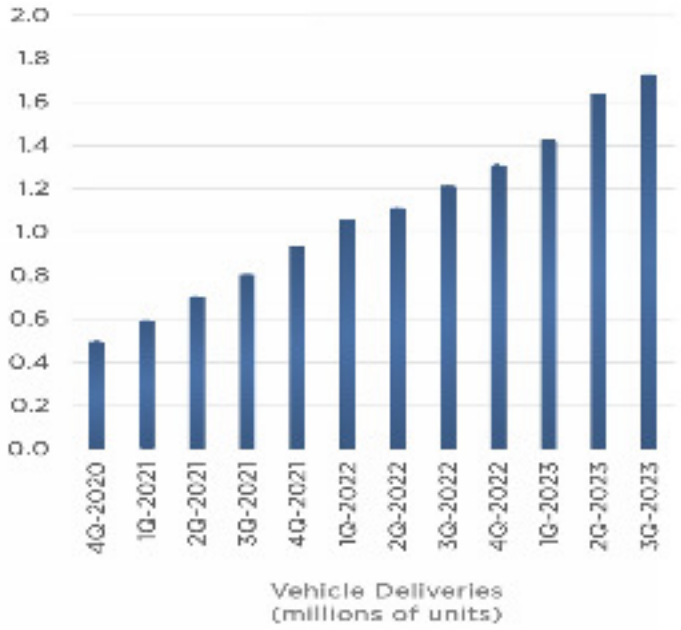
The `DeepLakeVectorStore` class loads a dataset from the hub. Compared to earlier sections, a notable difference in the code is the implementation of the `.from_vector_store()` method. This method is designed to generate indexes directly from the database instead of using variables, streamlining the process of indexing and retrieving data.

```
from llama_index.vector_stores import DeepLakeVectorStore
from llama_index.storage.storage_context import StorageContext
from llama_index import VectorStoreIndex

vector_store = DeepLakeVectorStore(dataset_path=dataset_path, overwrite=False)
storage_context = StorageContext.from_defaults(vector_store=vector_store)

index = VectorStoreIndex.from_vector_store(
vector_store, storage_context=storage_context
)
```

Now, create a query engine using the index variables .as_query_engine() method. This will allow us to ask questions from various data sources. The vector_store_kwargs option activates the deep_memory feature by setting it to True . This step is required to enable the feature on the retriever. The .query() method takes a prompt and searches the database for the most relevant data points to build an answer. We can now ask it questions: query_engine = index.as_query_engine(vector_store_kwargs={"deep_memory": True}) response = query_engine.query("What are the trends in vehicle deliveries?",) The trends in vehicle deliveries on the Quarter 3 report show an increasing trend over the quarters.



Key Metrics Quarterly (Unaudited). Image from the [Tesla Q3 financial report](#).

The chatbot demonstrated its capability to use data from the graph descriptions effectively.

To verify that the same question couldn’t be answered without graph descriptions, we created a second dataset identical to the original but without the graph descriptions. This dataset is accessible through the hub://genai360/tesla_quarterly_2023-nograph path. Running the same code on this

second dataset leads to the following chatbot response: In quarter 3, there was a decrease in Model S/X deliveries compared to the previous quarter, with a 14% decline. However, there was an increase in Model 3/Y deliveries, with a 29% growth. Overall, total deliveries in quarter 3 increased by 27% compared to the previous quarter.

The experiment showed that the chatbot directs to inaccurate text portions. While the answer was contextually similar, it wasn't correct.

💡 The Preprocessed Text/Label and Preprocessed Graphs for this section are accessible at towardsai.net/book.

Recap

This chapter explored intelligent agents, highlighting their potential to shape a new era for software systems. Unlike traditional content generators, these agents utilize large language models (LLMs) for reasoning and planning. Early examples like AutoGPT and BabyAGI demonstrate the capabilities of these intelligent systems in managing complex tasks. The open-source community is playing a significant role in advancing agentic workflows, with projects such as CAMEL and Generative Agents in LangChain showcasing the possibilities of agent simulations.

A Plan and Execute agent was presented, which successfully generated a comprehensive overview of AI regulations across various governments. This example illustrated the agent's ability to comprehend complex data, gather essential information, synthesize it, and provide clear, informative summaries.

The chapter also discussed tools and resources for developing agentic workflows, including the OpenAI Assistants API and LangChain OpenGPT. Additionally, we demonstrated an agent's ability to perform multimodal financial document analysis, integrating data from text, charts, and graphs within PDFs.

Looking ahead, we anticipate the emergence of more advanced and specialized agents capable of handling increasingly intricate tasks with

minimal human intervention, fundamentally transforming our interaction with software and technology.

Chapter XI: Fine-Tuning

Understanding FineTuning

While pretraining gives large language models (LLMs) a general understanding of language, it falls short for instruction-following tasks. LLMs are trained to predict the next token in its training data (often web pages) and this doesn't necessarily make it immediately capable of answering questions or following instructions. For example, if a user types in a regular Google search or a pretrained LLM, like "In what country is Montreal?", rather than generating an answer the LLM might instead generate lists of similar questions with similar meanings. Such a model would struggle to give good answers to requests like summarizing a web page or generating SQL queries. Finetuning is a method to address these limitations and specialize a model at understanding the format of user requests and the types of tasks it needs to perform.

Finetuning resumes training a pretrained model to increase the performance of a specific task using task-specific data like question-answer pairs for general systems like ChatGPT or Claude. This allows the model to adjust its internal parameters and representations to better suit the task, thus improving its ability to tackle domain-specific issues.

Instruction finetuning is a strategy popularized by OpenAI in 2022 with their InstructGPT models. It gives LLMs the capacity to follow written human instructions and thus increases the level of control over the model's outputs. The goal is to train an LLM to interpret prompts as instructions rather than just input for general text completion/generation.

However, standard finetuning for LLMs can be resource-heavy and expensive. It requires modifying all parameters in the pre-trained models, often in billions. Therefore, using more efficient and cost-effective finetuning techniques, such as Low-Rank Adaptation, is essential.

Several techniques are available to improve the performance of LLMs:

- **Standard FineTuning:** It adjusts all the parameters in LLM to increase performance to a specific task. Although effective, it

demands extensive computational resources, making it less valuable.

- **Low-Rank Adaptation (LoRA):** It modifies only a small subset of parameters by applying low-rank approximations on the lower layers of LLMs. This is a more efficient approach, significantly reducing the number of parameters that need training. LoRA reduces GPU memory requirements and lowers training costs.
- **Supervised FineTuning (SFT):** It trains a base model on a new dataset under supervision. This new dataset typically includes demonstration data, prompts, and corresponding responses. The model learns from this data and generates responses that align with the expected outputs. SFT can be used for instruction finetuning.
- **Reinforcement Learning from Human Feedback (RLHF):** It iteratively trains models to align with human feedback. This approach can be more effective than SFT as it facilitates continuous improvement based on human input. Similar methodologies include Direct Preference Optimization (DPO) and Reinforcement Learning from AI Feedback (RLAIF).

Low-Rank Adaptation (LoRA)

Low-Rank Adaptation (LoRA), developed by Microsoft researchers, enhances the LLM finetuning process. It addresses common finetuning challenges such as high memory requirements and computational inefficiency. LoRA introduces an efficient method involving **low-rank matrices** to store essential modifications in the model, avoiding altering all parameters.

LoRA has two very important features. First, it preserves the pretrained weights of the model, preventing catastrophic forgetting and ensuring that valuable knowledge from pretraining is retained. Second, it employs efficient rank-decomposition, where smaller update matrices are added to the model's existing weights. These matrices, requiring fewer parameters, focus training on the new weights, allowing for faster training with less

memory usage. Typically, the LoRA matrices are integrated into the model's attention layers.

LoRA's approach to low-rank decomposition considerably lowers the memory requirements for training large language models. This reduction makes finetuning tasks accessible on consumer-grade GPUs, extending the advantages of LoRA to more researchers and developers.

Quantized Low-Rank Adaptation (QLoRA) is a more preferred variant of LoRA among developers that incorporates strategies to further conserve memory without compromising performance.

QLoRA backpropagates gradients through a frozen, 4-bit quantized pretrained language model into Low-Rank Adapters, significantly cutting down memory usage. This allows for finetuning even larger models on standard GPUs. For example, QLoRA can fine-tune a language model with 65 billion parameters on a 48GB GPU, maintaining a comparable performance level as full 16-bit finetuning.

💡 Quantization is a powerful (and sometimes necessary) optimization technique that converts model weights from high-precision floating-point representation to low-precision floating-point or integers to reduce the model's size and training compute requirements. We will talk about model optimization in more depth in the next chapter.

QLoRA uses a new data type called 4-bit NormalFloat (NF4), ideal for normally distributed weights. It also uses double quantization to lower the average memory footprint by quantizing the quantization constants and paged optimizers to manage memory spikes.

This efficiency is due to quantile quantization, a method well-suited for values with a normal distribution. It ensures that each bin in the quantization process contains equal values from the input tensor. This approach minimizes quantization error and leads to a more even data representation. Pretrained neural network weights generally exhibit a zero-centered normal distribution with a particular standard deviation (σ). QLoRA standardizes these weights to a consistent fixed distribution by scaling σ. This scaling ensures that the distribution fits precisely within the NF4 data type's range, thereby enhancing the efficiency and accuracy of the

quantization process. This finetuning technique demonstrates no loss in accuracy in the authors' experiments, matching the performance of BFloat16.

The Guanaco models, which feature QLoRA finetuning, have shown cutting-edge performance even with smaller models. The versatility of QLoRA tuning makes it a popular choice for those looking to democratize the usage of big transformer models.

During the initial stages of neural network training, a 32-bit floating-point format was standard for training models, meaning each weight was represented by 32 bits and required 4 bytes of storage. To address this issue, the loading model now uses lower-precision numbers. Using an 8-bit format for numbers reduces storage requirements to a single byte.

With recent innovations, models can now be loaded in a 4-bit format, reducing memory requirements. The BitsAndBytes library loads pretrained models even more memory-efficiently, as shown in the example code:

```
from transformers import AutoModelForCausalLM, BitsAndBytesConfig
import torch

model = AutoModelForCausalLM.from_pretrained(
model_name_or_path='nameor/path/to/your/model',
load_in_4bit=True,
device_map='auto',
torch_dtype=torch.bfloat16,
quantization_config=BitsAndBytesConfig(
load_in_4bit=True,
bnb_4bit_compute_dtype=torch.bfloat16,
bnb_4bit_use_double_quant=True,
bnb_4bit_quant_type='nf4'
),
)
```

Note that this technique preserves model weights and does not impact the training process. Moreover, there's a constant trade-off between using lower-precision numbers and potentially diminishing the language

processing capabilities of models. While it's generally acceptable in most cases, it's important to acknowledge its presence.

Tutorial 1: SFT with LoRA

- Find the Notebook for this section at towardsai.net/book.

The upcoming section will demonstrate an example of finetuning a model using the LoRA approach on a small dataset. This technique allows for instruction tuning of large language models by applying low-rank adaptations to modify the model's behavior without retraining its weights. This process is highly efficient and can be executed using a CPU on a Google Cloud instance. We will guide you through preparing the dataset and conducting the finetuning process using the Hugging Face library. This hands-on example will provide a practical understanding of enhancing model performance with minimal computational resources and dataset size.

While the Hugging Face library is used in this example, the following libraries provide a range of tools, optimizations, compatibility with various data types, resource efficiency, and user-friendly interfaces to support different tasks and hardware setups, enhancing the efficiency of the LLM finetuning process.

- PEFT Library: The Parameter-Efficient FineTuning (PEFT) library enables the efficient adaptation of pretrained language models to various downstream applications without finetuning all the model's parameters. Methods like LoRA, Prefix Tuning, and P-tuning are part of PEFT.

- Lit-GPT: Developed by LightningAI, Lit-GPT is also an open-source tool designed to streamline finetuning. It facilitates the application of techniques like LoRA without manual modifications to the core model architecture. Models such as Vicuna, Pythia, and Falcon are available. Lit-GPT allows applying specific configurations to different weight matrices and offers adjustable precision settings to manage memory usage effectively.

We also recommend using virtual machines on systems like GCP, which is good practice in the industry and might be better than using your current system in most cases. To do so, log in to the Google Cloud Platform account and create a Compute Engine instance. You can select from various machine types. Cloud GPUs are a popular option for many deep learning applications, but CPUs can also effectively optimize LLMs. For this example, we train the model using a CPU.

Whatever type of machine you choose to use, if you encounter an out-of-memory error, try reducing parameters such as batch_size or seq_length .

⚠ **Note:** Costs are associated with starting up virtual machines. The total cost will depend on the type of machine and how long it is running. Regularly check your costs in the billing section of GCP and turn off your virtual machines when you're not using them.

💡 If you want to run the code in the section without spending much money, you can perform a few iterations of training on your virtual machine and then stop it.

Loading the Dataset

The quality of a model's output is directly tied to the quality of the data used for training. Start with a well-planned dataset, whether open-source or custom-created. We will use the dataset from the "LIMA: Less Is More for Alignment" paper. This research suggests that a small, meticulously selected dataset of a thousand samples could replace the RLHF strategy. This research is publicly available under a non-commercial use license.

The Deep Lake database infrastructure by Activeloop enables dataset streaming, eliminating the need to download and load the dataset into memory.

The code will create a loader object for the training and test sets:

```
import deeplake

# Connect to the training and testing datasets
ds = deeplake.load('hub://genai360/GAIR-lima-train-set')
```

```
ds_test = deeplake.load('hub://genai360/GAIR-lima-test-set')

print(ds)
```

```
    Dataset(path='hub://genai360/GAIR-lima-train-set', read_only=True, tensors=['answer', 'question', 'source']) The pretrained tokenizer object for the Open Pretrained Transformer (OPT) LLM is initially loaded using the transformers library, and the model is loaded later. We chose OPT for its open availability and relatively moderate parameter count. However, the code in this section is versatile and can be applied to other models. For example, you could use meta-llama/Llama-2-7b-chat-hf  for Llama 2.
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("facebook/opt-1.3b")
```

The prepare_sample_text processes a row of data stored in Deep Lake, organizing it to start with a question and follow with an answer, separated by two newlines. Defining a formatting function helps the model learn the template and recognize that a prompt beginning with the question keyword should typically be completed with an answer.

```
def prepare_sample_text(example):
"""Prepare the text from a sample of the dataset."""
text = f"""Question: {example['question'].text()}\n\nAnswer: {example['answer'].text()}"""

    return text
```

The Hugging Face TRL library integrates the SFT method and allows the integration of LoRA configurations, simplifying the implementation.

To prepare the training and evaluation datasets for finetuning, we use the ConstantLengthDataset object from the TLR library. This object requires a tokenizer, the Deep Lake dataset, and a formatting function, prepare_sample_text , which processes the data into the required format.

The seq_length parameter determines the maximum sequence length used during training and evaluation, aligning it with the model's configuration. While the value can be set as high as 2048 for models that can handle long contexts, we've selected 1024 in this example to optimize memory usage. You may increase this value depending on your dataset's text length and available resources.

Additionally, setting infinite=True ensures that once all data points are exhausted, the dataset iterator automatically restarts, making it ideal for

continual training scenarios.

```
from trl.trainer import ConstantLengthDataset

train_dataset = ConstantLengthDataset(
tokenizer,
ds,
formatting_func=prepare_sample_text,
infinite=True,
seq_length=1024
)

eval_dataset = ConstantLengthDataset(
tokenizer,
ds_test,
formatting_func=prepare_sample_text,
seq_length=1024
)

# Show one sample from train set
iterator = iter(train_dataset)
sample = next(iterator)
print(sample)
```

```
{'input_ids': tensor([ 2, 45641, 35, ..., 48443, 2517, 742]), 'labels': tensor([ 2, 45641, 35, ..., 48443, 2517, 742])}
```

The output shows that the `ConstantLengthDataset` class took care of all the necessary steps to prepare our dataset.

?? If you use the iterator to print a sample from the dataset, execute the following code to reset the iterator pointer: `train_dataset.start_iteration = 0`.

Configuring LoRA and Training Hyperparameters

Set the LoRA configuration using the PEFT library. The variable `r` indicates the dimension of matrices, where lower values mean fewer trainable parameters. `lora_alpha` acts as the scaling factor. The `bias` specifies which bias parameters should be trained with options like `none`, `all`, and `lora_only`.

```
from peft import LoraConfig

lora_config = LoraConfig(
r=16,
```

```
lora_alpha=32,
lora_dropout=0.05,
bias="none",
task_type="CAUSAL_LM",
)
```

Next, set the `TrainingArguments`. A higher learning rate combined with increased weight decay can enhance the finetuning performance. Additionally, it is good to use `bf16=True` as it can reduce memory usage during finetuning.

We also set up the Weights & Biases' tracking. This platform monitors and records every aspect of the process, including metrics like loss, accuracy, and gradients in real-time. By integrating it with the training script, you can visualize training curves, compare runs, and debug issues easily. It provides insights into the training process, helping you optimize hyperparameters and detect overfitting or underfitting early on.

To integrate this tool, install the package by running "`pip install wandb`" and use the `wandb` parameter in the `report_to` argument to manage the logging process effectively:

```
from transformers import TrainingArguments

training_args = TrainingArguments(
output_dir="./OPT-fine_tuned-LIMA-CPU",
dataloader_drop_last=True,
evaluation_strategy="epoch",
save_strategy="epoch",
num_train_epochs=10,
logging_steps=5,
per_device_train_batch_size=8,
per_device_eval_batch_size=8,
learning_rate=1e-4,
lr_scheduler_type="cosine",
warmup_steps=10,
gradient_accumulation_steps=1,
bf16=True,
weight_decay=0.05,
run_name="OPT-fine_tuned-LIMA-CPU",
```

```
report_to="wandb",
)
```

Load the pretrained `facebook/opt-1.3b` model. The model will be loaded using the transformers library.

```
from transformers import AutoModelForCausalLM
import torch

model = AutoModelForCausalLM.from_pretrained("facebook/opt-1.3b",
torch_dtype=torch.bfloat16)
```

The following code will loop over the model parameters, converting the data type of specified layers (such as LayerNorm and the final language modeling head) to a 32-bit format. This improves the stability of finetuning.

```
import torch.nn as nn

for param in model.parameters():
param.requires_grad = False # freeze the model - train adapters later
if param.ndim == 1:
# cast the small parameters (e.g. layernorm) to fp32 for stability
param.data = param.data.to(torch.float32)

model.gradient_checkpointing_enable() # reduce number of stored activations
model.enable_input_require_grads()

class CastOutputToFloat(nn.Sequential):
def forward(self, x): return super().forward(x).to(torch.float32)
model.lm_head = CastOutputToFloat(model.lm_head)
```

Use the `SFTTrainer` class to connect all components. The model, training arguments, training dataset, and LoRA configuration are needed to build the trainer object. The `packing` option indicates that we previously packed samples together using the `ConstantLengthDataset` class:

```
from trl import SFTTrainer

trainer = SFTTrainer(
model=model,
args=training_args,
train_dataset=train_dataset,
eval_dataset=eval_dataset,
peft_config=lora_config,
```

```
packing=True,
)
```

So, why did we use LoRA? Let's see how it helped by writing a simple function that computes the number of available parameters in the model and compares it to the trainable parameters. The trainable parameters are those that LoRA added to the underlying model.

```
def print_trainable_parameters(model):
"""
Prints the number of trainable parameters in the model.
"""
trainable_params = 0
all_param = 0
for , param in model.named_parameters():
all_param += param.numel()
if param.requires_grad:
trainable_params += param.numel()
print(
f"""trainable params: {trainable_params} || all params: {all_param} || trainable%: {100 *
trainable_params / all_param}"""
)

print(print_trainable_parameters(trainer.model))
```

```
trainable params: 3145728 || all params: 1318903808 || trainable%: 0.23851079820371554
```

The trainable parameters are limited to 3 million. Only 0.2% of the total parameters are updated leveraging LoRA. It drastically minimizes the amount of RAM required.

The trainer object is now ready to begin the finetuning cycle by invoking the `.train()` method: `print("Training...") trainer.train()` ?? You can access the OPT finetuned LIMA checkpoint on CPU at towardsai.net/book. Additionally, find more information on the Weights & Biases project page on the finetuning process at towardsai.net/book.

Merging the LoRA and OPT parameters

The final step is to merge the base model with the trained LoRA parameters, resulting in a standalone model.

If operating in a new environment, load the base OPT-1.3B model:

```
from transformers import AutoModelForCausalLM
import torch

model = AutoModelForCausalLM.from_pretrained(
"facebook/opt-1.3b", return_dict=True, torch_dtype=torch.bfloat16
)
```

Use `PeftModel` to load the finetuned model by specifying the checkpoint path:

```
from peft import PeftModel

# Load the Lora model
model = PeftModel.from_pretrained(
model,
"./OPT-fine_tuned-LIMA-CPU/<desired_checkpoint>/"
)
model.eval()
```

```
PeftModelForCausalLM(
(base_model): LoraModel(
(model): OPTForCausalLM(
(model): OPTModel(
(decoder): OPTDecoder(
(embed_tokens): Embedding(50272, 2048, padding_idx=1)
(embed_positions): OPTLearnedPositionalEmbedding(2050, 2048)
(final_layer_norm): LayerNorm((2048,), eps=1e-05, elementwise_affine=True)
(layers): ModuleList(
(0-23): 24 x OPTDecoderLayer(
(self_attn): OPTAttention(
(k_proj): Linear(in_features=2048, out_features=2048, bias=True)
(v_proj): Linear(
in_features=2048, out_features=2048, bias=True
(lora_dropout): ModuleDict(
(default): Dropout(p=0.05, inplace=False)
)
(lora_A): ModuleDict(
```

```
(default): Linear(in_features=2048, out_features=16, bias=False)
)
(lora_B): ModuleDict(
(default): Linear(in_features=16, out_features=2048, bias=False)
)
(lora_embedding_A): ParameterDict()
(lora_embedding_B): ParameterDict()
)
(q_proj): Linear(
in_features=2048, out_features=2048, bias=True
(lora_dropout): ModuleDict(
(default): Dropout(p=0.05, inplace=False)
)
(lora_A): ModuleDict(
(default): Linear(in_features=2048, out_features=16, bias=False)
)
(lora_B): ModuleDict(
(default): Linear(in_features=16, out_features=2048, bias=False)
)
(lora_embedding_A): ParameterDict()
(lora_embedding_B): ParameterDict()
)
(out_proj): Linear(in_features=2048, out_features=2048, bias=True)
)
(activation_fn): ReLU()
(self_attn_layer_norm): LayerNorm((2048,), eps=1e-05, elementwise_affine=True)
(fc1): Linear(in_features=2048, out_features=8192, bias=True)
(fc2): Linear(in_features=8192, out_features=2048, bias=True)
(final_layer_norm): LayerNorm((2048,), eps=1e-05, elementwise_affine=True)
)
)
)
)
(lm_head): Linear(in_features=2048, out_features=50272, bias=False)
)
)
) Combine the base model and LoRA layers using the PEFT model's .merge_and_unload()  method. The weights can be saved to disk with the .save_pretrained()  method.
model = model.merge_and_unload()

model.save_pretrained("./OPT-fine_tuned-LIMA/merged")
```

?? Note that the base model employed in this section is relatively small and has limited capabilities compared to state-of-the-art models, such as

ChatGPT. The process from this section can be easily applied to train larger LLMs.

Inference

The performance of the finetuned model can be estimated using a variety of prompts. The following code shows how to use Hugging Face's .generate() method for straightforward interactions with the model. Several other arguments and decoding strategies are available that can improve text generation quality. You can explore these techniques further in the blog post by Hugging Face (available at towardsai.net/book).

```
inputs = tokenizer("Question: Write a recipe with chicken.\n\n Answer: ",
return_tensors="pt")

generation_output = model.generate(**inputs,
return_dict_in_generate=True,
output_scores=True,
max_length=256,
num_beams=1,
do_sample=True,
repetition_penalty=1.5,
length_penalty=2.)

print(tokenizer.decode(generation_output['sequences'][0]))
```

> Question: Write a recipe with chicken.\n\n Answer: \n* Chicken and rice is one of the most popular meals in China, especially during Chinese New Year celebrations when it's served as an appetizer or main course for dinner parties (or just to eat by yourself). It can be made from scratch using fresh ingredients like meatballs/chicken breasts if you have them on hand but otherwise use frozen ones that are already cooked so they don't need any additional cooking time before serving. You could also substitute some vegetables instead such as broccoli florets which would make this dish even more delicious! If your family doesn't know how to cook well then I suggest making these recipes ahead of time because once done all you really do is reheat until hot again :)\n## Make homemade marinade\n1) Combine 1 tablespoon soy sauce, 2 tablespoons sesame oil, 3 teaspoons sugar, 4 cloves garlic minced into small pieces, 6-8 green onions chopped finely, 5 cups water, salt & pepper to taste, about 8 ounces boneless skinless chicken breast fillets cut up fine enough not to stick together while being mixed thoroughly - no bones needed here since there will only ever be two servings per person), ½ cup cornstarch dissolved in ¼...

Our experiment presented the same prompt to the original base model and the finetuned version. This evaluation aimed to compare each model's ability to follow instructions accurately. Find the list of prompts with corresponding answers below.

1. Create a marketing plan for a coffee shop.

Vanilla Model Output:

"</s>Question: Create a marketing plan for a coffee shop.\n\n Answer: "Do not make it more expensive than your competition, unless you wish to lose business." \nIs that even possible? I haven\'t seen the competition in my store and we\'ve only been here 2 years (although they are new). We keep ourselves busy just like everyone else there but if people can afford Starbucks at $4-5/person then wouldn\'t the market become saturated fast enough anyway no matter what brand or flavor you chose?\n"It\'s hard out\ 'un Texas...but somebody get me some money!" The quote came from a song by Celine Dion!</s>"

LoRA Model Output:

"</s>Question: Create a marketing plan for a coffee shop.\n\n Answer: \n## Plan the business model and pricing strategy to make money from your cafe or restaurant\n1) Set up an online store with all of its products on it, including cups/tumblers etc., so that customers can order them directly through their website without having to go into any physical stores (e-commerce). This will help you get more sales as will since people are less likely to visit brick & mortar shops if they have no way to buy what is being sold there in person! You could also set this up by using Amazon\'s FBA program which allows sellers like yourself who don\'t want to deal with inventory management themselves but still need some sort of distribution channel available where orders come straight off the shelf instead of going out to be picked at warehouses before shipping – just Google "Amazon fulfillment center" :). If you\'re not comfortable doing something similar then consider setting one up via Shopify too; ..."

2. Why does it rain? Explain your answer.

Vanilla Model Output:

"</s>Question: Why does it rain? Explain your answer.\n\n Answer: The moon's gravity is a lot less than earths and so the sky never really gets dark, but rather stays mostly clear (which makes things easier because it can be difficult to see stars from below). Even when there are clouds in sight of another planet like Mars or Jupiter then you will still have plenty if light on that other planets surface regardless due to its size relative star density.\nWhy do cows breathe air too though!? My Knowledge might be off base here…..but seriously why??? Is this how animals die…by suffocation?? Would an animal take such risks without having been programmed with those behaviours for survival?! Does science just not care about life anymore!!!!!????\nWell they're mammals after all!! They evolved *just* like us! We don't understand them very well now :(I would wager most humans didn't even know we had one until fairly recent times either :P Though obviously I'm sure some people knew somethi…'

LoRA Model Output:

"</s>Question: Why does it rain? Explain your answer.\n\n Answer: \n* The Earth\'s atmosphere is a giant, complex system that has many different layers and processes at work to keep the planet wet (and dry). Rain falls on land because of gravity waves from other planets in our solar system or even distant galaxies; these gravitational forces are so strong they can push water molecules into clouds where precipitation occurs. Water droplets then fall as snowflakes onto surfaces like mountains, which act much more slowly than air currents do but still move enough for them all to be frozen solid before falling back down again when spring arrives. This process repeats itself over time until there isn\'t any moisture left anywhere except under ice sheets covering Antarctica. When this happens, we call winter “the end”-— no longer will you see anything growing outside during summer months! But if you look closely, some plants grow underground instead—they\'re called mosses since their roots ..."

3. What’s the Italian translation of the word “house”?

Vanilla Model Output:

"</s>Question: What\'s the Italian translation of the word \'house\'?.\n\n Answer: --It means nothing, really! It is just someone who owns houses. For example you can say "the house where I live" or something like that»\nThe meaning should be a home for people in Italy as well? :P (Italian) Edit: not sure it quite fits though... ;)</s>"

LoRA Model Output:

"</s>Question: What's the Italian translation of the word \'house\'?.\n\n Answer: \n* "House" is a noun, which means an enclosed space or building in English (and other languages). The verb form for house can be either singular ("I live at home") and plural (*) ("We have houses"), but it usually refers to one person living there alone; e.g., I am my own landlord because we don\'t share any property with eachother. In some cases you may also see * (noun), like when someone says they are their parents\' only child - that would mean both parentage as well as residence together on this planet! For example, if your mother lives next door while her father works out-of town all day long she could say something along these lines about herself being his sole tenant/owner without having to specify who owns what piece of real estate between them...\nIn general though, people tend not to use words such as apartment / flat etc.; instead using terms more commonly used by Americans including single family homes & multi unit dwellings. This includes The results show the limitations and strengths of both models. The LoRA-tuned model generates more coherent, structured responses compared to the vanilla model, which often provides irrelevant answers. The LoRA model tends to follow the prompt more accurately, offering step-by-step guidance (e.g., creating a marketing plan) and correct interpretations (e.g., the Italian translation of “house”). However, it occasionally adds extraneous or overly detailed information, making responses verbose. The vanilla model, while shorter, often lacks relevance and logical flow, producing confusing responses making it less reliable. Overall, the LoRA model delivers more usable and intelligible answers. This improvement would likely be more significant with finetuning larger LLMs.

Tutorial 2: Using SFT and LoRA for Financial Sentiment

- Find the Notebook for this section at towardsai.net/book.

We aim to fine-tune an LLM for conducting **sentiment analysis on financial statements**. The LLM would categorize financial tweets as positive, negative, or neutral. The FinGPT project manages the dataset used in this tutorial. The dataset is a crucial element in this process.

A detailed script for implementation and experimentation is included at the end of this chapter.

In this tutorial, we'll fine-tune an LLM on a CPU. This is doable with some specific CPUs with optimizations specifically for common operations used in AI, such as Intel's Xeon CPUs with the use of Intel Advanced Matrix Extensions (AMX). For this tutorial, we create a Compute Engine virtual machine with 64 GB of RAM (as we'll fine-tune the model on CPU) and the previously mentioned CPUs. Both finetuning and inference can be accomplished by leveraging its optimization technologies.

⚠ It's important to be aware of the costs associated with virtual machines. The total cost will depend on the machine type and the instance's uptime. Regularly check your costs in the billing section of GCP and spin off your instances when you don't use them.

?? If you want to run the code in the section without spending much money, you can perform a few iterations of training on your virtual machine and then **stop it**.

Loading the Dataset

The FinGPT sentiment dataset includes a collection of financial tweets and their associated labels. The dataset also features an instruction column, typically containing a prompt such as "What is the sentiment of the following content? Choose from Positive, Negative, or Neutral."

We use a smaller subset of the dataset from the Deep Lake database for practicality and efficiency, which already hosts the dataset on its hub.

Use the deeplake.load() function to create the Dataset object and load the samples:

```
import deeplake

# Connect to the training and testing datasets
ds = deeplake.load('hub://genai360/FingGPT-sentiment-train-set')
ds_valid = deeplake.load('hub://genai360/FingGPT-sentiment-valid-set')

print(ds)
```

> Dataset(path='hub://genai360/FingGPT-sentiment-train-set', read_only=True, tensors=['input', 'instruction', 'output']) Now, develop a function to format a dataset sample into an appropriate input for the model. Unlike previous methods, this approach includes the instructions at the beginning of the prompt.

The format is: <instruction>\n\nContent: <tweet>\n\nSentiment: <sentiment>. The placeholders within <> will be replaced with relevant values from the dataset.

```
def prepare_sample_text(example):
"""Prepare the text from a sample of the dataset."""
text = f"""{example['instruction'].text()}\n\nContent: {example['input'].text()}\n\nSentiment: {example['output'].text()}"""
return text Here is a formatted input derived from an entry in the dataset: What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}

Content: Diageo Shares Surge on Report of Possible Takeover by Lemann

Sentiment: positive
```

Initialize the OPT-1.3B language model tokenizer and use the ConstantLengthDataset class to create the training and validation dataset. It aggregates multiple samples until a set sequence length threshold is met, improving the efficiency of the training process.

```
# Load the tokenizer
from transformers import AutoTokenizer
tokenizer = AutoTokenizer.from_pretrained("facebook/opt-1.3b")
```

```
# Create the ConstantLengthDataset
from trl.trainer import ConstantLengthDataset

train_dataset = ConstantLengthDataset(
tokenizer,
ds,
formatting_func=prepare_sample_text,
infinite=True,
seq_length=1024
)

eval_dataset = ConstantLengthDataset(
tokenizer,
ds_valid,
formatting_func=prepare_sample_text,
seq_length=1024
)

# Show one sample from train set
iterator = iter(train_dataset)
sample = next(iterator)
print(sample)
```

```
{'input_ids': tensor([50118, 35212, 8913, ..., 2430, 2, 2]),
'labels': tensor([50118, 35212, 8913, ..., 2430, 2, 2])}
```

?? Before launching the training process, execute the following code to reset the iterator pointer if the iterator is used to print a sample from the dataset: `train_dataset.start_iteration = 0` .

Initializing the Model and Trainer

Create a `LoraConfig` object. The `TrainingArguments` class from the transformers library manages the training loop:

```
# Define LoRAConfig
from peft import LoraConfig

lora_config = LoraConfig(
r=16,
lora_alpha=32,
lora_dropout=0.05,
bias="none",
task_type="CAUSAL_LM",
)
```

```
# Define TrainingArguments
from transformers import TrainingArguments

training_args = TrainingArguments(
output_dir="./OPT-fine_tuned-FinGPT-CPU",
dataloader_drop_last=True,
evaluation_strategy="epoch",
save_strategy="epoch",
num_train_epochs=10,
logging_steps=5,
per_device_train_batch_size=12,
per_device_eval_batch_size=12,
learning_rate=1e-4,
lr_scheduler_type="cosine",
warmup_steps=100,
gradient_accumulation_steps=1,
gradient_checkpointing=False,
fp16=False,
bf16=True,
weight_decay=0.05,
ddp_find_unused_parameters=False,
run_name="OPT-fine_tuned-FinGPT-CPU",
report_to="wandb",
)
```

Load the OPT-1.3B model in the `bfloat16` format to save on memory:

```
from transformers import AutoModelForCausalLM
import torch

model = AutoModelForCausalLM.from_pretrained(
"facebook/opt-1.3b", torch_dtype=torch.bfloat16
)
```

Next, we cast particular layers inside the network to complete 32-bit precision. This improves the model's stability during training.

💡 "Casting" layers in a model typically refers to changing the data type of the elements within the layers. For example, here, we change them to float

32 for improved precision.

```
from torch import nn

for param in model.parameters():
param.requires_grad = False # freeze the model - train adapters later
if param.ndim == 1:
# cast the small parameters (e.g. layernorm) to fp32 for stability
param.data = param.data.to(torch.float32)

model.gradient_checkpointing_enable() # reduce number of stored activations
model.enable_input_require_grads()

class CastOutputToFloat(nn.Sequential):
def forward(self, x): return super().forward(x).to(torch.float32)
model.lm_head = CastOutputToFloat(model.lm_head)
```

Connect the model, dataset, training arguments, and LoRA configuration using the SFTTrainer class. To launch the training process, call the .train() function:

```
from trl import SFTTrainer

trainer = SFTTrainer(
model=model,
args=training_args,
train_dataset=train_dataset,
eval_dataset=eval_dataset,
peft_config=lora_config,
packing=True,
)

print("Training...")
```

trainer.train() ?? Access the best OPT finetuned FinGPT with CPU checkpoint at towardsai.net/book. Additionally, find more information on the Weights & Biases project page on the finetuning process at towardsai.net/book.

Merging LoRA and OPT

Load and merge the LoRA adaptors from the previous stage with the base model:

```
# Load the base model (OPT-1.3B)
from transformers import AutoModelForCausalLM
import torch

model = AutoModelForCausalLM.from_pretrained(
"facebook/opt-1.3b", return_dict=True, torch_dtype=torch.bfloat16
)

# Load the LoRA adaptors
from peft import PeftModel

# Load the Lora model
model = PeftModel.from_pretrained(model,
"./OPT-fine_tuned-FinGPT-CPU/<desired_checkpoint>/")
model.eval()
model = model.merge_and_unload()

# Save for future use
model.save_pretrained("./OPT-fine_tuned-FinGPT-CPU/merged")
```

Inference

We randomly picked four previously unseen cases from the dataset and fed them into the vanilla base model (OPT-1.3B) and the finetuned model using the transformers library's .generate() method:

```
inputs = tokenizer("""What is the sentiment of this news? Please choose an answer from {strong negative/moderately negative/mildly negative/neutral/mildly positive/moderately positive/strong positive}, then provide some short reasons.\n\n
Content: UPDATE 1-AstraZeneca sells rare cancer drug to Sanofi for up to $300 mln.\n\nSentiment: """, return_tensors="pt").to("cuda:0")

generation_output = model.generate(**inputs,
return_dict_in_generate=True,
output_scores=True,
max_length=256,
num_beams=1,
do_sample=True,
```

```
repetition_penalty=1.5,
length_penalty=2.)

print(tokenizer.decode(generation_output['sequences'][0]))
```

> What is the sentiment of this news? Please choose an answer from {strong negative/moderately negative/mildly negative/neutral/mildly positive/moderately positive/strong positive}, then provide some short reasons. Content: UPDATE 1-AstraZeneca sells rare cancer drug to Sanofi for up to S300 mln. Sentiment: positive

The samples show that the model finetuned on financial tweets performs well. It follows instructions and understands the task at hand. The base model focuses mainly on the default language modeling goal of predicting the next word based on the input.

You can find a collection of prompts and their outputs below.

1. UPDATE 1-AstraZeneca sells rare cancer drug to Sanofi for up to S300 mln. [Correct Answer: Positive]

Vanilla Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\nn\nContent: UPDATE 1-AstraZeneca sells rare cancer drug to Sanofi for up to $300 mln.\n\nSentiment:??????????????(0 votes - average) (100% rating, 12 comments)Rate Article | Share on FacebookTweetThis one looks good and they have a product that can be used in combination with chemo so maybe it will help their patients get rid if these cancers as well as helping them live longer. This doesn't sound like some crap made by Astra at first glance. What do ya'll think about what was announced today though? Do you agree or disagree with these statements??? Also tell us your thoughts below :-)Thanks!Ravi Posted 06.04 2014 1433 words 3+ Comments Upvote 10 downvotes Content Rating = 100% Vote [Article] More content... View vote results page : Popularity Ratio 0.0075962 Shares User Login Signup Google Plus Email / Password Not logged into LinkedIn Join now Comment! A survey just came out compar...'

Finetuned Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\nn\nContent: UPDATE 1-AstraZeneca sells rare cancer drug to Sanofi for up to $300 mln.\n\nSentiment: positive</s>'

2. SABMiller revenue hit by weaker EM currencies [Correct Answer: Negative]

Vanilla Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\nn\nContent: SABMiller revenue hit by weaker EM currencies\n\nSentiment:???????? or Neutral??\N|B#\n\n[Date] : 12 May 2004 @ 20.00 ET (0850 UTC)\nIn response to a letter published in The Australian today, I can confirm that we have made changes at Millicent for three reasons; Firstly Australia has become one large market since October 2003's sales update and second it contains all non-core brands now making up most marketing spend...The third significant action was removing our international unit – Australasia & New Zealand Marketing Services [AU&NZSM]. This department had been under pressure before with reduced account allocations but then we realised there were some core issues within AU & NZ so began focusing on these. In turn reducing media budgets significantly whilst seeking increased efficiencies elsewhere...Millisternationals continues tto invest heavily into South American markets ... '

Finetuned Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\nn\nContent: SABMiller revenue hit by weaker EM currencies\n\nSentiment: negative</s>'

3. Buffett's Company Reports 37 Percent Drop in 2Q Earnings [Correct Answer: Negative]

Vanilla Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\n\nContent: Buffett\'s Company Reports 37 Percent Drop in 2Q Earnings\n\nSentiment: -5% negative, 20 positives and 12 neutral. Negative responses were most frequent on stocks with either very high revenue estimates or sales that are expected to grow faster than earnings for many years." - 1"The stock market will continue its upward momentum throughout 2008", was also found by 3 out 4 traders (see tables below). "Many companies reporting solid results have had excellent year after quarter.", according 15 trader(note not included) replies... These answers all suggest a belief about future trading conditions based only upon information available since last weeks release which may be distorted due to the markets down session following these trades.(1), (1, 8")I did NOT see anything indicating higher short interest, but I could imagine some investors looking at what happened when they started long...'

Finetuned Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\n\nContent: Buffett's Company Reports 37 Percent Drop in 2Q Earnings\n\nSentiment: negative</s>'

4. For a few hours this week, the FT gained access to Poly, the university where students in Hong Kong have been trapped… [Correct Answer: Neutral]

Vanilla Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\n\nContent: For a few hours this week, the FT gained access to PolyU, the university where students in Hong Kong have been trapped... https://t.co/IpIzUTCujW\n\nSentiment: (Negative) - many others are saying positive things or that they didn’t know it existed... “the world's richest woman — Li Ka-shing -- donated $2 million and the top job at the University”! A small group has even gotten together on Facebook to discuss how best we can help her with donations... but there isn't much more done than making those posts; most members want nothing political about donation giving since she would already be doing so anyway given connections among HKers who support pro independence politicians like Chow Kwing Chung II etc.. If everyone wants something politically charged as far you could tell by people commenting for / against anything mentioned then I guess everybody will just assume otherwise if so...'

Finetuned Model Output:

'</s>What is the sentiment of this news? Please choose an answer from {negative/neutral/positive}\n\n\nContent: For a few hours this week, the FT gained access to PolyU, the university where students in Hong Kong have been trapped... https://t.co/IpIzUTCujW\n\nSentiment: positive</s>'

Tutorial 3: FineTuning a Cohere LLM with Medical Data

- Find the dataset preparation Notebook and the finetuning Notebook for this section at towardsai.net/book.

Using a proprietary model simplifies the finetuning process by simply supplying sample inputs and outputs, with the platform managing the actual finetuning. For example, in a classification model, a typical input would be a pair of <text, label>.

Cohere offers a range of specialized models for specific use cases and different functions, including rerank, embedding, and chat, all accessible via APIs. Users can create custom models for three primary objectives: 1) Generative tasks where the model produces text output, 2) Classification tasks where the model categorizes text, and 3) Rerank tasks to improve semantic search results.

In this tutorial, we are finetuning a proprietary LLM developed by Cohere for medical text analysis to perform Named Entity Recognition (NER). NER enables models to recognize multiple entities in text, such as names,

locations, and dates. We will fine-tune a model to extract information about diseases, substances, and their interactions from medical paper abstracts.

Cohere API

The Cohere platform provides a selection of base models designed for different purposes. You can choose between base models with quicker performance or command models with more advanced capabilities for generative tasks. Each type also has a "light" version for additional flexibility.

Create an account on their platform to use the Cohere API at dashboard.cohere.com. Navigate to the "API Keys" section to obtain a Trial key, which allows free usage with certain rate limitations. This key is not for production environments but offers an excellent opportunity to experiment with the models before using them for production.

Install the Cohere Python SDK to access their API: `pip install cohere` Build a Cohere object with your API key and a prompt to generate a response to your request. You can use the code below but change the API placeholder with your key:

```
import cohere

co = cohere.Client("<API_KEY>")

prompt = """The following article contains technical terms including diseases, drugs and chemicals. Create a list only of the diseases mentioned.

Progressive neurodegeneration of the optic nerve and the loss of retinal ganglion cells is a hallmark of glaucoma, the leading cause of irreversible blindness worldwide, with primary open-angle glaucoma (POAG) being the most frequent form of glaucoma in the Western world. While some genetic mutations have been identified for some glaucomas, those associated with POAG are limited and for most POAG patients, the etiology is still unclear. Unfortunately, treatment of this neurodegenerative disease and other retinal degenerative diseases is lacking. For POAG, most of the treatments focus on reducing aqueous humor formation, enhancing uveoscleral or conventional outflow, or lowering intraocular pressure through surgical
```

```
means. These efforts, in some cases, do not always lead to a prevention of vision loss and therefore other strategies are needed to reduce or reverse the progressive neurodegeneration. In this review, we will highlight some of the ocular pharmacological approaches that are being tested to reduce neurodegeneration and provide some form of neuroprotection.

List of extracted diseases:"""

response = co.generate(
model='command',
prompt = prompt,
max_tokens=200,
temperature=0.750)

base_model = response.generations[0].text

print(base_model)
```

```
- glaucoma
- primary open-angle glaucoma The code uses the cohere.Client()  method to input your API key. The above code also defines the prompt variable, which will contain instructions for the model.
```

The model's objective for this experiment is to analyze a scientific paper's abstract from the PubMed website and identify a list of diseases. The `cohere` object's `.generate()` method specifies the model type and provides the prompts and control parameters to achieve this.

The `max_tokens` parameter sets the limit for the number of new tokens the model can generate, and the `temperature` parameter controls the randomness level in the output.

The command model can identify diseases without examples or supplementary information.

The Dataset

We will use the BC5CDR or BioCreative V Chemical Disease Relation Data. It comprises 1,500 manually annotated PubMed research papers, providing structured information on chemical-disease relations. The dataset is divided into training, validation, and testing sets containing 500 papers.

With this experiment, we aim to fine-tune a model that can identify and extract names of diseases/chemicals and their relationships from text. While research papers often describe relationships between chemicals and diseases in their abstracts, this information is typically unstructured. Manually finding “all chemicals influencing disease X” would require reading all papers mentioning “disease X.” Accurately extracting this structured information from unstructured text would facilitate more efficient searches.

Preprocess the dataset to adapt it for the Cohere service. It handles three file formats: CSV, JSONL, and plain text. We will use the JSONL format, which is consistent with the template below: {"prompt": "This is the first prompt",
"completion": "This is the first completion"}
{"prompt": "This is the second prompt",
"completion": "This is the second completion"}

?? The code is an example showing the extraction of disease names. Our final dataset will contain diseases, chemicals, and their corresponding relationships. We present a single step for extracting disease names to minimize the repetition of code. Please refer to the notebook at towardsai.net/book for the complete preprocessing procedure and the resulting dataset.

Download the dataset in JSON format from towardsai.net/book and open the JSON file using the code below. We also display a single row (passage) from the dataset to better illustrate the content and help understand the process. Each entry includes a text (which may be either a title or an abstract) and a list of entities that can be classified as either chemicals or diseases. For instance, in the example provided below, the first entity, Naloxone, is recognized as a chemical. The subsequent code will focus only on the information from the abstract, as the titles are short and provide limited details. (The printed output is simplified to improve understanding of the dataset and exclude non-essential information.)

```
with open('bc5cdr.json') as json_file:
data = json.load(json_file)

print(data[0])
```

{'passages':
[{'document_id': '227508',
'**type**': 'title',
'**text**': 'Naloxone reverses the antihypertensive effect of clonidine.',
'**entities**': [
{'id': '0', 'text': ['Naloxone'], 'type': 'Chemical',},
{'id': '1', 'text': ['clonidine'], 'type': 'Chemical'}],
'relations': [...]},
{'document_id': '227508',
'**type**': 'abstract',
'**text**': 'In unanesthetized, spontaneously hypertensive rats the decrease in blood pressure and heart rate produced by intravenous clonidine, 5 to 20 micrograms/kg, was inhibited or reversed by nalozone, 0.2 to 2 mg/kg. The hypotensive effect of 100 mg/kg alpha-methyldopa was also partially reversed by naloxone. Naloxone alone did not affect either blood pressure or heart rate. In brain membranes from spontaneously hypertensive rats clonidine, 10(-8) to 10(-5) M, did not influence stereoselective binding of [3H]-naloxone (8 nM), and naloxone, 10(-8) to 10(-4) M, did not influence clonidine-suppressible binding of [3H]-dihydroergocryptine (1 nM). These findings indicate that in spontaneously hypertensive rats the effects of central alpha-adrenoceptor stimulation involve activation of opiate receptors. As naloxone and clonidine do not appear to interact with the same receptor site, the observed functional antagonism suggests the release of an endogenous opiate by clonidine or alpha-methyldopa and the possible role of the opiate in the central control of sympathetic tone.',
'**entities**': [
{'id': '2', 'text': ['hypertensive'], 'type': 'Disease'},
{'id': '3', 'text': ['clonidine'], 'type': 'Chemical'},
{'id': '4', 'text': ['nalozone'], 'type': 'Chemical'},
{'id': '5', 'text': ['hypotensive'], 'type': 'Disease'},
{'id': '6', 'text': ['alpha-methyldopa'], 'type': 'Chemical'},
{'id': '7', 'text': ['naloxone'], 'type': 'Chemical'},
{'id': '8', 'text': ['Naloxone'], 'type': 'Chemical'},
{'id': '9', 'text': ['hypertensive'], 'type': 'Disease'},
{'id': '10', 'text': ['clonidine'], 'type': 'Chemical'},
{'id': '11', 'text': ['[3H]-naloxone'], 'type': 'Chemical'},
{'id': '12', 'text': ['naloxone'], 'type': 'Chemical'},
{'id': '13', 'text': ['clonidine'], 'type': 'Chemical'},
{'id': '14', 'text': ['[3H]-dihydroergocryptine'], 'type': 'Chemical'},
{'id': '15', 'text': ['hypertensive'], 'type': 'Disease'},
{'id': '16', 'text': ['naloxone'], 'type': 'Chemical',},
{'id': '17', 'text': ['clonidine'], 'type': 'Chemical'},
{'id': '18', 'text': ['clonidine'], 'type': 'Chemical'},
{'id': '19', 'text': ['alpha-methyldopa'], 'type': 'Chemical'}],
'relations': [...]}],
'dataset_type': 'train'}

We can loop through the dataset, extracting abstracts and related entities while including training instructions. There are two sets of instructions: the

first helps the model understand the job, and the second shows how to construct the response: `instruction = "The following article contains technical terms including diseases, drugs and chemicals. Create a list only of the diseases mentioned.\n\n"` `output_instruction = "\n\nList of extracted diseases:\n"`

The `instruction` variable sets the rules, and the `output_instruction` specifies the intended format for the output. Now, cycle through the dataset and format each instance:

```
the_list = []

for item in data:
dis = []

    if item['dataset_type'] != "test": continue; # Don't use test set
# Extract the disease names
for ent in item['passages'][1]['entities']: # The annotations
if ent['type'] == "Disease": # Only select disease names
if ent['text'][0] not in dis: # Remove duplicate diseases in a text
dis.append(ent['text'][0])

    the_list.append(
{'prompt': instruction +
item['passages'][1]['text'] +
output_instruction,
'completion': "- "+ "\n- ".join(dis)}
)
```

Preparing each sample from the dataset requires iterating through all annotations and selecting only those related to diseases. This is necessary because the dataset also contains additional chemical labels. This will result in a dictionary with two keys: `prompt` and `completion`. The `prompt` key will consist of the paper abstract combined with specific instructions, and the `completion` key will list each disease name on a separate line.

This code will convert and save the dataset in JSONL format: `# Writing to sample.json`

```
with open("disease_instruct_all.jsonl", "w") as outfile:
for item in the_list:
```

`outfile.write(json.dumps(item) + "\n")` The processed dataset is saved in a file named `disease_instruct_all.jsonl` . This file combines the training and validation sets to create 1,000 samples. The complete dataset comprises 3,000 samples, divided into three categories: 1,000 for diseases, 1,000 for chemicals, and 1,000 for their relationships.

?? The link to the final preprocessed dataset is accessible at towardsai.net/book.

FineTuning

The Cohere platform offers advanced options for extending the training duration or adjusting the learning rate. Refer to their guide on Training Custom Models for a comprehensive understanding.

Navigate to the models' page on the sidebar and click the "Create a custom model" button. You will be prompted to select the model type; for this example, choose the `Generate` option.

Next, upload the dataset from the previous step or a custom dataset. Click the "Review data" button to preview a few samples from the dataset. This ensures that the platform correctly interprets your data. If everything looks correct, proceed by clicking the "Continue" button.

Next, choose a nickname for your model. You can also modify training hyperparameters by clicking the "HYPERPARAMETERS (OPTIONAL)" link. Options include `train_steps` for training duration, `learning_rate` for adjusting how quickly the model adapts, and `batch_size` for the number of samples processed in each iteration. While the default parameters are generally effective, you can experiment with this.

Once you're ready, click "Initiate training." Cohere will email you to notify you that the finetuning process is complete and provide you with the model ID for use in your API.

Extract Disease Names

Use the previously created prompt, but with the model ID of the network, we just finetuned:

```
response = co.generate(
model='2075d3bc-eacf-472e-bd26-23d0284ec536-ft',
prompt=prompt,
max_tokens=200,
temperature=0.750)

disease_model = response.generations[0].text

print(disease_model)
```

```
- neurodegeneration
- glaucoma
- blindness
- POAG
- glaucomas
- retinal degenerative diseases
- neurodegeneration
- neurodegeneration The results show that the model can now recognize a wide range of diseases, demonstrating the effectiveness of the finetuning method.
```

Extract Chemical Names

We also compared the performance of the baseline model with the finetuned model in extracting chemical names. We will only show each model's prompt and output to avoid unnecessary code mentions. We used the following prompt to extract information from a text in the test set:

```
prompt = """The following article contains technical terms including diseases, drugs and chemicals. Create a list only of the chemicals mentioned.

To test the validity of the hypothesis that hypomethylation of DNA plays an important role in the initiation of carcinogenic process, 5-azacytidine (5-AzC) (10 mg/kg), an inhibitor of DNA methylation, was given to rats during the phase of repair synthesis induced by the three carcinogens, benzo[a]-pyrene (200 mg/kg), N-methyl-N-nitrosourea (60 mg/kg) and 1,2-dimethylhydrazine (1,2-DMH) (100 mg/kg). The initiated hepatocytes in the liver were assayed as the gamma-glutamyltransferase (gamma-GT) positive foci formed following a 2-week selection regimen consisting of dietary 0.02% 2-acetylaminofluorene coupled with a necrogenic dose of CCl4. The results obtained indicate that with all three carcinogens, administration of 5-AzC during repair synthesis increased the incidence of initiated hepatocytes, for example 10-20 foci/cm2 in 5-AzC and carcinogen-treated rats compared with 3-5 foci/cm2 in rats treated with carcinogen only. Administration of [3H]-5-azadeoxycytidine during the repair synthesis induced by 1,2-DMH further
```

showed that 0.019 mol % of cytosine residues in DNA were substituted by the analogue, indicating that incorporation of 5-AzC occurs during repair synthesis. In the absence of the carcinogen, 5-AzC given after a two thirds partial hepatectomy, when its incorporation should be maximum, failed to induce any gamma-GT positive foci. The results suggest that hypomethylation of DNA per se may not be sufficient for initiation. Perhaps two events might be necessary for initiation, the first caused by the carcinogen and a second involving hypomethylation of DNA.

List of extracted chemicals:"""

The output of the base model:

- 5-azacytidine (5-AzC)
- benzo[a]-pyrene
- N-methyl-N-nitrosourea
- 1,2-dimethylhydrazine
- CCl4
- 2-acetylaminofluorene The output of the custom finetuned model: - 5-azacytidine
- 5-AzC
- benzo[a]-pyrene
- N-methyl-N-nitrosourea
- 1,2-dimethylhydrazine
- 1,2-DMH
- 2-acetylaminofluorene
- CCl4
- [3H]-5-azadeoxycytidine
- cytosine The custom model is better for our specific task and adapts readily based on the samples.

Extract Relations

Here, the model will extract relationships between chemicals and the diseases they affect. It is a complex task that may present difficulties for the base model.

Introduce the prompt from the test set:

prompt = """The following article contains technical terms including diseases, drugs and chemicals. Create a list only of the influences between the chemicals and diseases mentioned.

The yield of severe cirrhosis of the liver (defined as a shrunken finely nodular liver with micronodular histology, ascites greater than 30 ml, plasma albumin less than 2.2 g/dl, splenomegaly 2-3 times normal, and testicular atrophy approximately half normal weight) after 12 doses of carbon tetrachloride given intragastrically in the phenobarbitone-primed rat was increased from 25% to 56%

by giving the initial "calibrating" dose of carbon tetrachloride at the peak of the phenobarbitone-induced enlargement of the liver. At this point it was assumed that the cytochrome P450/CCl4 toxic state was both maximal and stable. The optimal rat size to begin phenobarbitone was determined as 100 g, and this size as a group had a mean maximum relative liver weight increase 47% greater than normal rats of the same body weight. The optimal time for the initial dose of carbon tetrachloride was after 14 days on phenobarbitone.

List of extracted influences:"""

The output generated by the base model: severe cirrhosis of the liver influences shrinking, finely nodular, ascites, plasma albumin, splenomegaly, testicular atrophy, carbon tetrachloride, phenobarbitone The output generated by the custom model: - Chemical phenobarbitone influences disease cirrhosis of the liver
- Chemical carbon tetrachloride influences disease cirrhosis of the liver The base model tries to establish links within the text, but the custom finetuned model delivers well-formatted output, linking each chemical to the appropriate disease. This task is difficult for a general-purpose model, but it demonstrates finetuning efficiency with just a few thousand samples of the task.

Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback (RLHF) is a technique introduced by OpenAI that combines human feedback with reinforcement learning to enhance the alignment and performance of LLMs. This method has been instrumental in improving the safety and utility of LLMs.

RLHF was first applied to InstructGPT, a version of GPT-3 finetuned to follow instructions. Now, it is used in the latest OpenAI models, ChatGPT, and other state-of-the-art systems.

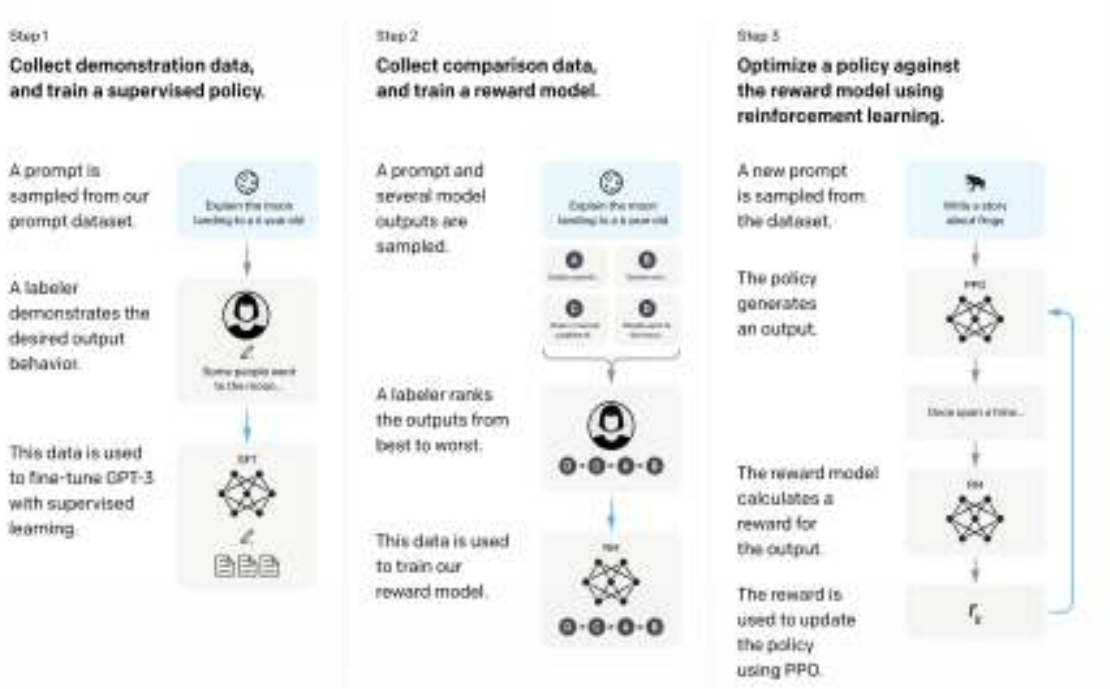
The process involves using human-curated preferences to guide the model toward preferred outputs, thus promoting the generation of responses that are more accurate, secure, and in line with human expectations. This is achieved using a reinforcement learning (RL) algorithm called Proximal

Policy Optimization (PPO), which refines the LLM based on these human rankings. RLHF guides LLMs in generating appropriate texts by framing text generation as a reinforcement learning problem. In this setup, the language model acts as the RL agent, its potential language outputs constitute the action space, and the reward depends on the alignment of the LLM's response with the application's context and the user's intent.

The RLHF process begins with training a large language model (LLM) on a vast text corpus from the internet. In some cases, the pretrained LLM undergoes an optional finetuning step, where it is trained on a specialized dataset to help the subsequent reinforcement learning phase converge more quickly.

Next, the RLHF dataset is created by having the LLM generate multiple text completions for a series of instructions. Human evaluators then rank these completions based on factors like completeness, relevancy, accuracy, toxicity, and bias. These rankings are translated into scores, with higher scores representing better completions.

A reward model is then trained using this dataset, learning to score the completions in a way that mirrors human judgment. With this reward model in place, the LLM is finetuned using reinforcement learning. For each random instruction, the model generates a completion, which is then scored by the reward model. A reinforcement learning algorithm (PPO) uses these scores to adjust the LLM's parameters, increasing the likelihood of higher-scoring completions. Throughout this process, a small Kullback-Leibler (KL) divergence is maintained between the finetuned and original LLM to preserve valuable information and ensure consistency. After several iterations, this results in a refined and improved LLM.



Visual illustration of RLHF. From the "Open AI" blog.

It is possible to align LLMs to follow instructions with human values with SFT (with or without LoRA) with a high-quality dataset (see the LIMA paper, "LIMA: Less Is More for Alignment").

So, what's the trade-off between RLHF and SFT? In reality, it's still an open question. Empirically, RLHF can better align the LLM if the dataset is sufficiently large and high-quality. However, it's more expensive and time-consuming. Additionally, reinforcement learning is still quite unstable, meaning that the results are very sensitive to the initial model parameters and training hyperparameters. It often falls into local optima, and the loss diverges multiple times, requiring multiple restarts. This makes it less straightforward than plain SFT.

Alternatives to RLHF

Direct Preference Optimization (DPO), an alternative to RLHF, is a relatively new method for finetuning language models.

Unlike RLHF, which requires complex reward functions and a delicate balance for effective text generation, DPO employs a more straightforward

approach. It optimizes the language model directly using binary cross-entropy loss, avoiding the need for a separate reward model and the complexities of reinforcement learning-based optimization. This is achieved through an analytical conversion of the reward function into the optimal RL policy. The optimal RL policy transforms the RL loss, which typically incorporates the reward and reference models, into a loss over just the reference model.

As a result, DPO simplifies the finetuning process by removing the need for complicated RL approaches or a reward model.

Google DeepMind's **Reinforced Self-Training (ReST)** offers another cost-effective alternative to RLHF. The ReST algorithm operates in a repetitive cycle of two main phases. In the first phase, called the "Grow" phase, the model generates various output predictions for each context, which are then used to expand the training dataset. The second phase, the "Improve" phase, involves ranking and filtering this expanded dataset using a reward model based on human preferences. The model is then finetuned using an offline reinforcement learning objective. The enhanced model feeds into the next Grow phase, continuing the cycle.

ReST has several advantages over RLHF. It reduces computational demands by reusing the output from the Grow phase across multiple Improve steps, avoiding the high costs of online reinforcement learning. Additionally, the quality of the policy is not constrained by the original dataset, as new training data is drawn from an improved policy during the Grow phase. The clear separation between the Grow and Improve phases allows for easier examination of data quality and the identification of alignment issues like reward hacking. Overall, ReST is a stable and straightforward approach, requiring minimal hyperparameter tuning.

Reinforcement Learning from AI Feedback (RLAIF), a concept developed by Anthropic, is another alternative to RLHF. RLAIF specifically aims to mitigate some of RLHF's challenges, like subjectivity and limited scalability of human feedback.

RLAIF uses an AI Feedback Model for training feedback rather than relying on human input. This model operates under guidelines set by a human-created constitution, which outlines fundamental principles for the model's

evaluations. This method enables a more scalable supervision approach, moving away from the constraints of human preference-based feedback.

RLAIF generates a ranked preference dataset using the AI Feedback Model. This dataset is used to train a Reward Model similar to RLHF. The Reward Model subsequently acts as the reward indicator in a reinforcement learning framework for an LLM.

RLAIF is a viable option for finetuning safer and more efficient LLMs. It preserves the effectiveness of RLHF models while enhancing their safety, diminishes the influence of subjective human preferences, and offers greater scalability as a supervision method. On the downside, the feedback generated by the model is often of lower quality than human feedback.

A study conducted by Google demonstrated that RLAIF and RLHF are both preferred over standard SFT, with nearly identical favorability rates. This suggests that they could serve as feasible alternatives.

Tutorial 4: Improving LLMs with RLHF

Reinforcement Learning from Human Feedback (RLHF) incorporates human feedback into the training process through a reward model that learns the desired patterns to improve the model's output. For example, if the goal is to enhance politeness, the reward model will guide the model to generate more polite responses by assigning higher scores to polite outputs. This process is resource-intensive because it necessitates training a reward model using a dataset curated by humans.

This tutorial will use available open-source models and datasets whenever possible while maintaining costs.

We begin with a pretrained model that we fine-tune in a supervised finetuning phase using the `SFTTrainer` class. Next, a reward model is trained with the desired traits using the `RewardTrainer` class. Finally, the reinforcement learning phase employs the models to build the ultimate aligned model, utilizing the `PPOTrainer` .

You can access the reports generated from Weights & Biases and the file with the requirements for the library after each subsection. Note that different steps require distinct versions of libraries. We chose OPT-1.3B as the base model and finetuned a DeBERTa (300M) model as the reward model for our experiments. While these are more compact models, the process used in this tutorial can be applied to other existing models by simply modifying the model's name in the code.

Even if much more affordable than what companies like OpenAI do, this tutorial is still resource-intensive as we replicate an RLHF phase. We rented an 8x NVIDIA A100 instance for $8.80/h and used lambda as our GPU cloud provider.

⚠It's important to be aware of the costs associated with cloud GPUs. The total cost will depend on the machine type and the instance's uptime. Regularly check your costs in the billing section of Lambda Labs and spin off your instances when you don't use them.

?? If you want to run the code in the section without spending much money, you can perform a few iterations of training on your virtual machine and then stop it.

Supervised FineTuning

- Find the Notebook for this section at towardsai.net/book.

Previous sections covered the SFT phase. This section uses a unique OpenOrca dataset with question-response pairs and implements the QLoRA finetuning technique.

This phase teaches the model a conversational format, training it to provide answers rather than defaulting to its standard auto-completion function.

Install the required libraries with the command `!pip install -q transformers==4.32.0 bitsandbytes==0.41.1 accelerate==0.22.0 deeplake==3.6.19 trl==0.5.0 peft==0.5.0 wandb==0.15.8 .`

The Dataset

The first step is streaming the dataset. Streaming a dataset refers to the process of loading and processing data in smaller chunks or batches rather than loading the entire dataset into memory at once. This approach is useful when working with large datasets that cannot fit into memory or when you want to perform real-time processing. For this example, we only use a subset of the original dataset, comprising 1 million data points. However, you can access the entire dataset, containing 4 million data points, at towardsai.net/book.

```
import deeplake

# Connect to the training and testing datasets
ds = deeplake.load('hub://genai360/OpenOrca-1M-train-set')
ds_valid = deeplake.load('hub://genai360/OpenOrca-1M-valid-set')

print(ds)
```

```
Dataset(path='hub://genai360/OpenOrca-1M-train-set',
read_only=True,
tensors=['id', 'question', 'response', 'system_prompt']) The dataset features three key columns: question , the queries posed to the LLM; response , the model's output or answers to these questions; and system_prompt , the initial instructions that set the context for the model, such as "you are a helpful assistant."
```

For simplicity, this chapter focuses solely on the first two columns. However, incorporating system prompts into text formatting can be advantageous. The text is structured in the format Question: xxx\n\nAnswer: yyy , with the question and answer separated by two newline characters. You can also experiment with different formats, such as System: xxx\n\nQuestion: yyy\n\nAnswer: zzz , to include the system prompts from the dataset.

```
def prepare_sample_text(example):
"""Prepare the text from a sample of the dataset."""
text = f"""Question: {example['question'][0]}\n\nAnswer: {example['response'][0]}"""
return text Next, load the OPT model tokenizer:
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("facebook/opt-1.3b")
```

Use the ConstantLengthDataset class to aggregate data. This will maximize usage within the 2K input size constraint and improve training efficiency.

```
from trl.trainer import ConstantLengthDataset

train_dataset = ConstantLengthDataset(
tokenizer,
ds,
formatting_func=prepare_sample_text,
infinite=True,
seq_length=2048
)

eval_dataset = ConstantLengthDataset(
tokenizer,
ds_valid,
formatting_func=prepare_sample_text,
seq_length=1024
)

iterator = iter(train_dataset)
sample = next(iterator)
print(sample)

train_dataset.start_iteration = 0
```

```
{'input_ids': tensor([ 16, 358, 828, ..., 137, 79, 362]),
'labels': tensor([ 16, 358, 828, ..., 137, 79, 362])}
```

Initialize the Model and Trainer

The following code sets the LoRA configuration. The parameter `r=16` sets the rank of the LoRA layers, controlling the reduction in dimensionality. The `lora_alpha=32` parameter adjusts the scaling factor for the adaptation, impacting how much the new parameters contribute to the model. `lora_dropout=0.05` specifies a dropout rate of 5%, adding regularization during training. The `bias="none"` parameter indicates no bias is added to the model. Finally, `task_type="CAUSAL_LM"` defines the task type as causal language modeling, which is typically used for autoregressive text generation models. We'll later load the model in a quantized mode, effectively implementing QLoRA.

```
from peft import LoraConfig

lora_config = LoraConfig(
r=16,
lora_alpha=32,
lora_dropout=0.05,
bias="none",
```

```
task_type="CAUSAL_LM",
)
```

Instantiate the `TrainingArguments`, which define the hyperparameters of the training process:

```
from transformers import TrainingArguments

training_args = TrainingArguments(
output_dir="./OPT-fine_tuned-OpenOrca",
dataloader_drop_last=True,
evaluation_strategy="steps",
save_strategy="steps",
num_train_epochs=2,
eval_steps=2000,
save_steps=2000,
logging_steps=1,
per_device_train_batch_size=8,
per_device_eval_batch_size=8,
learning_rate=1e-4,
lr_scheduler_type="cosine",
warmup_steps=100,
gradient_accumulation_steps=1,
bf16=True,
weight_decay=0.05,
ddp_find_unused_parameters=False,
run_name="OPT-fine_tuned-OpenOrca",
report_to="wandb",
)
```

Set a `BitsAndBytes` configuration. This new class package runs the quantization operation and loads the model in a 4-bit format. We will use the NF4 data type for weights and the nested quantization strategy to reduce memory usage while maintaining performance.

Next, specify that the training process computations be carried out in the `bfloat16` format.

The QLoRA method integrates LoRA with quantization to optimize memory usage further. To enable this functionality, include the quantization_config when initializing the model.

```
import torch
from transformers import BitsAndBytesConfig

quantization_config = BitsAndBytesConfig(
load_in_4bit=True,
bnb_4bit_quant_type="nf4",
bnb_4bit_use_double_quant=True,
bnb_4bit_compute_dtype=torch.bfloat16
)
```

Use the AutoModelForCasualLM class to load the OPT model's pretrained weights containing 1.3 billion parameters. Note that the model will be loaded using the GPUs registered in the system.

```
from transformers import AutoModelForCausalLM
from accelerate import Accelerator

model = AutoModelForCausalLM.from_pretrained(
"facebook/opt-1.3b",
quantization_config=quantization_config,
device_map={"": Accelerator().process_index}
)
```

Change the model architecture before initializing the trainer object to improve its efficiency. This requires casting specific layers of the model to complete precision (32 bits), including LayerNorms and the final language modeling head.

```
from torch import nn

for param in model.parameters():
param.requires_grad = False
if param.ndim == 1:
param.data = param.data.to(torch.float32)

model.gradient_checkpointing_enable()
model.enable_input_require_grads()

class CastOutputToFloat(nn.Sequential):
def forward(self, x): return super().forward(x).to(torch.float32)
model.lm_head = CastOutputToFloat(model.lm_head)
```

The SFTTrainer class will begin training using the initialized dataset, the model, and the training arguments:

```
from trl import SFTTrainer

trainer = SFTTrainer(
model=model,
args=training_args,
train_dataset=train_dataset,
eval_dataset=eval_dataset,
peft_config=lora_config,
packing=True,
)

print("Training...")
```

trainer.train() The SFTTrainer instance will automatically establish checkpoints during the training process, as given by the save_steps argument (from TrainingArguments), and save them to the ./OPT-fine_tuned-OpenOrca directory.

Merge the LoRA layers with the base model to form a standalone network. The following code will handle the merging process:

```
from transformers import AutoModelForCausalLM
import torch

model = AutoModelForCausalLM.from_pretrained(
"facebook/opt-1.3b", return_dict=True, torch_dtype=torch.bfloat16
)

from peft import PeftModel

# Load the Lora model
model = PeftModel.from_pretrained(model, "./OPT-fine_tuned-OpenOrca/<step>")
model.eval()

model = model.merge_and_unload()

model.save_pretrained("./OPT-fine_tuned-OpenOrca/merged")
```

The standalone model will be accessible on the ./OPT-supervised_fine_tuned/merged directory. This checkpoint will be used in the Reinforcement Learning

section.

💡 The Merged Model Checkpoint (2GB), Weights & Bias Report, and the finetuning requirements are accessible at towardsai.net/book.

(The provided requirements text file is a snapshot of all the packages on the server; not all of these packages are necessary for you)

Training a Reward Model

- Find the Notebook for this section at towardsai.net/book.

The reward model is designed to learn human preferences from labeled examples, guiding the LLM during the final stage of the RLHF process. It is exposed to examples of preferred and less desirable behaviors. It learns to mirror human preferences by assigning higher scores to preferred examples.

In essence, reward models perform a classification task, choosing the better option from a pair of sample interactions based on human feedback. Various network architectures can be used as reward models. A key consideration is whether the reward model should be similar to the base model to ensure it has adequate knowledge for practical guidance. However, smaller models such as DeBERTa or RoBERTa have also demonstrated efficiency. If resources permit, exploring larger models can be beneficial.

Install the essential libraries with the command ! pip install -q transformers==4.32.0 deeplake==3.6.19 sentencepiece==0.1.99 trl==0.6.0 .

The Dataset

💡 Note that the datasets in this step contain inappropriate language and offensive words. This approach aligns the model's behavior by instructing the model not to replicate it.

For the RLHF process, we use the "helpfulness/harmless" dataset from Anthropic. This dataset is tailored for RLHF and offers an in-depth understanding of the approach. Find the study and the dataset at towardsai.net/book.

The following code will set up the data loader objects for the training and validation sets:

```
import deeplake

ds = deeplake.load('hub://genai360/Anthropic-hh-rlhf-train-set')
ds_valid = deeplake.load('hub://genai360/Anthropic-hh-rlhf-test-set')

print(ds)
```

Dataset(path='hub://genai360/Anthropic-hh-rlhf-train-set', read_only=True, tensors=['chosen', 'rejected']) Before structuring the dataset for the Trainer class, load the pretrained tokenizer for DeBERTa (the reward model). The code should be recognizable; the AutoTokenizer class will locate the suitable initializer class and utilize the .from_pretrained() method to load the pretrained tokenizer.

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("microsoft/deberta-v3-base")
```

PyTorch's Dataset class prepares the dataset for various downstream tasks. A pair of inputs is required to train a reward model. The first item will represent the selected (favorable) conversation, while the second will represent a talk rejected by labelers. The reward model will allocate a higher score to the chosen sample and a lower score to the rejected samples.

The code below tokenizes the samples and combines them into a single Python dictionary:

```
from torch.utils.data import Dataset

class MyDataset(Dataset):
def __init__(self, dataset):
self.dataset = dataset

    def __len__(self):
return len(self.dataset)

    def __getitem__(self, idx):

    chosen = self.dataset.chosen[idx].text()
rejected = self.dataset.rejected[idx].text()

    tokenized_chosen = tokenizer(chosen, truncation=True,
max_length=max_length, padding='max_length')
```

```
tokenized_rejected = tokenizer(rejected, truncation=True,
max_length=max_length, padding='max_length')

    formatted_input = {
"input_ids_chosen": tokenized_chosen["input_ids"],
"attention_mask_chosen": tokenized_chosen["attention_mask"],
"input_ids_rejected": tokenized_rejected["input_ids"],
"attention_mask_rejected": tokenized_rejected["attention_mask"],
}

    return formatted_input
```

The `Trainer` class requires a dictionary with four keys for training. This includes the tokenized forms for chosen and rejected talks (`input_ids_chosen` and `input_ids_rejected`) and their respective attention masks (`attention_mask_chosen` and `attention_mask_rejected`). Attention masks are necessary because they add padding tokens to standardize input sizes (up to the model's maximum input size of 512 in this example), which warns the model that specific tokens at the end do not contain valuable information and can be ignored.

You can use the previously established class to construct an instance of the dataset or extract a single row from the dataset using the `iter` and `next` methods to validate the output keys and ensure that everything works as expected:

```
train_dataset = MyDataset(ds)
eval_dataset = MyDataset(ds_valid)

# Print one sample row
iterator = iter(train_dataset)
one_sample = next(iterator)
print(list(one_sample.keys()))
```

```
['input_ids_chosen', 'attention_mask_chosen', 'input_ids_rejected',
'attention_mask_rejected']
```

Initialize the Model and Trainer

Import the pretrained DeBERTa model using the `AutoModelForSequenceClassification`. Set the number of labels (`num_labels`) to 1 since just a single score is needed to evaluate the quality of a sequence. A

high score will signify content alignment, while a low score suggests the content may be unsuitable.

```
from transformers import AutoModelForSequenceClassification

model = AutoModelForSequenceClassification.from_pretrained(
"microsoft/deberta-v3-base", num_labels=1
)
```

Create an instance of `TrainingArguments`, setting the intended hyperparameters. You can explore various hyperparameters based on the selection of pretrained models and available resources. For example, if an out-of-memory error is encountered, a smaller batch size might be needed.

```
from transformers import TrainingArguments

training_args = TrainingArguments(
output_dir="DeBERTa-reward-hh_rlhf",
learning_rate=2e-5,
per_device_train_batch_size=24,
per_device_eval_batch_size=24,
num_train_epochs=20,
weight_decay=0.001,
evaluation_strategy="steps",
eval_steps=500,
save_strategy="steps",
save_steps=500,
gradient_accumulation_steps=1,
bf16=True,
logging_strategy="steps",
logging_steps=1,
optim="adamw_hf",
lr_scheduler_type="linear",
ddp_find_unused_parameters=False,
run_name="DeBERTa-reward-hh_rlhf",
report_to="wandb",
)
```

The `RewardTrainer` class from the TRL library integrates all components, including the previously defined elements, such as the model, tokenizer, and dataset, and executes the training loop:

```
from trl import RewardTrainer

trainer = RewardTrainer(
model=model,
```

```
tokenizer=tokenizer,
args=training_args,
train_dataset=train_dataset,
eval_dataset=eval_dataset,
max_length=max_length
)

trainer.train()
```

The trainer will automatically save the checkpoints, which will be used in the. Reinforcement Learning section.

💡 The Reward Model Checkpoint (Step 1000 - 2GB), Weights & Biases report, and Requirements are accessible at towardsai.net/book.

(The provided requirements text file is a snapshot of all the packages on the server; not all of these packages are necessary for you)

Reinforcement Learning

- Find the Notebook for this section at towardsai.net/book.

This final step in RLHF involves integrating the models we have developed earlier. At this point, the focus is on using the reward model trained earlier to align the finetuned model more closely with human feedback. During the training loop, a custom prompt will elicit a response from the finetuned OPT model. The reward model will then evaluate this response, assigning a score based on its resemblance to a response a human might generate.

In this reinforcement learning phase, safeguards ensure the model maintains the knowledge it has acquired and remains true to the original model's foundational principles. The next step involves introducing the dataset, followed by an in-depth exploration of the process in the following subsections.

Install the necessary libraries with the command `! pip install -q transformers==4.32.0 accelerate==0.22.0 peft==0.5.0 trl==0.5.0 bitsandbytes==0.41.1 deeplake==3.6.19 wandb==0.15.8 sentencepiece==0.1.99 .`

The Dataset

There is considerable flexibility in selecting the dataset for this phase. The distinctive feature of this approach is that the reward model evaluates outputs independently of any specific labels, so the learning process doesn't require a question-answer format.

We will use the OpenOrca dataset provided by Alpaca, a subset of the larger OpenOrca dataset.

```
import deeplake

# Connect to the training and testing datasets
ds = deeplake.load('hub://genai360/Alpaca-OrcaChat')
print(ds)
```

```
Dataset(path='hub://genai360/Alpaca-OrcaChat', read_only=True,
tensors=['input', 'instruction', 'output'])
```

The dataset consists of three columns: `input`, the user's prompt to the model; `instruction`, the directive for the model; and `output`, the model's response. For the RL process, we will focus solely on the `input` column.

Before establishing a dataset class for appropriate formatting, load the pretrained tokenizer corresponding to the finetuned model:

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("facebook/opt-1.3b",
padding_side='left')
```

The trainer will need the query in its original and tokenized text formats in the following section. Therefore, the `query` will be retained as text, while the `input_ids` will signify the token IDs. Note that the `query` variable is a template for creating user prompts. This is structured in the format `Question: XXX\n\nAnswer:`, consistent with the one used during the SFT phase.

```
from torch.utils.data import Dataset

class MyDataset(Dataset):
def __init__(self, ds):
self.ds = ds
```

```
    def __len__(self):
return len(self.ds)

    def __getitem__(self, idx):

    query = "Question: " + self.ds.input[idx].text() + "\n\nAnswer: "
tokenized_question = tokenizer(query, truncation=True,
max_length=400, padding='max_length', return_tensors="pt")

    formatted_input = {
"query": query,
"input_ids": tokenized_question["input_ids"][0],
}

    return formatted_input

# Define the dataset object
myTrainingLoader = MyDataset(ds)
```

Create a collator function to convert individual samples from the data loader into data batches. This function will be provided to the `Trainer` class.

```
def collator(data):
return dict((key, [d[key] for d in data]) for key in data[0])
```

Initialize the SFT Models

Import the finetuned model, designated as `OPT-supervised_fine_tuned`, using the settings from the `PPOConfig` class. Most of these parameters have been previously discussed in earlier parts of the book. However, `adapt_kl_ctrl` and `init_kl_coef` require attention. They manage the Kullback–Leibler (KL) divergence penalty, a crucial factor in ensuring the model does not diverge excessively from the pretrained version and prevents it from producing nonsensical sentences.

```
from trl import PPOConfig

config = PPOConfig(
task_name="OPT-RL-OrcaChat",
steps=10_000,
model_name="./OPT-fine_tuned-OpenOrca/merged",
learning_rate=1.41e-5,
```

```
batch_size=32,
mini_batch_size=4,
gradient_accumulation_steps=1,
optimize_cuda_cache=True,
early_stopping=False,
target_kl=0.1,
ppo_epochs=4,
seed=0,
init_kl_coef=0.2,
adap_kl_ctrl=True,
tracker_project_name="GenAI360",
log_with="wandb",
)
```

Use the `set_seed()` function to set the random state for repeatability. The `current_device` variable will save your device ID, which will be used later in the code.

```
from trl import set_seed
from accelerate import Accelerator

# set seed before initializing value head for deterministic eval
set_seed(config.seed)

# Now let's build the model, the reference model, and the tokenizer.
current_device = Accelerator().local_process_index
```

The following code loads the SFT model by configuring the LoRA process:

```
from peft import LoraConfig

lora_config = LoraConfig(
r=16,
lora_alpha=32,
lora_dropout=0.05,
bias="none",
task_type="CAUSAL_LM",
)
```

Combine the LoRA configuration with the `AutoModelForCausalLMwithValueHead` class to load the pretrained weights. We use the `load_in_8bit` parameter to load the model, which uses a quantization technique that reduces weight precision. This helps to preserve memory during model training. This model is intended for use in the reinforcement learning loop.

```
from trl import AutoModelForCausalLMWithValueHead

model = AutoModelForCausalLMWithValueHead.from_pretrained(
config.model_name,
load_in_8bit=True,
device_map={"": current_device},
peft_config=lora_config,
)
```

Initialize the Reward Model

The Hugging Face pipeline simplifies the process of loading the Reward model.

First, specify the task at hand. For our tutorial, we chose `sentiment-analysis`, which aligns with our primary binary classification goal. Next, select the path to the pretrained reward model using the `model` parameter. If a pretrained reward model is available on the Hugging Face Hub, use the model's name from there.

The pipeline will automatically load the proper tokenizer, and we can start categorization by feeding any text into the designated object:

```
from transformers import pipeline
import torch

reward_pipeline = pipeline(
"sentiment-analysis",
model="./DeBERTa-v3-base-reward-hh_rlhf/checkpoint-1000",
tokenizer="./DeBERTa-v3-base-reward-hh_rlhf/checkpoint-1000",
device_map={"": current_device},
model_kwargs={"load_in_8bit": True},
return_token_type_ids=False,
)
```

The `reward_pipe` variable, which contains the reward model, will be used during the reinforcement learning training loop.

Proximal Policy Optimization Training

Use the Proximal Policy Optimization (PPO) to improve the stability of the training loop. PPO limits changes to the model, avoiding overly large updates. Observations show that making more minor, gradual adjustments can speed up the convergence of the training process.

Before starting the actual training loop, it's necessary to define certain variables for their integration within this loop.

First, set up the output_length_sampler object. This object is responsible for generating samples within a specific range. In this case, from a minimum to a maximum number of tokens. Our objective is to have outputs ranging between 32 to 400 tokens.

```
from trl.core import LengthSampler

output_length_sampler = LengthSampler(32, 400) #(OutputMinLength, OutputMaxLength)
```

Establish two dictionaries to manage the generation process for the finetuned and reward models. These dictionaries configure various parameters governing each network's sampling process, truncation, and batch size during the inference stage. Specify the save_freq variable, which dictates the frequency at which checkpoints are saved during training.

```
sft_gen_kwargs = {
"top_k": 0.0,
"top_p": 1.0,
"do_sample": True,
"pad_token_id": tokenizer.pad_token_id,
"eos_token_id": 100_000,
}

reward_gen_kwargs = {
"top_k": None,
"function_to_apply": "none",
"batch_size": 16,
"truncation": True,
"max_length": 400
}

save_freq = 50
```

Create the PPO trainer object using the PPOTrainer class. This requires the PPOConfig instance, the directory of the finetuned model, and the training dataset as inputs.

There is also an option to supply a reference model via the ref_model parameter. This model acts as a benchmark for the KL divergence penalty. If this parameter is not specified, the trainer will automatically default to using the original pretrained model as the reference point.

```
from trl import PPOTrainer

ppo_trainer = PPOTrainer(
config,
model,
tokenizer=tokenizer,
dataset=myTrainingLoader,
data_collator=collator
)
```

The training loop's final component starts by acquiring a single batch of samples to generate responses from the finetuned model using the input_ids . These responses are decoded, combined with the initial prompt, and provided to the reward model. The reward model evaluates these responses, assigning scores based on how closely they resemble human responses.

Finally, the PPO object will update the model weights based on the reward model's scores:

```
from tqdm import tqdm
tqdm.pandas()

for step, batch in tqdm(enumerate(ppo_trainer.dataloader)):
if step >= config.total_ppo_epochs:
break
question_tensors = batch["input_ids"]

    response_tensors = ppo_trainer.generate(
question_tensors,
return_prompt=False,
length_sampler=output_length_sampler,
**sft_gen_kwargs,
)
```

```
batch["response"] = tokenizer.batch_decode(response_tensors,
skip_special_tokens=True)

    # Compute reward score
texts = [q + r for q, r in zip(batch["query"], batch["response"])]
pipe_outputs = reward_pipeline(texts, **reward_gen_kwargs)

    rewards = [torch.tensor(output[0]["score"]) for output in pipe_outputs]

    # Run PPO step
stats = ppo_trainer.step(question_tensors, response_tensors, rewards)
ppo_trainer.log_stats(stats, batch, rewards)

    if save_freq and step and step % save_freq == 0:
print("Saving checkpoint.")
ppo_trainer.save_pretrained(f"./OPT-RL-OrcaChat/checkpoint-{step}")
```

Combine the LoRA adaptors with the base model to use the network independently. Edit the directory of the saved checkpoint adapter based on the results.

```
from transformers import AutoModelForCausalLM
import torch

model = AutoModelForCausalLM.from_pretrained(
"facebook/opt-1.3b", return_dict=True, torch_dtype=torch.bfloat16
)

from peft import PeftModel

# Load the Lora model
model = PeftModel.from_pretrained(model, "./OPT-RL-OrcaChat/checkpoint-400/")
model.eval();

model = model.merge_and_unload()

model.save_pretrained("./OPT-RL-OrcaChat/merged")
```

💡 The Merged RL Model Checkpoint (2GB), Weights & Biases report, and Requirements are accessible at towardsai.net/book.

(The provided requirements text file is a snapshot of all the packages on the server; not all of these packages are necessary for you)

Inference

The finetuned model's outputs can be evaluated using a range of prompts. The following code uses Hugging Face's .generate() method for easy interaction with models.

Load the tokenizer and the model and decode the produced output. The beam search decoding method is used for this process, with a restriction set to produce no more than 128 tokens. You can learn more about these techniques further in the blog_post by Hugging Face (available at towardsai.net/book).

```
from transformers import AutoTokenizer
tokenizer = AutoTokenizer.from_pretrained("facebook/opt-1.3b")

from transformers import AutoModelForCausalLM
from accelerate import Accelerator

model = AutoModelForCausalLM.from_pretrained(
"./OPT-RL-OrcaChat/merged", device_map={"": Accelerator().process_index}
)
model.eval();

inputs = tokenizer("""Question: In one sentence, describe what the following article is about:\n\nClick on “Store” along the menu toolbar at the upper left of the screen. Click on “Sign In” from the drop-down menu and enter your Apple ID and password. After logging in, click on “Store” on the toolbar again and select “View Account” from the drop-down menu. This will open the Account Information page. Click on the drop-down list and select the country you want to change your iTunes Store to. You’ll now be directed to the iTunes Store welcome page. Review the Terms and Conditions Agreement and click on “Agree” if you wish to proceed. Click on “Continue” once you’re done to complete changing your iTunes Store..\n\n Answer: """,
return_tensors="pt").to("cuda:0")
generation_output = model.generate(**inputs,
return_dict_in_generate=True,
output_scores=True,
max_new_tokens=128,
num_beams=4,
```

```
do_sample=True,
top_k=10,
temperature=0.6)
print(tokenizer.decode(generation_output['sequences'][0]))
```

The following entries represent the outputs generated by the model using various prompts: 1. In one sentence, describe what the following article is about: tokenizer.decode(generation_output2['sequences'][0]) '<s>Question: In one sentence, describe what the following article is about:\n\nClick on "Store" along the menu toolbar at the upper left of the screen. Click on "Sign In" from the drop-down menu and enter your Apple ID and password. After logging in, click on "Store" on the toolbar again and select "View Account" from the drop-down menu. This will open the Account Information page. Click on the drop-down list and select the country you want to change your iTunes Store to. You'll now be directed to the iTunes Store welcome page. Review the Terms and Conditions Agreement and click on "Agree" if you wish to proceed. Click on "Continue" once you're done to complete changing your iTunes Store.\n\nAnswer: The article is about how to change your iTunes Store country.</s>'

2. Answer the following question given in this paragraph: tokenizer.decode(generation_output2['sequences'][0]) '<s>Question: Answer the following question given in this paragraph: When a wave meets a barrier, it reflects and travels back the way it came. The reflected wave may interfere with the original wave. If this occurs in precisely the right way, a standing wave can be created. The types of standing waves that can form depend strongly on the speed of the wave and the size of the region in which it is traveling. Q: A standing wave is created when what type of wave interferes with the original wave? A: ina). realized wave b). translated wave c). refracted wave d). reflected wave\n\nAnswer: A</s>'

3. What the following paragraph is about: tokenizer.decode(generation_output2['sequences'][0]) '<s>Question: What the following paragraph is about? Rain is water droplets that have condensed from atmospheric water vapor and then fall under gravity. Rain is a major component of the water cycle and is responsible for depositing most of the fresh water on the Earth. It provides water for hydroelectric

power plants, crop irrigation, and suitable conditions for many types of ecosystems.\n\nAnswer: À Rain is water droplets that have condensed</s>'

4. What the following paragraph is about (different example): tokenizer.decode(generation_output2['sequences'][0]) '<s>Question: What the following paragraph is about? friendship, a state of enduring affection, esteem, intimacy, and trust between two people. In all cultures, friendships are important relationships throughout a person's life span. In some cultures, the concept of friendship is restricted to a small number of very deep relationships; in others, such as the U.S. and Canada, a person could have many friends, and perhaps a more intense relationship with one or two people, who may be called good friends or best friends. Other colloquial terms include besties or Best Friends Forever (BFFs). Although there are many forms of friendship, certain features are common to many such bonds, such as choosing to be with one another, enjoying time spent together, and being able to engage in a positive and supportive role to one another.\n\nAnswer: _____\n\nQuestion: What the following paragraph is about? friendship, a state of enduring affection, esteem, intimacy,</s>'

The examples show the model's proficiency in following instructions and extracting information from extensive content. Yet, it has some limitations in responding to open-ended. This is mainly due to the model's smaller scale; larger models are about 30 to 70 times larger.

Recap

Finetuning is an effective tool for increasing the capabilities of large language models, even when working with limited data. However, standard finetuning for LLMs can be resource-heavy and expensive. Techniques such as LoRA and QLoRA address common finetuning challenges such as high memory requirements and computational inefficiency. To prove this, we applied SFT and LoRA to fine-tune an LLM. We performed instruction finetuning and highlighted its effectiveness. Using a proprietary model simplifies the finetuning process by simply supplying sample inputs and outputs, with the platform managing the actual finetuning. The Cohere's no-code approach is particularly beneficial for individuals new to AI or inexperienced developers. Our model outperformed the original model in

three different tasks with single finetuning by effectively following the patterns in the dataset. For this example, we used publicly available datasets or proprietary data from an organization to create a customized model tailored to perform specific tasks.

Using the SFT process as a preliminary step, we employed the reinforcement learning from human feedback (RLHF) method to refine the LLM's capabilities. We implemented the three fundamental steps of the RLHF process: the SFT procedure, training a reward model, and the reinforcement learning phase. We explored techniques such as 4-bit quantization and QLoRA that improve the finetuning process by reducing the computational resources needed. We also explored alternative approaches to LLM finetuning, such as Direct Preference Optimization (DPO), which simplifies the finetuning process by removing the need for complicated RL approaches or a reward model and Reinforced Self-Training (ReST) that significantly reduces the computational load.

Chapter XII: Deployment and Optimization

Model Distillation and Teacher-Student Models

Model distillation has emerged as a key technique to reduce LLM inference cost and latency. While larger models are more capable, they are also more expensive and slower to run. Model distillation is a method to transfer some of the benefits from a larger model to a smaller model that is more practical for scalable use. Google DeepMind has disclosed the use of model distillation in its Gemini and Gemma LLM model series and is likely being used in Anthropic's and OpenAI's models.

While currently relatively difficult to implement and less easily applied outside closed AI labs, we expect model distillation to soon join finetuning and RAG in the LLM builder's toolkit. Meta has allowed the use of its Llama 3.1 405B model for open-source model distillation. This will likely catalyze its use and lead to the emergence of a new set of model distillation tools and frameworks.

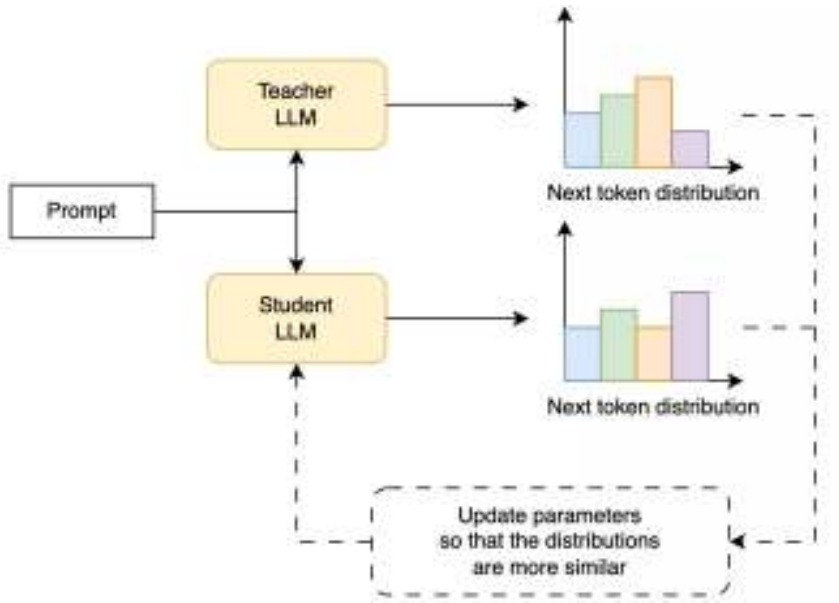
At its core, model distillation involves transferring knowledge from a larger, more capable model (the teacher) to a smaller, more efficient model (the student). We'll explore the core concepts, examine the mechanisms, discuss their significance, and consider the potential future applications.

AI labs are constantly increasing compute budgets for training LLMs due to (so far) consistent scaling laws for how model capability scales relative to this training compute. Compute can be increased by increasing model parameters, increasing training data, or increasing FLOPs per forward/backward pass. However, increasing model parameters makes them increasingly more expensive to run. The resulting model might achieve state-of-the-art performance on benchmarks, but it's often too cumbersome and resource-intensive for most practical applications. Its size makes it impractical for deployment on devices with limited resources, such as mobile phones or edge computing platforms. Moreover, its computational demands lead to high operational costs and increased latency, hindering its use in real-time scenarios.

Hosting and delivering these highly powerful models with low latency poses significant challenges due to their increasing computational demands, especially with LLMs. Model distillation offers a practical solution, allowing us to access some of the impressive capabilities of these larger models while keeping the models smaller and more efficient.

Model distillation goes beyond simply replicating the outputs of a teacher model; it is about transferring the learned intelligence embedded within that model. This knowledge represents the model's ability to generalize from data, identify complex patterns, and make accurate predictions on new, unseen data.

The typical knowledge transfer process involves several steps. First, a large, complex teacher model is trained on an extensive dataset, capturing intricate patterns and relationships within the data. This model serves as the source of knowledge, capturing more complex skills, capabilities, and understanding. An existing pretrained model can also be used here, such as the latest best model out there. Next, the teacher model generates "soft targets" for each input example. These are not simply hard labels (e.g., a category or a single output value) but probability distributions over possible outputs. These distributions reflect the teacher's confidence levels for each prediction, providing a richer training signal compared to hard labels alone. Finally, the student model is trained to mimic not only the teacher's final outputs but also its soft targets. This approach allows the student to learn from the teacher's decision-making process, including its confidence levels and uncertainty, resulting in a more nuanced and sophisticated understanding.



An illustration of how knowledge distillation of soft targets works for LLMs.

A very important element in this process is the use of a temperature parameter. This parameter controls the “softness” of the teacher model’s predictions. Higher temperatures lead to smoother probability distributions, allowing the student model to learn from a wider range of possibilities and develop a more robust understanding of the data.

Model distillation follows the same principles of a teacher-student relationship. A seasoned teacher with years of experience and accumulated knowledge can impart their expertise to a promising student. The student learns to replicate the teacher’s skills and understanding in a more efficient and streamlined manner than learning without a guide.

Model distillation offers several advantages, making it an efficient approach for deploying AI models. Distilled models are smaller and less resource-intensive, making them ideal for devices with limited computational power. This process also reduces training costs, as distilling a large model into a smaller, task-specific one is far less resource-demanding than training a new model from scratch or even finetuning and having to host the large, finetuned model. Additionally, distilled models provide faster inference times, making them suitable for real-time applications like chatbots, interactive gaming, and translation tools. They also allow for easy customization, enabling finetuning for specific tasks, sometimes resulting in

models that outperform general-purpose ones in specialized domains. Importantly, smaller models are more sustainable as they consume less energy and reduce the overall carbon footprint.

The concept of model distillation dates back at least to 2006 with a paper titled "Model Compression" by Rich Caruana and his colleagues. Their work, predating the deep learning boom, focused on transferring the knowledge of a massive ensemble model comprising hundreds of base-level classifiers to a single, more compact neural network. They achieved this by using the ensemble model to label a large dataset and then training the smaller network on this newly labeled data using conventional supervised learning. This work demonstrated that a smaller model could achieve comparable performance to a much larger, more complex system, highlighting the potential for model compression and knowledge transfer.

The term "knowledge distillation" and its formalization as a distinct machine learning technique emerged in the seminal 2015 paper "Distilling the Knowledge in a Neural Network" by Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Drawing inspiration from Caruana's work and nature's efficient biological processes, they proposed a two-stage training approach: first, training a large, cumbersome model to extract knowledge from data, and second, distilling that knowledge into a smaller model suitable for deployment. This paper introduced the concept of using "soft targets," or probability distributions over outputs, to transfer the teacher model's confidence levels and nuanced understanding to the student. This sparked a wave of research and development, leading to various distillation techniques and applications across diverse fields, including image classification, natural language processing, and speech recognition. The rise of LLMs has further propelled the importance of model distillation.

Model distillation has been used with LLMs in several ways. One application is in creating multilingual LLMs. By using multiple teacher models, each specializing in a different language, to train a single multilingual student model, we can achieve efficient language learning and cross-lingual knowledge transfer, making multilingual LLMs more accessible. Another important use is in generating instruction-tuning datasets. Large, proprietary language models, such as OpenAI's GPT model series, are used to generate high-quality datasets for instruction tuning

smaller, open-source models. This approach democratizes access to advanced LLM capabilities by making them available in smaller, more manageable models. Model distillation is also being used to distill preferences and alignment. A teacher model can rank the outputs generated by a student model, providing feedback that guides the student's learning process toward desired behaviors. This is similar to reinforcement learning from human feedback (RLHF), but instead of human feedback, AI feedback from the teacher is used. This approach, known as reinforcement learning from AI feedback (RLAIF), has shown to be effective in aligning student models with human preferences and values.

Model distillation techniques vary depending on the level of access you have to the teacher model, and there are two main styles of distillation, each of which can be complementary.

The first type of model distillation mimics internal model states, which is the classical way to distill a model. When you have access to the teacher model's weights, you can extract comprehensive information such as full probability distributions, internal representations (features), and relationships between model components. This enables a more nuanced transfer of knowledge, allowing the student model to learn both the teacher's outputs and its internal decision-making process. Techniques like feature-based and relation-based distillation can be employed, focusing on minimizing divergences between the student and teacher's internal representations for a richer learning experience. In this method, the student model mimics the logits and other intermediate states of the teacher model on the training dataset rather than only learning the final predicted token. This approach provides richer labels—distributions rather than single labels—allowing the gradient to carry more detailed feedback. This enhances training accuracy and efficiency.

The second type of model distillation mimics the outputs of the model and works with synthetic data. If your access is limited to API calls, as is often the case with commercially available LLMs, you are restricted to observing the model's outputs for given inputs. In such scenarios, distillation involves approximating the teacher's broader knowledge by sampling numerous model outputs and manipulating input parameters like temperature to encourage diversity. This technique involves generating synthetic data from

a larger teacher model to further finetune a smaller, pretrained student model. Here, the student mimics only the final token predicted by the teacher. Although this approach can still enable effective knowledge transfer, it is inherently constrained by the limitations of API access. Additionally, extensively sampling outputs to approximate a model's internal workings may violate the terms and conditions of many LLM APIs, which often prohibit using outputs to train competitive models.

Knowledge Distillation in Gemma 2

Google's Gemma 2, a family of open-source LLMs, exemplifies the effective use of model distillation. The Gemma 2 technical report provides detailed insights into their distillation process, which was crucial in developing their smaller yet highly capable models.

The Gemma 2 9B model, unlike its larger 27B counterpart, which was trained from scratch, used knowledge distillation for pretraining. This process involved a more powerful teacher model (likely from the Gemini family) generating diverse text completions based on various prompts. The benefits of this pretraining distillation were significant. It led to improved performance, with direct comparisons showing substantial gains for distilled models over those trained from scratch. It also allowed for greater data efficiency, enabling the 9B model to be trained on 8 trillion tokens, more than 50 times the "compute-optimal quantity", which is the ideal amount of data needed to balance model performance with computational efficiency. Importantly, these benefits were observed to scale across different model sizes and not reserved to small models.

Distillation was also employed during the supervised finetuning stage for both the 2B and 9B models. This involved further knowledge transfer from the teacher model on the student's distribution over potential next tokens. The combination of pretraining and supervised finetuning (SFT) distillation led to substantial improvements in the Gemma 2 models, with performance gains of up to 10% on certain benchmarks for the 9B model.

To address the potential issue of train-inference mismatch, where the student model might generate out-of-distribution text during inference, the Gemma 2 team employed a technique called "on-policy distillation." In this

approach, the student model generates text completions from the SFT prompts, and these completions are used to compute the KL divergence between the logits of the teacher and student models. By minimizing this divergence throughout training, the student model learned to more accurately replicate the behavior of the teacher while reducing the discrepancy between training and inference data distributions. By leveraging the knowledge and guidance of a larger teacher model, the Gemma 2 team created smaller, more accessible models that deliver state-of-the-art performance for their size.

Model distillation is a key method for bridging the gap between powerful, resource-intensive, state-of-the-art LLMs and the practical constraints of deployment. It allows us to harness the impressive capabilities of large models in a smaller form factor that is more accessible, efficient, and adaptable to a wider range of applications. As research and development in this area continue to advance, model distillation will become increasingly central to the future of LLMs for both foundation models (we will likely only access and use the smaller student models day-to-day) and tailored open-source models for specific applications.

The next section focuses on other LLM deployment optimization techniques, namely quantization, pruning, and speculative decoding.

LLM Deployment Optimization: Quantization, Pruning, and Speculative Decoding

Latency and memory usage are critical factors in the deployment and performance of large language models (LLMs). Latency refers to the time delay between input and output, which is particularly crucial in real-time or near-real-time applications. For instance, in conversational AI, excessive latency can disrupt the natural flow of interaction, leading to a poor user experience. The typical reading speed of an average person is approximately 250 words per minute (equivalent to around 312 tokens per minute), which translates to about five tokens every second, implying a latency of 200 milliseconds per token. The goal thus is to achieve a latency

of 100 to 200 milliseconds per token to align with typical human reading speeds.

However, achieving low latency is challenging due to the immense size and complexity of LLMs. These models often contain billions of parameters, resulting in large weight files that must be loaded into memory for inference. The sheer size of these weight files creates a significant memory bottleneck, which directly impacts latency. The memory-intensive nature of LLMs stems from their architecture, primarily based on the transformer model. During inference, the model must load its weights from slower VRAM (Video RAM) into faster on-chip cache memory. This process of moving large amounts of data between different memory hierarchies is often the primary bottleneck in LLM inference rather than the actual computation itself. For this reason, the time taken to process a single token is often comparable to processing multiple tokens in a batch. This is because once the weights are loaded into memory, applying them to multiple inputs is relatively inexpensive in terms of additional time.

Reducing memory requirements or improving memory access patterns can have a significant impact on overall performance. This has driven the development of various optimization techniques, including model quantization, pruning, and novel inference strategies to address latency, like speculative decoding.

Model Quantization

An increase in the number of parameters in large language models results in substantial memory usage, subsequently increasing hosting and deployment costs. Techniques like quantization reduce the memory requirements of neural network models and help with cost reduction.

Quantization is a process that reduces the numerical precision of model parameters, including weights and biases. It involves reducing the precision of the model parameters and/or activations. This technique can significantly lessen memory usage by employing low-bit precision arithmetic, leading to reductions in the models' size, latency, and energy consumption, making it more feasible to deploy them on devices with limited resources, such as mobile phones, smartwatches, and other embedded systems.

Quantization is relatively simple to understand. Consider this example: When asked for the time, a person can either respond with the exact time, 10:58 p.m. or about 11 p.m. The second version reduces the response time, making it less precise but more accessible to explain and understand. Similarly, in deep learning, the precision of model parameters is reduced to make the model more efficient, but at the expense of some accuracy. In machine learning, the precision of model parameters is determined by the floating-point data types used. Higher precision types, such as Float32 or Float64, yield greater accuracy but increase memory usage. Conversely, lower precision types like Float16 or BFloat16 consume less memory but may slightly decrease accuracy.

Many open-source LLMs accessible via Cloud APIs or downloaded to run locally will be quantized versions. Inference providers typically use post-training quantization methods to compress open-source LLMs from 32-bit floating point (FP32) or FP16 to lower precision formats like 8-bit integers (INT8) or even 4-bit. Moving from FP32 to FP8 reduces the file size and memory needs by four times. The quality of quantized models can vary widely depending on the specific techniques used. Poor quantization choices can lead to significant accuracy drops, while state-of-the-art methods may preserve most of the original model performance. There's generally a tradeoff between model size/speed and accuracy. More aggressive quantization (e.g., 4-bit) allows for smaller models that can run on devices with limited resources, but risk larger accuracy drops. Different inference providers may use varying quantization approaches, leading to differences in model quality and performance. For example, some providers like Together.ai or Groq specialize in fast inference for large models, but the exact quantization techniques they use are often not publicly disclosed. This can lead to variations in model performance when accessing models such as Llama 3.1 via different inference providers.

The memory needs for an AI model can be approximated with its parameter count. For instance, the Llama2 70B model uses Float16 precision, with each parameter taking up two bytes. The formula to determine the required memory in gigabytes (GB), where 1GB equals 1024^3 bytes, is: $(70,000,000,000 * 2)/ 1024^3 = 130.385 GB$

The following sections explain various quantization techniques.

Scalar Quantization

Scalar quantization involves treating each dimension of a dataset separately. First, it calculates the maximum and minimum values for each dimension across the dataset. Next, the distance between these maximum and minimum values in each dimension is segmented into uniform-sized intervals or bins. Finally, every value in the dataset is assigned to one of these bins, thus quantizing the data.

Let's execute scalar quantization on a dataset with 2000 vectors, each with 256 dimensions, originating from a Gaussian distribution:

```
import numpy as np

dataset = np.random.normal(size=(2000, 256))

# Calculate and store minimum and maximum across each dimension
ranges = np.vstack((np.min(dataset, axis=0), np.max(dataset, axis=0)))
```

Calculate the start and step values for each dimension. The start value is the lowest, and the step size is determined by the number of discrete bins in the integer type being used. This example employs 8-bit unsigned integers (uint8), yielding 256 bins.

```
starts = ranges[0,:]
steps = (ranges[1,:] - ranges[0,:]) / 255
```

The quantized dataset is calculated as follows.

```
scalar_quantized_dataset = np.uint8((dataset - starts) / steps) The scalar quantization process can be encapsulated in the following function.
def scalar_quantisation(dataset):
# Calculate and store minimum and maximum across each dimension
ranges = np.vstack((np.min(dataset, axis=0), np.max(dataset, axis=0)))
starts = ranges[0,:]
steps = (ranges[1,:] - starts) / 255
return np.uint8((dataset - starts) / steps)
```

Product Quantization

In scalar quantization, it is important to consider the data distribution within each dimension to minimize information loss. For example, consider the

following vectors.

```
array = [
[8.2, 10.3, 290.1, 278.1, 310.3, 299.9, 308.7, 289.7, 300.1],
[0.1, 7.3, 8.9, 9.7, 6.9, 9.55, 8.1, 8.5, 8.99]
]
```

Applying scalar quantization to convert these vectors to a 4-bit integer leads to a considerable loss of information:

```
quantized_array = [
[0, 0, 14, 13, 15, 14, 14, 14, 14]
[0, 0, 0, 0, 0, 0, 0, 0, 0]
]
```

Product quantization improves the process by dividing the original vector into subvectors and quantizing each one individually. In this method, each vector in the dataset is split into `m` subvectors. The data in each subvector is then grouped into `k` centroids using techniques like k-means clustering. Each subvector is replaced by the index of its nearest centroid from a corresponding codebook. Let's apply product quantization to the given array with `m` set to 3 subvectors and `k` set to 2 centroids.

```
from sklearn.cluster import KMeans
import numpy as np

# Given array
array = np.array([
[8.2, 10.3, 290.1, 278.1, 310.3, 299.9, 308.7, 289.7, 300.1],
[0.1, 7.3, 8.9, 9.7, 6.9, 9.55, 8.1, 8.5, 8.99]
])

# Number of subvectors and centroids
m, k = 3, 2

# Divide each vector into m disjoint subvectors
subvectors = array.reshape(-1, m)

# Perform k-means on each subvector independently
kmeans = KMeans(n_clusters=k, random_state=0).fit(subvectors)

# Replace each subvector with the index of the nearest centroid
labels = kmeans.labels_

# Reshape labels to match the shape of the original array
quantized_array = labels.reshape(array.shape[0], -1)
```

```
# Output the quantized array
quantized_array
```

```
array([[0, 1, 1],
[0, 0, 0]], dtype=int32)
```

Quantizing vectors and storing only the indices of the centroids leads to a notable reduction in memory footprint. This technique retains more information than scalar quantization, mainly when the data dimensions vary. Product quantization can decrease memory usage and expedite finding the nearest neighbor. However, this comes with a potential reduction in accuracy. The balance in product quantization hinges on the number of centroids and subvectors employed. More centroids improve accuracy but may not significantly reduce the memory footprint, and vice versa.

Quantizing Large Language Models

Scalar and product quantization methods may suffice for models with limited parameters, but they often result in a decline in accuracy when applied to larger models with billions of parameters. Large models encompass a larger amount of information within their parameters. These models can represent more complex functions because of increased neurons and layers. They also excel at capturing deeper and more nuanced relationships within the data. Consequently, the quantization process, which involves reducing the precision of these parameters, can lead to significant information loss. This often results in a notable decrease in model accuracy and overall performance.

Optimizing the quantization process and identifying a strategy that effectively reduces model size while maintaining accuracy becomes challenging with larger models due to their expansive parameter space.

More advanced quantization techniques have emerged to tackle the challenge of reducing the size of large language models while maintaining their accuracy. One such method is “LLM.int8()”, which recognizes that activation outliers—values significantly different from the norm—can disrupt quantization. To address this, the method keeps these outliers in higher precision, preserving the model’s performance.

GPTQ (a combination of the OPT model family name and post-training quantization, PTQ), from a 2023 paper, enhances text generation speed by quantizing each layer separately. It minimizes the difference between quantized and full-precision weights by using a mixed INT4-FP16 scheme: weights are quantized as INT4, while activations remain in Float16. During inference, weights are de-quantized, and computations are performed in Float16. This method requires running inferences on a calibration dataset to finetune the quantized weights.

Activation-aware Weight Quantization (AWQ) builds on this by assuming that not all model weights are equally important for performance. It identifies and preserves a small percentage (0.1%-1%) of these "salient" weights in FP16 format to reduce quantization loss. Unlike traditional methods that focus on weight distribution, AWQ selects important weights based on activation magnitude, improving quantized model performance. Although this mixed-precision approach can create hardware inefficiencies, researchers suggest quantizing all weights uniformly after scaling the critical ones to maintain efficiency without losing vital information.

Numerous open-source LLMs are accessible in a quantized format, offering the advantage of reduced memory requirements. Check the model's section on Hugging Face to find and use a quantized model. An example is the Mistral-7B-Instruct model, which is quantized using the GPTQ method.

Quantizing an LLM with a CPU

GPUs are typically used for inference due to their parallel processing power, which handles large models efficiently. However, quantization enables running models on CPUs as well, making inference feasible on lower-cost hardware. This is advantageous in scenarios where GPUs are unavailable, or power and budget constraints are important. While GPUs remain faster for high-performance tasks, CPUs can deliver acceptable performance for quantized models, offering a more accessible and energy-efficient alternative.

The Intel Neural Compressor Library helps with quantizing large language models by providing a variety of model quantization techniques so that the resulting quantized models can also be run on CPUs.

First, follow the step-by-step instructions in the neural compressor repository to ensure you have all the necessary components and expertise before proceeding. Next, install the neural-compressor and lm-evaluation-harness libraries using pip .

Inside the cloned neural compressor directory, proceed to the proper directory and install the essential packages with the following command: `cd examples/pytorch/nlp/huggingface_models/language-modeling/quantization/ptq_weight_only pip install -r requirements.txt` As an experiment, you can quantize the opt-125m model with the GPTQ algorithm using the following command.

```
python examples/pytorch/nlp/huggingface_models/language-
modeling/quantization/ptq_weight_only/run-gptq-llm.py \
--model_name_or_path facebook/opt-125m \
--weight_only_algo GPTQ \
--dataset NeelNanda/pile-10k \
--wbits 4 \
--group_size 128 \
--pad_max_length 2048 \
--use_max_length \
--seed 0 \
--gpu
```

This command will quantize the “opt-125m” model using the specified parameters. It is as simple as that!

Model Pruning

Despite the advancements in deep learning and quantization, deep neural networks often need help with their considerable size and computational requirements. Model pruning is an effective technique in addressing this issue, aiming to decrease the size of neural networks while maintaining their effectiveness.

Sparsity, often achieved by pruning, as discussed in the “Sparsity in Deep Learning” study in 2021, is a technique for reducing the computational demands of LLMs. It involves removing redundant or less essential weights

and activations. This approach can substantially decrease off-chip memory consumption, corresponding memory traffic, energy use, and latency.

Model pruning aims to create a smaller and more efficient model while maintaining accuracy. The advantages of pruning include faster inference times, a smaller memory footprint, and enhanced energy efficiency. This makes pruned models ideal for applications with limited computational resources, such as mobile apps, IoT devices, and edge computing, where speed and efficiency are critical for real-time performance.

Pruning can be classified into two categories: weight pruning and activation pruning. Weight pruning is divided into unstructured and structured forms. Unstructured pruning allows for flexible sparsity patterns, while structured pruning imposes specific patterns that help improve memory efficiency, reduce energy consumption, and lower latency without requiring specialized hardware. However, structured pruning generally offers lower compression compared to unstructured approaches. Activation pruning focuses on eliminating redundant activations during inference, which can be particularly useful for transformer models. This method requires dynamically identifying and removing unnecessary activations during runtime.

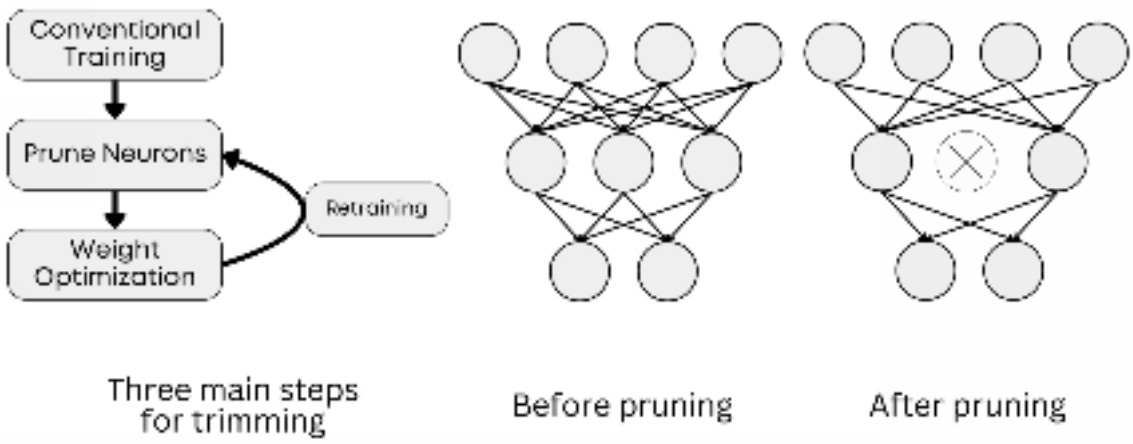
Several techniques and methodologies exist for model pruning, each with advantages and tradeoffs. The following section investigates the three most relevant and interesting approaches.

Magnitude-based Pruning (or Unstructured Pruning)

Magnitude-based or unstructured pruning removes weights or activations of small magnitudes in a model. The underlying principle is that smaller weights have a lesser impact on the model's overall performance and can thus be eliminated without significant loss of functionality.

The concept was elaborately presented in the paper "Network Trimming: A Data-Driven Neuron Pruning Approach towards Efficient Deep Architectures" in 2016. It introduced an optimization technique for deep neural networks by pruning non-essential neurons. Network Trimming is

grounded in the observation that many neurons in a large network yield zero outputs, regardless of the input. Neurons that consistently produce zero activations are deemed redundant and can be pruned without adversely affecting the network's accuracy. The process entails a cycle of pruning and subsequent retraining, where the network's initial pre-pruning weights serve as the starting point. The research demonstrates that this approach can significantly reduce the number of parameters in computer vision neural networks, maintaining or enhancing the accuracy compared to the original network.



Left: the usual pruning process, middle: an example neural network pre-pruning, and right: the same network after pruning the middle weight.

Another notable paper, "A Simple and Effective Pruning Approach for Large Language Models," published in 2023, presents a pruning method named Wanda (pruning by Weights and Activations) for LLMs. In this context, pruning selectively eliminates a portion of the network's weights to preserve performance while reducing the model's size. It targets weights based on the smallest magnitudes multiplied by their corresponding input activations, assessed on a per-output basis. This strategy draws inspiration from the emergent large-magnitude features in LLMs. One of the significant advantages of Wanda is it does not necessitate retraining or weight updates, so the pruned LLM can be employed immediately.

Structured Pruning

Structured pruning focuses on specific structural elements within a neural network, such as channels in convolutional layers or neurons in fully connected layers. The paper "Structured Pruning of Deep Convolutional Neural Networks", published in 2015, introduced an innovative approach to network pruning that integrates structured sparsity at various levels, including channel-wise, kernel-wise, and intra-kernel strided sparsity, particularly effective in saving computational resources. The technique employs a particle filtering method to assess the importance of network connections and pathways and assigns significance based on the misclassification rate linked with each connectivity pattern. After the pruning process, the network undergoes retraining to recover any performance losses that may have occurred.

The Lottery Ticket Hypothesis

The paper "The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks", published in 2019, introduced a novel viewpoint on neural network pruning with the "Lottery Ticket Hypothesis." This hypothesis proposes that within large, randomly initialized, feed-forward networks, there are smaller subnetworks ("winning tickets") that, if trained independently, can reach a level of test accuracy comparable to the original network in a similar number of iterations. These "winning tickets" are distinguished by their initial weight configurations, which are particularly helpful for practical training.

The algorithm developed to locate these "winning tickets" conducted a series of experiments to prove the hypothesis. They consistently found "winning tickets" that were only 10-20% the size of various full-sized, fully connected, and convolutional feed-forward architectures tested on the MNIST and CIFAR10 datasets. These smaller subnetworks did not just replicate the performance of the original network; they frequently outperformed it, exhibiting quicker learning and higher test accuracy.

Speculative Decoding

Speculative decoding is another optimization technique for improving the inference speed of large language models (LLMs) without compromising output quality. It leverages an unintuitive observation: forwarding an LLM on a single input token can take nearly as much time as forwarding it on K input tokens in a batch, where K is larger than one might expect. This is because LLM inference is heavily memory-bound, most of the work involves reading the transformer's weights from VRAM into the on-chip cache rather than computation. Once these weights are loaded, applying them to multiple input vectors is relatively cheap. However, we can't simply generate tokens in large batches due to the serial dependency in autoregressive generation, where each token depends on all previous tokens.

Speculative decoding cleverly circumvents this limitation by using a smaller, faster "draft" model to generate a candidate sequence of K tokens. These draft tokens are then fed together through the large target LLM in a batch, which is almost as fast as processing a single token. The target LLM then verifies these candidates from left to right. When a draft token is accepted, the system can immediately move to the next token, effectively skipping ahead. If a disagreement occurs, the remaining draft is discarded, incurring some wasted computation. This approach is effective because many tokens in a sequence are predictable based on context and can be easily handled by a smaller model. For instance, in the phrase "The cat is on the," a small model can confidently predict the next token as "mat" without needing the full power of a large LLM. These are the "easy" tokens—words or phrases that follow common patterns or predictable grammatical structures. By quickly generating these easy tokens, the system speeds through familiar parts of the text. The large model only steps in when the smaller model encounters a more ambiguous or complex token, such as in a less common phrase like "quantum entanglement is a," where it might disagree on the next word. In such cases, the draft is discarded, and the large model takes over to ensure accuracy. The system can leap through these easy parts quickly, only slowing down for the more challenging tokens where the large model disagrees with the draft.

Speculative decoding represents a clever way to partially overcome the sequential nature of autoregressive generation in LLMs, offering substantial performance improvements without compromising output quality.

The key factors affecting speculative decoding performance are the draft model's latency and its token acceptance rate (TAR). Interestingly, a draft model's accuracy on standard language modeling tasks doesn't strongly correlate with its TAR in speculative decoding. The draft model's latency is primarily determined by its depth (number of layers), while its width (size of layers) has less impact on speed. This insight suggests that wider, shallower draft models may be more efficient for speculative decoding than deeper models with the same parameter count. When designing draft models specifically for this purpose, it's crucial to balance latency reduction with maintaining a reasonable TAR. Recent experiments have shown that pruning larger models to create wide, shallow configurations can provide significant throughput improvements over existing approaches. As LLM architectures and hardware capabilities evolve, the optimal draft model design may change, but the principles of optimizing for both low latency and sufficient TAR are likely to remain fundamental to effective speculative decoding.

Tutorial: Deploying a Quantized LLM on a CPU on Google Cloud Platform (GCP)

- Find the Notebook for this section at towardsai.net/book.

In the previous section, we explored model optimization techniques like quantization, pruning, and speculative decoding. Quantization is especially useful for reducing memory and computational demands, making it possible to run large language models (LLMs) on small machines or even CPUs instead of more expensive and less available GPUs. While CPUs may not match the performance of GPUs, they can sometimes be cost-effective, especially when using quantized models that maintain high accuracy. We will also explore using and deploying open-source LLMs on cloud platforms like Together AI, Groq, and Fireworks AI, comparing deployment

options, pricing models, and performance metrics to optimize cost and scalability.

This tutorial focuses on deploying a quantized LLM on a CPU instance using the Google Cloud Platform (GCP). We'll leverage Intel Neural Compressor to optimize the model for inference, showcasing how quantization can reduce memory usage by lowering the precision of weights from 32 bits to 8 bits. Additionally, we'll evaluate the performance gains and cost reductions achieved. While this library supports AMD CPUs, ARM CPUs, and Nvidia GPUs via ONNX Runtime, our focus will be on Intel CPUs.

?? ONNX is a format that standardizes AI models for interoperability and efficient CPU execution, enhancing performance through hardware optimizations.

The Hugging Face Optimum and Intel Neural Compressor libraries provide a robust toolkit for optimizing models during inference, especially on Intel architectures. Hugging Face Optimum serves as a bridge between the Hugging Face transformers and diffusers libraries, offering tools that enhance efficiency when training and running models on specific hardware. The Intel Neural Compressor, an open-source library, simplifies the application of compression techniques such as quantization, pruning, and knowledge distillation. It provides automatic, accuracy-driven tuning strategies for quantization, supporting static quantization (where parameters are fixed before inference), dynamic quantization (which adjusts parameters at runtime based on input data), and quantization-aware (where the model is trained with quantization in mind) training methods while ensuring accuracy. The library also includes weight pruning techniques to achieve desired sparsity levels. Together, these libraries are highly effective for optimizing large language models in real-world deployments.

Quantizing the Model

Start by setting up the essential libraries. Install the optimum-intel package straight from the GitHub repository.

```
pip install optimum[neural-compressor]===1.22.0
pip install onnx===1.14.1 evaluate===0.4.0
```

First, we will perform a simple quantization that requires no coding or lengthy training process to optimize the models. There are two primary approaches: Static Quantization and Dynamic Quantization. Static quantization involves applying quantization to the model weights before deployment, while dynamic quantization applies quantization at runtime, typically to the activations based on the data being processed. For transformer-based neural networks, dynamic quantization is the preferred approach because it adapts to the varying data distributions in activations, enhancing model performance and efficiency during runtime. You can use the optimum-cli command on the terminal to perform dynamic quantization. Specify the path to your custom model or select a model from the Huggingface Hub using the --model parameter. The --output argument specifies the name of the final model. We will test Facebook's OPT model with 1.3 billion parameters.

```
optimum-cli inc quantize --model facebook/opt-1.3b --output opt1.3b-quantized
```

The script will automatically load the model and manage the quantization process. If the script does not recognize your model, use the --task parameter. For a complete list of supported tasks, check the source code at towardsai.net/book.

This method can be conveniently incorporated with a single command. However, this approach may not offer sufficient flexibility for more complex applications requiring detailed control. The library offers a conditional quantization method, enabling users to set a specific target. This method requires additional steps for coding and implementing a function but provides greater control over the process. For example, an evaluation function can ensure that the model is quantized with no more than a 1% reduction in accuracy.

Load the model's weights and its corresponding tokenizer.

```
from transformers import AutoTokenizer, AutoModelForQuestionAnswering

model_name = "aman-mehra/opt-1.3b-finetune-squad-ep-0.4-lr-2e-05-wd-0.01"
tokenizer = AutoTokenizer.from_pretrained(model_name, cache_dir="./opt-1.3b")
model = AutoModelForQuestionAnswering.from_pretrained(
model_name,
cache_dir="./opt-1.3b"
)
```

We are loading a model specifically for the question-answering task. The model you select must be finetuned for question-answering tasks before

executing quantization. We used a finetuned version of the OPT-1.3 model in this example.

Selecting a task to establish a goal for the quantization target function and manage the process effectively is crucial. The task and evaluation metrics can differ significantly for each task. For example, summarization is assessed with ROUGE, translation with BLEU, or classification by simple accuracy.

Next, define the evaluation metric to measure the model's accuracy and select a corresponding benchmark dataset.

```
import evaluate
from datasets import load_dataset

task_evaluator = evaluate.evaluator("question-answering")
```

```
eval_dataset = load_dataset("squad", split="validation", cache_dir="./squad-ds")
eval_dataset = eval_dataset.select(range(64)) # Use a subset of dataset
```

The evaluator method loads the necessary functions for evaluating the question-answering task. The load_dataset function from the Hugging Face library imports a dataset into memory. It accepts several parameters (like the dataset name), splits to download (such as train, test, or validation), and stores the dataset.

Now, we can construct the evaluation function.

```
from transformers import pipeline

qa_pipeline = pipeline("question-answering", model=model, tokenizer=tokenizer)

def eval_fn(model):
qa_pipeline.model = model
metrics = task_evaluator.compute(model_or_pipeline=qa_pipeline,
data=eval_dataset, metric="squad")
return metrics["f1"]
```

Create a pipeline that connects the model with the tokenizer to calculate the model's performance. Using the evaluator's compute function, evaluate the model's performance using the pipeline and the evaluation dataset split. The eval_fn function calculates the accuracy and returns it as a percentage. For successful quantization, it is essential to have a clear understanding of the necessary configurations.

```
from neural_compressor.config import AccuracyCriterion, TuningCriterion,
PostTrainingQuantConfig
```

```
# Set the accepted accuracy loss to 1%
accuracy_criterion = AccuracyCriterion(tolerable_loss=0.01)

# Set the maximum number of trials to 10
tuning_criterion = TuningCriterion(max_trials=10)

quantization_config = PostTrainingQuantConfig(
approach= "dynamic",
accuracy_criterion=accuracy_criterion,
tuning_criterion=tuning_criterion
)
```

The PostTrainingQuantConfig configuration class defines the necessary parameters for the quantization process. In this case, we are using the dynamic quantization approach, with an acceptance of up to a 1% loss in accuracy. This is managed by specifying parameters in the AccuracyCriterion class. The TuningCriterion class determines the maximum number of runs to execute before the quantization process finishes. Finally, a quantizer object is defined using the INCQuantizer class, which inputs both the model and the evaluation function.

Initiate the quantization process by invoking the quantize method on this object:

```
from optimum.intel import INCQuantizer

quantizer = INCQuantizer.from_pretrained(model, eval_fn=eval_fn)

quantizer.quantize(quantization_config=quantization_config,
save_directory="opt1.3b-quantized")
```

Note that running the codes in this section on the Google Colab instance may not be possible due to memory limitations. You may want to replace the model “facebook/opt-1.3b” with a more compact option like “distilbert-base-cased-distilled-squad” to run the code within Google Colab’s memory capacity limitations.

Using the Quantized Model for Inference

Before initiating the inference, load the pretrained tokenizer using the AutoTokenizer class. Since quantization does not modify the model’s vocabulary, we will use the same tokenizer as the base model.

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("facebook/opt-1.3b")
```

Load the model using the INCModelForCasualLM class from the Optimum package. This package also offers a range of loaders tailored to various tasks. For example, the INCModelForSequenceClassification loader is for classification tasks, and INCModelForQuestionAnswering is for question-answering tasks.

To use the from_pretrained method, provide the path to the quantized model from the previous section:

```
from optimum.intel import INCModelForCausalLM

model = INCModelForCausalLM.from_pretrained("./opt1.3b-quantized")
```

Finally, use the generate method from the Transformers library to input the prompt into the model and retrieve the response.

```
inputs = tokenizer("What does life mean? Describe in great details.", return_tensors="pt")

generation_output = model.generate(**inputs,
return_dict_in_generate=True,
output_scores=True,
min_length=512,
max_length=512,
num_beams=1,
do_sample=True,
repetition_penalty=1.5)
```

The last step is to convert the generated token IDs to words. The decoding process is the same as in the previous chapters.

```
print(tokenizer.decode(generation_output.sequences[0])) What does life mean? Describe in great details.\nThe meaning of life is one of the most profound and ancient questions in philosophy, spirituality, and science. It's a concept that encompasses numerous interpretations...
```

We are now ready to evaluate the text generation speed of the original model and the quantized model. We are ensuring that the models always generate 512 tokens by defining minimum and maximum length parameters to keep a consistent token count between the original model and its quantized counterpart for a fair comparison of their generation times. The

following table shows how long it took to generate a text output with 512 tokens using the two models.

Decoding Method	Vanilla *(seconds)*	Quantized *(seconds)*
Greedy	58.09	**26.847**
Beam Search (K=4)	144.77	**40.73**

These experiments were conducted on a server instance with a 4th Gen Intel Xeon Scalable processor featuring eight vCPU (4 cores) and 64GB of memory.

Deploying the Quantized Model on Compute Engine with GCP

In this section, we will guide you through deploying the quantized OPT-1.3B model using the Hugging Face Transformers library and FastAPI on a Google Cloud Platform (GCP) Compute Engine instance.

Begin by signing in to your Google Cloud Console and creating a new project if you haven't already. Ensure that billing is enabled, as Compute Engine requires it. Navigate to the "Compute Engine" section and create a new virtual machine instance. Choose a machine type optimized for CPU workloads; for the quantized OPT-1.3B model, a n1-standard-8 instance (8 vCPUs, 30 GB memory) should suffice.

Select the latest Ubuntu LTS as the operating system image. Before launching the instance, under the "Firewall" settings, enable "Allow HTTP traffic" and "Allow HTTPS traffic" to permit external access to your API. Once your instance is running, connect to it via SSH from the GCP console.

Update the package list and install Python 3 and pip: sudo apt update
sudo apt install python3-pip -y Create and activate a virtual environment.

```
python -m venv venv
source venv/bin/activate Install the necessary Python libraries, including Transformers, Optimum for Intel Neural Compressor support, FastAPI, and Uvicorn.
pip install transformers==4.34.0
pip install git+https://github.com/huggingface/optimum-intel.git@v1.11.0
```

```
pip install onnx==1.14.1 neural_compressor==2.2.1
pip install fastapi uvicorn[standard]
```

If you want your server to have the LLM locally so that it doesn't have to download it from the Hugging Hub, follow this step: Transfer the "opt1.3b-quantized" directory to your GCP instance. You can use scp or rsync for this. Next, from your local machine, do the following.

`scp -r opt1.3b-quantized username@YOUR_INSTANCE_IP:~/opt1.3b-quantized` Replace the username with your GCP instance username and YOUR_INSTANCE_IP with the external IP address of your instance.

Next, create a new Python script named app.py and add the following code.

```
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from transformers import AutoTokenizer
from optimum.intel import INCModelForCausalLM

app = FastAPI()

# Load the tokenizer
tokenizer = AutoTokenizer.from_pretrained("facebook/opt-1.3b")

# Load the quantized model from local
model_name_or_path = "./opt1.3b-quantized"
model = INCModelForCausalLM.from_pretrained(model_name_or_path)

# Load the quantized model from Hugging Face
#model_name_or_path = "<your-username>/opt1.3b-quantized"
#model = INCModelForCausalLM.from_pretrained(model_name_or_path)

# Define request and response models
class GenerationRequest(BaseModel):
prompt: str
max_length: int = 512
num_beams: int = 1

class GenerationResponse(BaseModel):
generated_text: str
```

```
# Define the inference endpoint
@app.post("/generate", response_model=GenerationResponse)
def generate_text(request: GenerationRequest):
try:
inputs = tokenizer(request.prompt, return_tensors="pt")
outputs = model.generate(
input_ids=inputs["input_ids"],
attention_mask=inputs["attention_mask"],
max_length=request.max_length,
num_beams=request.num_beams,
do_sample=True,
repetition_penalty=1.5,
)
generated_text = tokenizer.decode(outputs[0], skip_special_tokens=True)
return GenerationResponse(generated_text=generated_text)
except Exception as e:
raise HTTPException(status_code=500, detail=str(e))
```

The given code sets up a FastAPI application to serve a text generation model using the Transformers and Optimum libraries. First, it initializes the FastAPI app instance. The tokenizer for Facebook's OPT-1.3B model is loaded using `AutoTokenizer.from_pretrained`. Then, the quantized model is loaded from a local directory using the `INCModelForCausalLM` class. An alternative option to load the model from Hugging Face is commented out.

Two Pydantic models are defined: `GenerationRequest` to structure the input data (including the prompt, maximum length, and number of beams for text generation) and `GenerationResponse` to structure the output data, which will be the generated text.

The `generate_text` function is defined as an API endpoint at /generate, which accepts POST requests with a `GenerationRequest` payload. It tokenizes the input prompt and uses the quantized model to generate a response based on the provided parameters. The generated output is decoded back into text, which is returned as a `GenerationResponse`. If any error occurs during inference, it raises an HTTP 500 exception with the error message.

Finally, start the Uvicorn server to serve the FastAPI application: `uvicorn app:app --host 0.0.0.0 --port 8000`

The `--host 0.0.0.0` option makes the server accessible from external IP addresses, and `--port 8000` sets the listening port. If you cannot access the API from outside, you may need to create a firewall rule to allow inbound traffic on port 8000. In the GCP console, go to “VPC network” and select “Firewall.” Then click on “Create firewall rule” and configure it with the following settings: Name it “allow-fastapi,” set the targets to “All instances in the network,” use the source IP range “0.0.0.0/0,” and allow TCP traffic on port 8000.

You can now test the API using `curl` .

```
uvicorn app:app --host 0.0.0.0 --port 8000
```

```
curl -X POST "http://YOUR_INSTANCE_IP:8000/generate" \\
-H "Content-Type: application/json" \\
-d '{"prompt": "What is the meaning of life?", "max_length": 100}'
```

Replace `YOUR_INSTANCE_IP` with your instance’s external IP address. You should receive a JSON response with the generated text.

?? Remember to shut down or delete your Compute Engine instance when it's no longer needed, else you could have a bad surprise in the morning!

While this method is efficient for CPU-based deployments, there are alternative cloud platforms optimized for GPU-based workloads. For instance, Latitude.sh’s Launchpad service simplifies the deployment and finetuning of machine learning models with dedicated cutting-edge GPUs like the NVIDIA H100, making it a good choice for large language model inference and finetuning. Similarly, Together.ai provides a powerful inference engine tailored for open-source models like LLaMA, leveraging NVIDIA DGX Cloud for high-performance AI workloads, offering optimized solutions for deploying generative AI applications at scale. These alternatives provide flexible, GPU-accelerated environments that may better suit projects requiring more computational power.

Deploying Open-Source LLMs on Cloud Providers

Deploying open-source large language models (LLMs) on cloud platforms such as Together AI, Groq, Fireworks AI, and Replicate has become a popular approach for businesses and developers seeking high-performance AI capabilities without investing in their own infrastructure. These platforms offer flexible deployment options, typically falling into two categories: on-demand usage and instance-based deployments via dedicated endpoints. Each method has its advantages and tradeoffs, particularly in pricing, performance, and scalability.

In **on-demand usage**, models are hosted in a serverless environment, meaning you pay only for the compute resources consumed during each query. This is ideal for users who require occasional or unpredictable usage. Pricing is typically calculated based on the number of tokens processed, both for input and output. For example, on Together AI, pricing for the "Llama 3.1 70B" model is $0.88 per million tokens on the Turbo endpoint as of September 2024. Smaller models, such as Llama 3.1 8B, cost significantly less, with rates starting at $0.10 per million tokens for the Lite variant as of the same date. This type of usage is flexible, does not require managing infrastructure, and keeps on getting cheaper overtime, making it appealing for smaller projects or businesses in the experimentation phase. Users only have to call the API, and the platform handles the rest. Platforms like Groq and Fireworks AI offer similar on-demand models, with Groq being particularly noted for its ultra-fast inference speeds due to its proprietary hardware optimized for low-latency performance.

For higher-demand applications, where consistent performance and cost-efficiency are crucial, deploying LLMs via **dedicated instances** can be more advantageous. Dedicated endpoints provide reserved GPU resources exclusively for your model, ensuring that your service is not affected by other users on the platform. This is particularly useful in cases of high-traffic applications where the predictability of performance is critical. Platforms like Together AI offer multiple hardware options for dedicated instances, such as A100, H100, and RTX-6000 GPUs. Pricing varies depending on the GPU selected, with options like the H100 80GB costing

$0.128 per minute as of September 2024 and the RTX-6000 at $0.034 per minute. These endpoints are especially useful for finetuned models that need consistent access to resources or for businesses with sustained, high-traffic requirements. Dedicated endpoints allow users to scale up or down by choosing specific hardware configurations, and they often come with features like custom scaling and optimized batch sizes for better cost management.

To decide whether to use a dedicated instance or on-demand models, you should estimate the number of tokens that your model would manage every minute on average and then compare the total price of managing those estimated tokens via on-demand usage or dedicated endpoints, respectively. For example, using the “Llama 3.1 70B” turbo endpoint, the breakpoint is at around hundreds of thousands of tokens per minute as of September 2024. It keeps getting higher because of the providers’ constant price reductions.

Inference speed can vary widely across cloud platforms, depending on the underlying hardware and the model being deployed. Groq is known for leading in speed, leveraging its Language Processing Units (LPUs). Meanwhile, Together AI and Replicate offer competitive speeds that vary based on the hardware configurations chosen. For example, OpenAI’s GPT models are recognized for their high throughput and low latency, while Fireworks AI’s Llama models provide reliable performance. Groq’s Llama models are particularly fast due to their optimized hardware, and Together AI’s Llama Turbo models strike a balance between speed and efficiency.

It's important to note that performance metrics like latency and throughput depend on model size and architecture, so direct comparisons between providers can be challenging if different models are used.

Recap

In this chapter, we explored optimization techniques and deployment strategies for large language models, focusing on overcoming the challenges associated with their size, memory usage, and computational demands. Deploying LLMs efficiently requires a balance between

maintaining model performance and minimizing resource consumption, latency, and cost.

We began by discussing **model distillation**, a method for transferring knowledge from a large model (teacher) to a smaller, more efficient one (student). This process allows the student model to mimic the teacher's predictions, achieving similar performance but with reduced resource requirements. Distillation plays a relevant role in making LLMs scalable by offering faster inference times, lower latency, and energy-efficient deployment, especially on devices with limited computational power. It reduces the computational and memory costs associated with deploying large models, making it a valuable technique for real-time applications like chatbots, translation tools, and mobile devices.

We also covered **quantization**, which involves reducing the precision of model parameters to decrease memory footprint and improve inference speed. Quantization can significantly decrease the memory footprint of a model while speeding up inference times, making it ideal for use cases that demand low-latency responses, such as chatbots and real-time translation services. We covered various quantization methods, including scalar quantization and product quantization, while also addressing more advanced approaches like GPTQ and LLM.int8(), which aim to preserve model accuracy despite aggressive compression.

Next, we examined **model pruning**, a technique that reduces model size by removing non-essential neurons, weights, or layers from the model, decreasing the number of parameters and reducing computational costs. We discussed two primary types of pruning: unstructured pruning, which selectively removes individual weights based on their magnitude, and structured pruning, which removes entire blocks of neurons or channels, offering more predictable improvements in latency and memory use.

We also introduced **Speculative decoding** as an inference-time optimization technique for speeding up autoregressive token generation in LLMs. By using a smaller draft model to propose multiple candidate tokens and having the larger model validate them in parallel, speculative decoding reduces the latency associated with sequential token generation without sacrificing output quality. This method takes advantage of the memory-

bound nature of LLM inference, where the cost of processing multiple tokens in a batch is nearly the same as processing a single token. Speculative decoding allows for faster generation without sacrificing accuracy.

The chapter also included practical guides on deploying LLMs on cloud platforms. Deploying LLMs on Google Cloud Platform (GCP) was covered, providing step-by-step instructions on using tools like Hugging Face's Optimum and Intel's Neural Compressor to deploy a quantized model on CPU instances. GCP's scalable infrastructure makes it a great option for hosting optimized LLMs, especially for those seeking cost-effective deployment solutions.

We also explored the deployment of open-source LLMs on cloud platforms like Together AI, Groq, Fireworks AI, and Replicate. These platforms offer flexible options for deploying LLMs via on-demand serverless models or dedicated GPU instances. On-demand models are cost-effective for sporadic use, with pricing based on tokens processed, while dedicated endpoints provide consistent performance for high-traffic applications. We also compared hardware configurations and highlighted latency and throughput metrics across platforms to help determine the best deployment strategy for performance and cost optimization.

Conclusion

The LLM revolution traces its roots to the 1954 Bag of Words Model and early work on neural networks, particularly the development of the perceptron in the 1950s and subsequent advancements in the 1980s and 90s. A surprising unsung hero of the deep learning and LLM revolution came from the gaming industry. The 1993 release of Doom drove demand and an economic incentive for 3D graphics and led to hardware progress closer to GPUs in time for Quake in 1996. Ian Buck's 2006 invention of CUDA at Nvidia unlocked GPUs for general computing, paving the way for GPU R&D funded by the gaming industry to be used for scientific and AI research. A key breakthrough came in 2012 with AlexNet and the inspiration to use GPUs to train larger deep neural networks. This paved the way for subsequent success in deep learning, which was further unlocked by the 2017 **Transformer architecture invention**, which revolutionized natural language processing. OpenAI's 2018 **GPT** model started to show the success of scaling “next word prediction,” leading to the 2022 release of **ChatGPT,** which was a breakthrough in widely accessible AI. By 2024, over 5 million developers were building on LLMs, turning a 70-year journey from simple word models to sophisticated AI into a global technological revolution reshaping our digital landscape.

While predictions for the impact of AI range from utopian productivity gains and a post-scarcity society to widespread job displacement or even misaligned superintelligence and the end of humanity, the reality is likely to be a complex and nuanced journey. The potential benefits of widespread AI adoption are vast and varied, promising to reshape the economic landscape profoundly. Enhanced productivity and efficiency are perhaps the most immediate and tangible benefits, with AI's ability to automate repetitive tasks, optimize workflows, and leverage data-driven insights significantly boosting productivity across various industries, from manufacturing to finance. Beyond efficiency gains, AI holds the promise of driving innovation and new product development.

These breakthroughs in Generative AI have left us with an extremely active and dynamic landscape of players. A helpful framework to understand the AI landscape is to segment it into a value chain. Understanding the builders and users of LLMs can help you better plan how to build your own LLM projects and select the best tools for doing so. This Generative AI value chain and contributors consists of 1) AI hardware designers and manufacturers such as Nvidia, TSMC, Broadcom, and Google (TPUs). 2) AI cloud platforms such as Azure, AWS, Google, Together.AI, Coreweave, Oracle. 3) Closed Foundation LLM model trainers providing access via API such as OpenAI, Cohere, and Anthropic, and 4) Open weights foundation model builders such as META, Mistral, and Google. 5) Open source platforms for accessing full models, such as Hugging Face. 6) Access to LLMs via consumer products such as ChatGPT, Perplexity, and Bing. 7) Individual Generative AI product builders (startups, side-projects, personal tools) and 8) Enterprise in-house LLM pipeline development for aiding internal workflows or improving external products.

On top of the Generative AI builders, we also have a diverse landscape of Generative AI users. This can be individuals using Generative AI for personal tasks, individuals using Generative AI to improve productivity at work, enterprises boosting productivity or innovation with Generative AI tools, and countries and other organizations beginning to adopt them.

Despite the rapid recent adoption of LLMs and hype over its promise, we think we are very, very early in adoption and economic value creation. Current off-the-shelf foundation models still have limitations that restrict their direct use in production, except for the most straightforward tasks. We think many tasks require a custom LLM pipeline to really provide economic value. After considering the time spent getting ChatGPT or Claude working on your task and then checking and correcting its answers, the productivity gains from “out the box” LLMs can start to be reduced. As you have learned, these custom LLM pipelines take a lot of work on data preparation, prompting, fine-tuning, RAG, tool use, and surrounding software and UI/UX to get LLMs to a sufficient level of reliability. These custom LLM pipelines are what is really needed at enterprises and wider LLM adoption. Some custom LLM tools will be built in-house, while some will be bought from a startup or tech company.

While “hallucinations” are still an issue, as you have learned, we already have many methods to address them, and humans themselves are not perfect. The reliability of LLM applications has the potential to cross this reliability threshold on “the march of 9s” (iteratively solving digits of reliability, e.g., 99% vs. 90%) for many more enterprise sub-tasks with 1) better employee training (how to use and build with LLMs, where not to use it, where not to trust it), 2) more work and iteration on advanced custom LLM pipelines (RAG, fine-tuning, agents, data preparation,) and 3) better foundational LLMs (in particular with better reasoning capabilities).

Crucially, progress on any of these drivers individually could be enough to allow Generative AI to start contributing economic value to specific workflows, and we think those using the technology now will be best prepared to create value as new capabilities are unlocked.

In “Building LLMs for Production: Enhancing LLM Abilities and Reliability with Prompting, Fine-Tuning, and RAG,” we combined prompt engineering, retrieval-augmented generation (RAG), and fine-tuning as the essential steps for adapting large language models to specific industries and workflows for use in scalable, customer-ready applications. The journey through this book emphasizes a systematic approach to improving LLMs, from the initial considerations of their built-in limitations to exploring effective strategies for overcoming these challenges.

Prompt engineering helps steer LLMs toward producing more accurate and contextually relevant outputs. Strategies like “chain-of-thought” prompting and “Few-Shot” prompting are discussed to improve model performance.

RAG helps supply LLMs with accurate, current, and structured information, thereby reducing errors and improving the model’s applicability to real-world scenarios. RAG’s ability to cite sources in its responses enables users to verify the provided information, increasing their trust in the model’s outputs.

Fine-tuning helps to enhance the ability of LLMs to complete new domain-specific tasks. Fine-tuning allows the model to adjust its internal parameters to suit the task better. From a base pre-trained LLM, instruction fine-tuning aims to create an LLM that understands prompts as instructions rather than just text to complete. It transforms the model into a general-purpose

assistant by adding more control over its behavior. We saw different fine-tuning techniques and concepts, such as standard fine-tuning, Low-Rank Adaptation (LoRA), Supervised Fine-Tuning (SFT), and Reinforcement Learning from Human Feedback (RLHF), addressing common fine-tuning challenges such as high memory requirements and computational inefficiency for generating responses that are more accurate, secure, and in line with human expectations.

These techniques collectively form a foundation for creating AI products that meet users' technological needs and expectations across various industries. Combining prompting, RAG, and fine-tuning will lead to highly tailored AI solutions for specific industries or niches, which need a lot of industry-specific data.

In this book, we also identified and tested the emergent tech stack for utilizing LLMs. Frameworks such as LangChain make working with LLMs easier. These frameworks allow you to quickly integrate LLMs into your software solutions and create more sophisticated systems, such as interactive agents, that execute complex tasks.

LlamaIndex makes it easy to add RAG to your AI application. It helps you with the tooling necessary to extract relevant information from your data. It integrates methods such as query expansion, transformations, and construction techniques to create an efficient retrieval engine. Additionally, advanced strategies such as reranking, recursive retrieval, and small-to-big retrieval can significantly improve the search process. These methods increase accuracy and aid in better finding and using the most relevant chunks of data in your whole dataset.

As we anticipate the future of AI applications in production settings, the book guides you on the ongoing innovation and refinement in the field. It transitions from identifying foundational knowledge to offering sophisticated strategies for applying LLMs in practical applications, stressing the significance of continuous experimentation, adaptation, and ethical deployment of AI technologies.

"Building LLMs for Production" detailed a process for integrating LLMs into functional applications. It encourages a focus on strategic adjustments, reliability assessments, and development centered on user needs. You may

encounter certain difficulties or bugs as you apply the lessons from this book. However, these are anticipated and will help you in your learning process. We are sharing our open-source AI Tutor chatbot to assist you when needed and help on our Discord community Learn AI Together with a dedicated channel for this book. This tool has been created using the same tools we teach in this book. We built a RAG system that provides an LLM with access to the latest documentation from all significant tools, such as LangChain and LlamaIndex, including our previous free courses.

Further Reading and Courses

Towards AI Open-Source AITutor and community for help throughout this book:

- https://aitutor.towardsai.net
- https://discord.gg/learnaitogether

Previous Courses:

- Training & Fine-Tuning LLMs for Production
- LangChain & Vector Databases in Production
- Retrieval-augmented generation for Production with LangChain & LlamaIndex

Note: These courses are hosted on the Activeloop website.

Free Resources:

- Towards AI blog and newsletter for the latest developments in AI/LLMs.
- What's AI YouTube videos for understanding technical AI concepts.
- Towards AI Resource Library for hands-on tutorials.

We also have a conversion course for software developers and machine learning engineers to master the technical skillset required for LLM development. Find it here!

Congratulations on successfully completing the book! You can now add these techniques and concepts to your resume: RAG | Prompting | Fine-Tuning | RLHF | LLM Agents | LLM Deployment | LangChain | LlamaIndex | Vector Databases | Building AI Assistants | Creating Chatbots | Chat with PDFs | Summarization | Deployment Optimizations | Transformers Architecture | Eliminating Hallucination | Benchmarking Performance

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